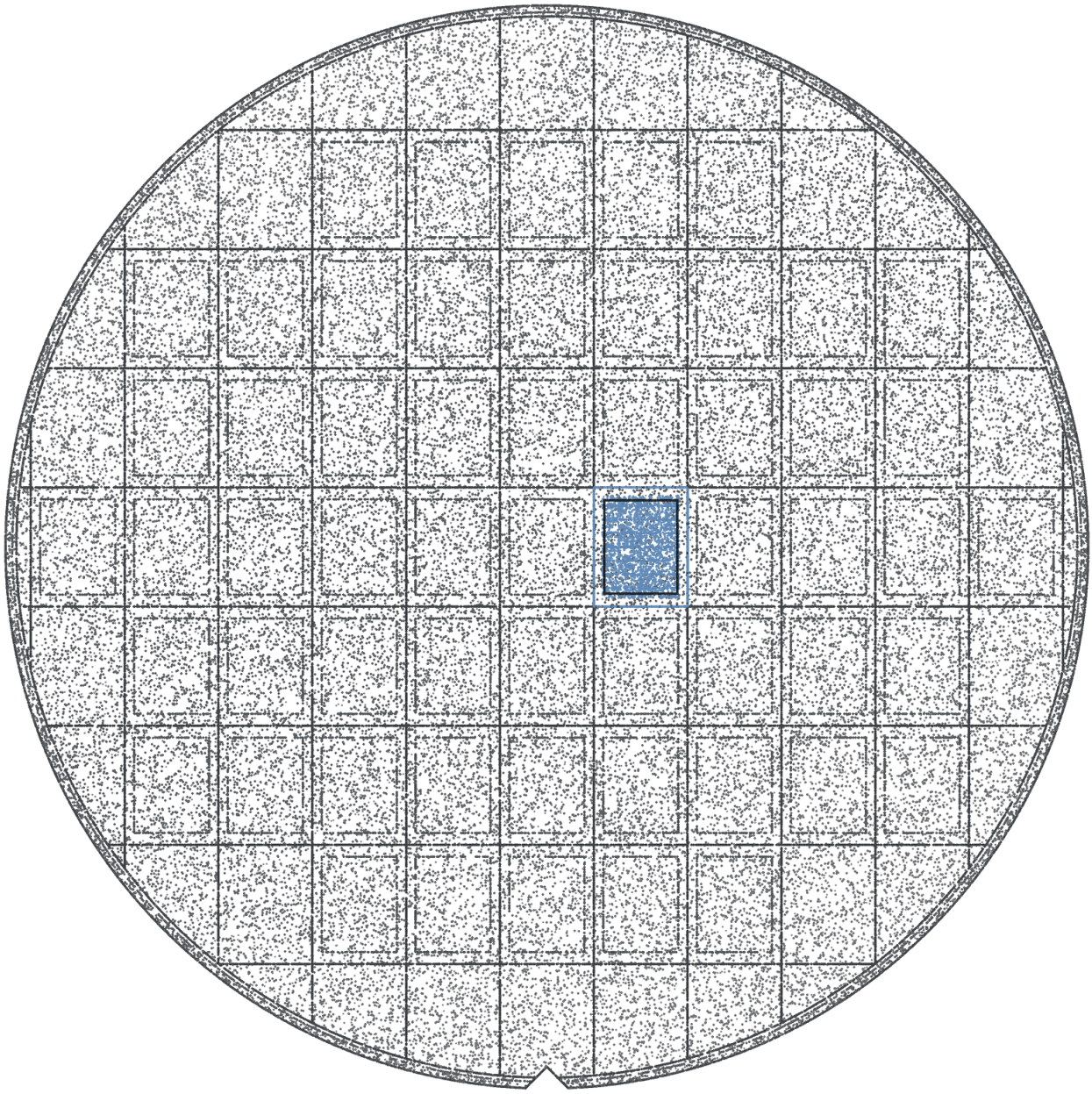


CHIP SUPPLY CHAIN

How AI chips are made

Every stage of the supply chain, and who controls it.

Yashvardhan Sharma with Claude

The Chain: seventeen chapters.

Edition of 20 October 2026 · chipsupplychain.org

The Chain

This page covers each of the fifteen stages of making an AI chip in detail, one chapter per stage, in the order a chip is made, from design software to the data center. Two more chapters at the end look across all the stages, at what a chip costs and who controls the chain. To help readers learn, each chapter has a 3D model, most have an animation, and every one ends with flashcards.

HOW TO READ A CHAPTER

Each stage chapter opens with a chokepoint card. It rates the stage on two scales, gives a price or market size, and says where China and the United States stand.

Concentration. How many firms could supply the stage today, rated Extreme, High, Moderate or Low.

Substitutability. How hard it would be for a well-funded state or firm to build an alternative, rated Easy, Moderate, Hard or Very hard.

Price or market size. What one unit costs, or what the stage earns in a year.

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ACROSS THE CHAIN

These two chapters are not stages. They look across all fifteen, at what a chip costs and at the contest over who controls each stage.

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CHAPTER
01
STAGE 1 OF 15

Chip Design, EDA and IP

Synopsys, Cadence and Siemens EDA sell the software used to design every leading-edge AI accelerator. In 2025 they took over 85 percent of an $18 billion market, licensed blocks included.

IN PLAIN TERMS

Chip design turns an idea for a new chip into a complete blueprint. A chip holds billions of transistors, tiny electrical switches. The blueprint fixes where each one sits and how it is wired to the others. The chip runs on a clock, and a signal has to cross those wires within one tick, less than a billionth of a second. Software from three companies, Synopsys, Cadence and Siemens, lays out the blueprint, checks it and signs it off. A foundry, a factory that makes chips for other companies, accepts only designs that have passed that software's checks. The foundry prints each layer of the design onto silicon through a stencil called a mask. Changing the blueprint once the masks are made takes new masks and a few months[1].

IN SHORT

The US government required a license to sell design software to anyone in China from late May to 2 July 2025[2]. Since October 2022 a narrower requirement has covered software specially made for chips with gate-all-around transistors, a transistor design the rule calls key to processes at 3 nm and below[3].

As of 2025, the think tank CSIS found Chinese design tools still far from replacing the full range of advanced capabilities US firms offer, especially for 5 nm and 3 nm chip design[4]. But Chinese design-software companies were winning share at older manufacturing processes, the research firm SemiAnalysis reported[5].

1 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due... SemiAnalysis, 24 July 2022. newsletter.semianalysis.com

2 CADENCE DESIGN SYSTEMS INC, Form 8-K current report for the period ended... U.S. Securities and Exchange Commission, 3 July 2025. sec.gov

3 Implementation of Certain 2021 Wassenaar Arrangement Decisions on Four Section... Federal Register, 15 August 2022. federalregister.gov

4 The Double-Edged Sword of Semiconductor Export Controls: Electronic Design... Center for Strategic and International Studies, 6 October 2025, p. 33. csis-website-prod.s3.amazonaws.com

5 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

CHOKEPOINT CARD EXTREME CONCENTRATION | HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	**Hard** Only three complete sets of chip design software exist[6]. A fourth would have to be written from nothing and approved by a foundry[5].	**$18B** world market for design software, simulation software and licensed circuit blocks in 2025[5]. SemiAnalysis put a single respin at leading-edge nodes at $50-100M in May 2026[5]

WHO LEADS

- US **Synopsys** 90%+ of the timing-signoff market at advanced nodes as of 2026[5]
- US **Cadence** 55-60% of the hardware emulator market at advanced nodes as of 2026[5]
- US **Siemens EDA** 85%+ of physical verification, the final layout check, at advanced nodes as of 2026[5]
- GB **Arm** About 50% of processor compute at the top cloud firms as of May 2026[7]
- US **Broadcom** $16.7B of AI semiconductor revenue in the third quarter of fiscal 2026, custom accelerators and networking combined[8]

WHERE IT IS MADE

- US **United States** Synopsys, Cadence, Broadcom, Marvell and Nvidia
- GB **United Kingdom** Arm processor and interconnect designs licensed from Cambridge, and since 2026 its own server chip
- TW **Taiwan** Alchip, Global Unichip and MediaTek: design services that carry a chip to the factory
- IL **Israel** Annapurna Labs, the AWS design house behind Trainium and Graviton
- IN **India** Nearly 20% of the world's chip design engineers, on the Indian government's count

WHY SUBSTITUTION IS SLOW

Changing one tool, such as the synthesis tool, means re-running the later steps of the design[5]. The foundry that makes the chip also specifies which signoff tools its customers must use[5].

WHERE CHINA STANDS

China designs its own AI accelerators, but Huawei's most powerful chip in 2026, the Ascend 950, delivers about half the computing performance of Nvidia's H100 from 2022, on Epoch AI's estimate[9]. CSIS found that as of 2025 Chinese design tools were still far from replacing the full range of advanced capabilities US firms offer, especially for 5 nm and 3 nm chips[4].

WHERE THE US STANDS

Synopsys and Cadence, the two largest design-software vendors by 2025 revenue, are American[5].

Design is the only stage of the supply chain that happens entirely in software, and the US government has already restricted it by letter. In late May 2025 the government required a license to sell chip design software to anyone in

4 The Double-Edged Sword of Semiconductor Export Controls: Electronic Design... Center for Strategic and International Studies, 6 October 2025, p. 33. csis-website-prod.s3.amazonaws.com

5 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

6 The Double-Edged Sword of Semiconductor Export Controls: Electronic Design... Center for Strategic and International Studies, 6 October 2025, p. 13. csis-website-prod.s3.amazonaws.com

7 Arm delivers record-breaking quarter and full-year results. Arm, 6 May 2026. newsroom.arm.com

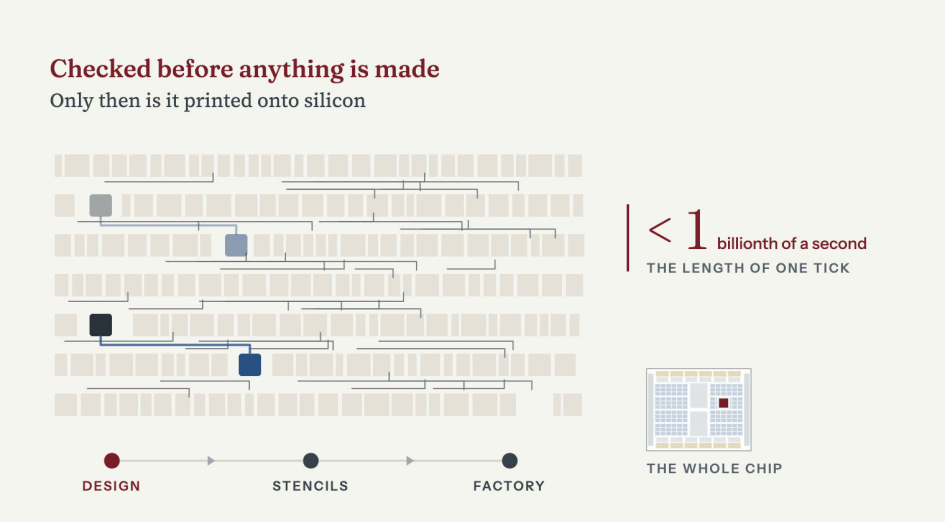
8 Broadcom Inc., Form 8-K current report for the period ended 2026-09-02 (8-K). U.S. Securities and Exchange Commission, 2 September 2026. sec.gov

9 Will Huawei catch up to Nvidia by 2030? Epoch AI, 24 September 2026. epoch.ai

China[2]. Cadence cut off its Chinese customers until the government lifted the requirement on 2 July[2].

How it works

How a chip is designed



1. **A blueprint.** Making a chip starts with design, which turns an idea into a complete blueprint. The blueprint fixes where each of the chip's billions of switches sits and how each one is wired to the others.
2. **One tick.** The chip works in ticks, each less than a billionth of a second. On every tick a signal leaves a register, a small store that holds one number, and it has to cross its wire to the next register before the next tick.
3. **Too far.** If the next register sits far away, the wire is long. The signal is still on its way when the next tick comes, so it arrives too late.
4. **Placed closer.** Design software from three companies, Synopsys, Cadence and Siemens, places the parts and wires them together. Here it moves the register closer, and the signal arrives in time.
5. **Checked.** The software checks that every signal arrives in time and that every shape can be printed, and a chip factory accepts only designs that pass the printing checks. The factory then prints the blueprint onto silicon through stencils, and changing it after that costs new stencils and a few months.

Simplified. Not to scale. The floorplan follows a GPU die. One tick, less than a billionth of a second, is slowed to about two seconds.

Design runs in five stages, all in software, and each stage's output is the next one's input.

- **Architecture.** Engineers fix the shape of the chip: the width of the block that multiplies numbers, the amount of fast on-chip memory beside it and the number of high-bandwidth memory stacks along the chip's edges. The output is a plan.
- **RTL.** Engineers write the plan as register-transfer-level code, RTL, in the hardware language SystemVerilog. A register is a small store on the chip that holds one number. On each tick of the chip's clock, every register takes

2 CADENCE DESIGN SYSTEMS INC, Form 8-K current report for the period ended... U.S. Securities and Exchange Commission, 3 July 2025. sec.gov

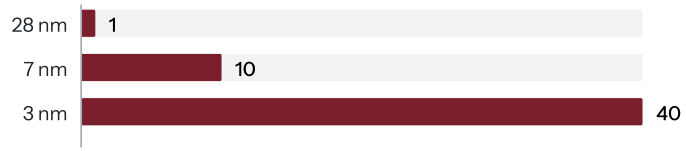
its next value at the same moment. The code says what every register holds on every tick. The output is a text description of the whole chip.

- **Verification.** A large team checks that the code does what the plan says. They run it in simulation and prove parts of it correct with mathematics. They also load it into emulators, cabinets of reprogrammable chips that behave like the design, so the chip's own software can boot before any silicon exists.
- **Physical design.** Software swaps each piece of the code for a logic cell, a small ready-made circuit the foundry has already proved it can make. It then places the cells on the chip, wires them together and proves that every signal arrives in time.
- **Tape-out.** Engineers hand the finished layout to the foundry, a step called tape-out. Software then turns the layout into pattern files for the mask shop, in steps that run after tape-out[10]. First it pre-distorts each shape so that the shape prints as drawn, a step called optical proximity correction. Then it cuts the layout into the files a mask writer reads, a step called mask data preparation.

All that checking exists because each revision after tape-out, called a respin, needs at least some new masks[1]. In 2022 SemiAnalysis put a full mask set at more than $1 million at 28 nm and more than $10 million at 7 nm[1]. The nanometer labels name process generations, called nodes, each finer than the last, and no longer measure anything. That year SemiAnalysis also expected 3 nm sets to push into the $40 million range[1]. Its May 2026 primer still puts a leading-edge mask set at $40 million, and a single leading-edge respin at $50 million to $100 million[5].

The same 2022 SemiAnalysis article reported that startups had taken chips on TSMC 7 nm, then a leading-edge process, from design to tape-out for $50 million to $75 million, everything included[1].

What one photomask set costs, by node, 2022 ($M). The 28 nm and 7 nm figures are minimums $M



Source: SemiAnalysis, The Dark Side of the Semiconductor Design Renaissance, July 2022. The 3 nm figure was a 2022 forecast. SemiAnalysis's May 2026 EDA Market Primer puts a leading-edge mask set at $40M

Variants and trade-offs

Merchant GPU versus custom ASIC

Nvidia designs its Blackwell GPU once and sells it to everyone, so the buyer gets mature software and a resale market. A custom accelerator, an ASIC, is built for one company's own workload. It handles a narrower range of tasks

1 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due... SemiAnalysis, 24 July 2022. newsletter.semianalysis.com

5 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

10 Unlocking post-tapeout flow scalability and performance with cloud computing. Siemens Digital Industries Software, 8 January 2025. blogs.sw.siemens.com

and is tuned to that workload. Ironwood, the seventh-generation tensor processing unit (TPU) Google announced in April 2025, carries 192 GB of memory per chip, six times its predecessor's, and delivers twice the performance per watt[11]. In April 2026 Google announced its eighth generation as two chips, the TPU 8t for training and the TPU 8i for inference, with up to twice Ironwood's performance per watt[12].

The EDA flow

Chip design software is known as electronic design automation, or EDA. Three complete EDA flows exist, each a set of tools that covers every step from code to signoff[6]. SemiAnalysis wrote in May 2026 that no startup can replicate one by writing it from scratch[5]. Changing the synthesis tool, which swaps the code for logic cells, means re-running the steps that follow: placing and wiring the cells, signoff and physical verification[5]. Foundries also specify which signoff tools their customers must use for tape-out[5].

For chips at advanced nodes, 7 nm and below, one vendor holds most of the market inside some steps of the flow. Synopsys alone has more than 90 percent of the market for static timing analysis, the check that every signal arrives in time at the chip's rated speed[5]. In May 2026 SemiAnalysis put Siemens EDA's Calibre at more than 85 percent of physical verification, the final layout check[5]. It put Cadence's Palladium at 55 to 60 percent of hardware emulation[5].

Each leading vendor's share of its own design-software segment at advanced nodes, 7 nm and below. Lower end of each reported range %



Source: SemiAnalysis, EDA Market Primer, May 2026, table of tool-level shares at advanced nodes

Buy the IP or build it

Design houses license ready-made circuit blocks, known as intellectual property or IP, including the processor cores, the interface controllers and the blocks that drive high-bandwidth memory. Arm licenses processor designs. In fiscal 2026 it reported $4.92 billion of revenue, $2.61 billion of it royalties[7]. It said it held about 50 percent of processor compute at the top cloud firms[7].

In 2026 Arm also launched the Arm AGI CPU, its first production silicon product[7]. By May 2026, servers built on it could be ordered from ASRock, Lenovo, Quanta and Supermicro[7].

5 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com
6 The Double-Edged Sword of Semiconductor Export Controls: Electronic Design... Center for Strategic and International Studies, 6 October 2025, p. 13. csis-website-prod.s3.amazonaws.com
7 Arm delivers record-breaking quarter and full-year results. Arm, 6 May 2026. newsroom.arm.com
11 Ironwood: The first Google TPU for the age of inference. Google, 9 April 2025. blog.google
12 Our eighth generation TPUs: two chips for the agentic era. Google, 22 April 2026. blog.google

The memory interface is hard to design, because its signals have to stay on time across thousands of microscopic solder bumps into a stacked memory die.

RISC-V is an open instruction set, the basic list of commands a processor understands, and it needs no license. Small RISC-V processors handle control tasks on the chip, and Nvidia uses them in every GPU[13].

Chiplets

The reticle limit caps the size of a single die, one piece of silicon[14]. That limit is the largest area a scanner, the machine that prints the chip's pattern, can expose in one shot. Many large accelerators are built from chiplets, several smaller dies wired together, and UCIe is an open industry standard for the links between them inside a package[15]. Version 3.0 of UCIe adds faster data rates, 48 and 64 gigatransfers per second[15]. In a December 2024 presentation, Broadcom said its 3.5D XDSiP platform packs more than 6,000 mm2 of silicon and 12 high-bandwidth memory stacks into one package[14].

Who makes it

Design software, simulation software and licensed blocks made up an $18 billion market in 2025, on SemiAnalysis's count[5]. Synopsys, Cadence and Siemens EDA took more than 85 percent of it, counting Ansys as part of Synopsys[5]. No other vendor had more than 5 percent of any core category[5]. Siemens EDA is a US-based subsidiary of the German firm Siemens[16].

Synopsys is the largest. Its own results put revenue at $7.054 billion for the fiscal year to 31 October 2025, with Ansys, bought that July, contributing $756.6 million[17]. Cadence, whose year ends in December, took $5.297 billion in calendar 2025[18].

The two fiscal years do not line up, so any total for this market depends on which twelve months are counted. It also depends on how much of Ansys it includes, because Ansys sells to other industries too. SemiAnalysis counts Synopsys at $8 billion for calendar 2025 with Ansys included[5]. On the same count Cadence is second, at $5.30 billion, and Siemens EDA third, at an estimated $2.2 billion to $2.5 billion[5]. It puts 31 percent of Ansys's business in semiconductors and high technology, 22 percent in aerospace and 18 percent in automotive[5].

5 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

13 Behind The Scenes of SHD Group's 2026 RISC-V Market Forecast. RISC-V International, 19 June 2026. riscv.org

14 3.5D XDSiP Platform Technology. Broadcom, 4 December 2024. docs.broadcom.com

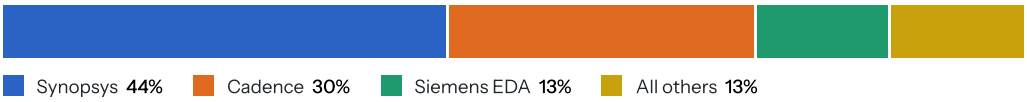
15 UCIe specifications. UCIe Consortium, accessed 9 September 2026. uciexpress.org

16 The Double-Edged Sword of Semiconductor Export Controls: Electronic Design... Center for Strategic and International Studies, 6 October 2025, p. 7. csis-website-prod.s3.amazonaws.com

17 SYNOPSYS INC, Form 8-K current report for the period ended 2025-12-10 (8-K). U.S. Securities and Exchange Commission, 10 December 2025. sec.gov

18 CADENCE DESIGN SYSTEMS INC, Form 8-K current report for the period ended... U.S. Securities and Exchange Commission, 17 February 2026. sec.gov

Share of revenue from design software, simulation software and licensed blocks, 2025. Synopsys includes part of Ansys's revenue %



Source: SemiAnalysis, EDA Market Primer, May 2026, table of calendar 2025 vendor shares

Some chip firms also build custom accelerators to a customer's design and sell the finished chips. Broadcom booked $16.7 billion of AI semiconductor revenue in the third quarter of fiscal 2026, up 221 percent year over year[8]. That figure covers both its custom AI accelerators and its AI networking chips[8].

In October 2025 OpenAI and Broadcom said they would deploy 10 gigawatts of OpenAI-designed accelerators[19]. Broadcom would turn the design into silicon and supply the Ethernet networking around it[19]. In April 2026 Broadcom and Meta extended their partnership to multiple gigawatts of MTIA, Meta's custom chip, including what they called the industry's first 2 nm AI accelerator[20].

Marvell, another US firm, also builds custom AI chips to each customer's specifications[21]. Below Broadcom and Marvell are the Taiwanese design-service houses. They carry a customer's circuit list all the way to a finished mask set on TSMC's newest nodes. One of them, Alchip, reported $992 million of revenue in 2025[22].

Chip design engineers are spread across more countries than these firms are. India alone has nearly 20 percent of the world's chip design engineers, the Indian government said in July 2025[23].

The chokepoint

Before the 2025 letters, a narrower control already applied. Since 14 October 2022, design software specially made for chips with gate-all-around transistors has needed a US export license for national-security reasons[3]. The rule calls that transistor design key to processes at 3 nm and below[3]. That license requirement covers exports to China[24]. The control carried out a December 2021 decision of the Wassenaar Arrangement, a group of countries that coordinate export controls[3].

3 Implementation of Certain 2021 Wassenaar Arrangement Decisions on Four Section... Federal Register, 15 August 2022. federalregister.gov

8 Broadcom Inc., Form 8-K current report for the period ended 2026-09-02 (8-K). U.S. Securities and Exchange Commission, 2 September 2026. sec.gov

19 OpenAI and Broadcom announce strategic collaboration to deploy 10 gigawatts of... Broadcom, 13 October 2025. investors.broadcom.com

20 Broadcom Announces Extended Partnership with Meta to Deploy Technology to... Broadcom, 14 April 2026. investors.broadcom.com

21 Marvell Technology, Inc., Form 10-K annual report for the period ended... U.S. Securities and Exchange Commission, 11 March 2026. sec.gov

22 Alchip Technologies, financials page. Alchip Technologies, accessed 9 September 2026. alchip.com

23 India's Semiconductor Vision Gathers Momentum with 3nm Chip Design and... Press Information Bureau, Government of India, 25 July 2025. pib.gov.in

24 Supplement No. 1 to Part 738, Title 15 -- Commerce Country Chart. Electronic Code of Federal Regulations, 30 September 2025. ecfr.gov

In late May 2025 the US Bureau of Industry and Security, the agency that issues export licenses, told the design-software vendors that sales to China now needed a license[2]. Cadence got its letter on 23 May[2]. The letter required a license to export design software and technology classified 3D991 and 3E991 to any party in China, or to a Chinese military end user anywhere[2]. Synopsys got its letter on 29 May and a second letter lifting the requirement on 2 July[25].

Unlike the 2022 control, the 2025 requirement came from the United States alone and took effect immediately. Synopsys's China revenue fell from 16 percent of its total in fiscal 2024 to 12 percent in fiscal 2025[5]. SemiAnalysis links the decline to tighter export restrictions and to local design-software companies winning share at mature nodes[5]. SemiAnalysis also quoted Synopsys's management saying that companies Synopsys cannot sell to look for alternatives, typically local design-software and licensed-block companies[5].

In October 2025 CSIS found that only a few companies, Cadence, Synopsys and Siemens, make design software that supports the entire chip design flow of more than 40 steps[6]. In a January 2021 report built on 2019 market data, CSET at Georgetown University found that one Chinese firm, Empyrean, could reportedly run a complete design flow only for certain analog and mixed-signal chips[26]. Mixed-signal chips combine analog and digital circuits. For all other chips, Chinese designers then relied on US tools[26]. The report found that other Chinese tools covered only narrow parts of the design flow or supplemented US tools, and largely did not support processes at 14 nm and below[26].

In its 2025 annual report, filed in April 2026, Empyrean says it now has complete design flows for analog, memory, radio-frequency and flat-panel display circuits[27]. It says its digital tools cover 80 percent of the main tools for digital design[28]. The same report lists foundry certifications for its circuit simulator at 8, 5 and 4 nm and for its physical verification tool, the final layout check, at 28 nm[29]. But CSIS found that as of 2025 Chinese design tools were still far from replacing the full range of advanced capabilities US firms offer, especially for 5 nm and 3 nm chip design[4].

China can design accelerators, but it cannot finish one at the leading edge without foreign tools, or build what it designs without foreign equipment. SMIC, China's leading foundry, makes chips for Huawei[9]. In June 2024 CSET

2 CADENCE DESIGN SYSTEMS INC, Form 8-K current report for the period ended... U.S. Securities and Exchange Commission, 3 July 2025. sec.gov

4 The Double-Edged Sword of Semiconductor Export Controls: Electronic Design... Center for Strategic and International Studies, 6 October 2025, p. 33. csis-website-prod.s3.amazonaws.com

5 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

6 The Double-Edged Sword of Semiconductor Export Controls: Electronic Design... Center for Strategic and International Studies, 6 October 2025, p. 13. csis-website-prod.s3.amazonaws.com

9 Will Huawei catch up to Nvidia by 2030? Epoch AI, 24 September 2026. epoch.ai

25 SYNOPSYS INC, Form 8-K current report for the period ended 2025-07-02 (8-K). U.S. Securities and Exchange Commission, 3 July 2025. sec.gov

26 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 50. cset.georgetown.edu

27 北京华大九天科技股份有限公司2025年年度报告全文. CNINFO, 27 April 2026, p. 18. static.cninfo.com.cn

28 北京华大九天科技股份有限公司2025年年度报告全文. CNINFO, 27 April 2026, p. 14. static.cninfo.com.cn

29 北京华大九天科技股份有限公司2025年年度报告全文. CNINFO, 27 April 2026, p. 15. static.cninfo.com.cn

found that SMIC's limited 7 nm capacity forced Huawei to choose between making its most advanced phone chips and its AI chips[30].

On Epoch AI's estimates, Huawei's flagship chip for 2026, the Ascend 950, delivers about half the computing performance of Nvidia's H100, which began shipping in 2022[9]. Epoch puts Nvidia's B300 at 6.75 times the 950's theoretical peak performance[9]. Epoch's median estimate is that Huawei will make about 1.5 million Ascend chips in 2026, against 5.9 million for Nvidia[9]. On those estimates, Huawei's 2026 output would carry less than 4 percent of the computing power Nvidia ships that year[9].

Epoch puts Huawei's chips three to four years behind Nvidia's until at least 2030, for three reasons[9]. SMIC cannot pack transistors as densely as TSMC, which makes Nvidia's. Chinese memory makers are only starting to produce high-bandwidth memory. And Huawei's software for programming its chips is far less mature than Nvidia's[9].

FIVE THINGS TO REMEMBER

- Chip design turns an idea into a blueprint, and a factory builds only designs checked in approved software.
- Synopsys, Cadence and Siemens EDA sell the software used to design every leading-edge AI chip.
- To compete with them, a new vendor would have to write a complete set of tools and win a foundry's approval.
- China designs its own AI chips, but in October 2025 CSIS found its design software still far from replacing US tools, especially for 5 nm and 3 nm chips.
- Huawei's best AI chip trails Nvidia's by three to four years, on Epoch AI's estimate.

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CHAPTER
02
STAGE 2 OF 15

Silicon and Wafers

Firms based in Japan, Taiwan, Germany and South Korea make most 300 mm wafers used for advanced chips. China makes most polysilicon, almost all of it for solar cells.

IN PLAIN TERMS

Every chip is built on a wafer, a thin round slice of silicon about 30 centimeters across that carries hundreds of chips at once. Making one starts with quartz rock, which is silicon bound to oxygen[1]. The silicon is refined until metal impurities are no more than a few parts per billion[2]. It is then melted, and a small seed crystal is lowered to the surface and drawn slowly back up. The melt freezes onto the seed as one long cylinder, a single crystal with its atoms lined up in one unbroken pattern from end to end. Saws cut the cylinder into discs, and each disc is polished to a mirror finish and made extremely flat across its whole face[2]. In 2025 the top five wafer makers served around 75 percent of the world market, and none of them is American or Chinese[3].

IN SHORT

Japanese companies held 53 percent of the world's silicon wafer sales in 2023, by a Fuji Keizai survey that Japan's trade ministry reproduced[4]. Firms based in Taiwan, South Korea and Europe held most of the rest[4]. China's largest maker of 300 mm wafers, Eswin, says it ranked sixth in the world with about 7.7 percent of global 300 mm shipments in the first half of 2026[5]. Chips also need electronic-grade polysilicon, the raw material for wafers, and none of the five major suppliers of it that the German wafer maker Siltronic named for 2025 is Chinese[6].

GlobalWafers of Taiwan opened a $3.5 billion plant in Texas in May 2025, the first production line of its kind built in the United States in over twenty years[7]. US Presidential Proclamation 11052 sets a minimum import price of $100 per kilogram for polysilicon ingots and wafers from 4 December 2026[8].

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8 Adjusting Imports of Polysilicon and Its Derivatives Into the United States. Federal Register, 11 August 2026. federalregister.gov

CHOKEPOINT CARD HIGH CONCENTRATION | MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
High	Moderate Several firms already make 300 mm wafers that fabs have approved, but customers have to test and approve a new wafer plant before they buy from it[9].	$11.4bn world wafer market in 2025, on 12,973 million square inches shipped[10]

WHO LEADS

- JP **Shin-Etsu Handotai** Its parent's electronics materials segment, wafers included, had ¥1,015.7bn of sales in the year to March 2026[11]
- JP **SUMCO** ¥409,670m of net sales in 2025[12]
- TW **GlobalWafers** NT$60.6bn revenue in 2025[13]
- KR **SK Siltron** One of the top five wafer makers by revenue in the first three quarters of 2025[3]
- DE **Siltronic** €1,346.7m revenue in 2025[14]

WHERE IT IS MADE

- JP **Japan** Shin-Etsu and SUMCO crystal growth and polishing
- TW **Taiwan** GlobalWafers headquarters and plants
- DE **Germany** Siltronic plants in Burghausen and Freiberg, and Wacker semiconductor-grade polysilicon
- KR **South Korea** SK Siltron
- US **United States** Hemlock polysilicon in Michigan, and GlobalWafers plants in Sherman, Texas and St Peters, Missouri

WHY SUBSTITUTION IS POSSIBLE

Losing one of the top five wafer makers leaves four, and a new wafer plant can still be built[3]. Customer approval of a new plant is called qualification[9]. Siltronic completed key customer qualifications at its new Singapore fab in July 2025[9].

WHERE CHINA STANDS

China made about 93 percent of the world's polysilicon in 2023, an estimated 98 percent of it solar-grade[15]. Eswin, mainland China's largest maker of 300 mm wafers, says it had about 7.7 percent of world 300 mm shipments in the first half of 2026[5].

WHERE THE US STANDS

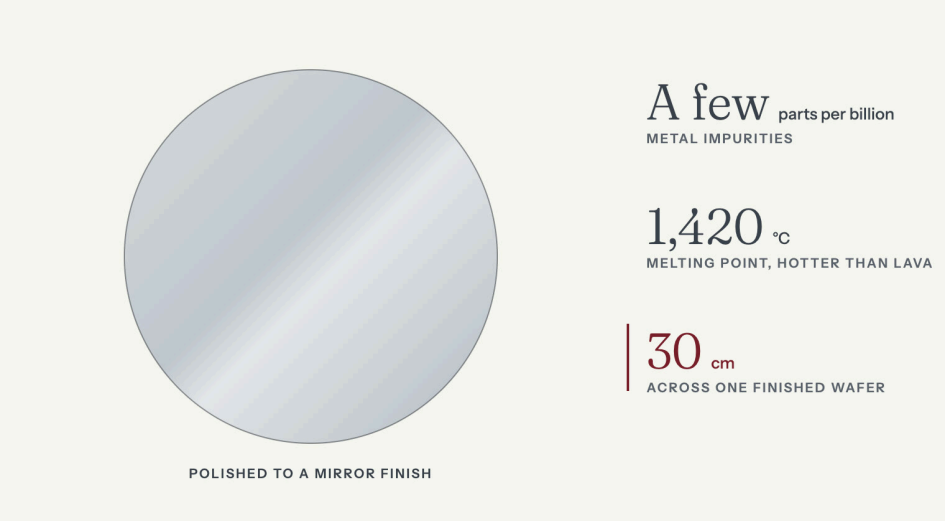
None of the five largest wafer makers in 2025 was American[3]. GlobalWafers of Taiwan won CHIPS awards of up to $406 million in December 2024 for wafer plants in Texas and Missouri[16].

The world market for silicon wafers was $11.4 billion in 2025, on 12,973 million square inches shipped[10]. That is a small fraction of Nvidia's data center revenue, which reached $193.7 billion in the fiscal year to 25 January 2026[17].

3 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 6. siltronic.com
5 西安奕斯伟材料科技股份有限公司2026年半年度报告. CNINFO, 25 August 2026, p. 18. static.cninfo.com.cn
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How it works

From sand to wafer



1. **Refine.** Silicon starts as quartz rock. It is refined into rods so pure that metal impurities are no more than a few parts per billion.
2. **Grow.** The rods are melted. A seed crystal touches the melt and is drawn slowly up, turning, and the silicon freezes onto it as one long crystal.
3. **Slice.** A wire saw slices the crystal, called an ingot, into thin discs.
4. **Polish.** Each disc is ground, etched and polished to a mirror finish, flat right out to the edge.

Simplified. The ingot and the machines are not drawn to scale.

Silicon starts as quartzite, a rock that is silicon bound to oxygen. An arc furnace melts it with carbon, and the carbon takes the oxygen away as gas, leaving rough silicon metal. Refiners turn the metal into a gas, trichlorosilane, and distill it, because the impurities boil at other temperatures and separate out.

In the next step, the Siemens process, the clean gas flows over silicon filaments heated inside a bell jar. On the hot surface the gas breaks apart and its silicon settles onto the filaments. Over days the filaments thicken into gray rods of polysilicon. Everything downstream is made from those rods. Polysilicon comes in two ultra-high-purity grades, one for semiconductors and one for solar cells, and chips use the semiconductor grade[1].

A grower melts the polysilicon at about 1,420 degrees Celsius, hotter than lava, and pulls a single crystal out of it by the Czochralski method[2]. The grower lowers a small seed crystal on the end of a rod to the surface of the melt and draws it back up, slowly turning. Silicon freezes onto the seed in the seed's own atomic pattern, and a cylinder, the ingot, grows beneath it. The pull starts with a thin neck so that flaws from the seed's first contact with the melt stay in the neck and never reach the ingot. Boron or phosphorus stirred into the melt, a step called doping, sets how readily the wafer conducts and whether negative or positive charges carry its current.

1 Mineral Commodity Summaries 2026: Silicon. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov
2 SUMCO, production processes page. SUMCO, accessed 9 September 2026. sumcosi.com

A wire saw, a web of fine wire running through abrasive, slices the ingot into discs. The maker then laps each disc, grinding both faces flat between rotating plates. It etches off the damaged surface with chemicals, then polishes the disc to a mirror with colloidal silica, a slurry of silica particles far too small to see. Almost every step throws material away.

Epitaxial wafers get one more layer of near-perfect silicon, grown on top at about 1,200 degrees Celsius[2]. That is still hot enough to glow orange.

Variants and trade-offs

Polished versus epitaxial

A polished wafer is a doped single crystal polished to a mirror finish[2]. An epitaxial wafer is a polished wafer with a thin layer of single-crystal silicon grown on top[2]. The transistors are built in that layer, so flaws in it can make them leak current. SUMCO says an epitaxial wafer suits customers who need a near-perfect crystal or layers that differ in electrical resistance[2]. SEMI, the industry association, credits the 2025 rise in shipments to AI-driven demand for advanced epitaxial wafers for logic chips and polished wafers for high-bandwidth memory[10].

Silicon on insulator

In a silicon-on-insulator (SOI) wafer, a buried layer of oxide, which is glass, separates a thin working layer of silicon from the bulk beneath. Soitec says nearly all SOI wafers sold worldwide are made with its Smart Cut process[18]. Soitec's SOI product lines include RF-SOI for radio chips and fully depleted SOI (FD-SOI), and its markets run from mobile communications to cloud AI[18]. But SEMI tied the AI-driven growth in logic chips to advanced epitaxial wafers[10].

Who makes it

The Center for Security and Emerging Technology (CSET) at Georgetown University named five key wafer firms in a January 2021 brief[19]. They were Shin-Etsu and SUMCO of Japan, GlobalWafers of Taiwan, Siltronic of Germany and SK Siltron of South Korea[19]. CSET also listed nine Chinese-owned producers, among them Zhonghuan, Simgui, ZingSEMI and Finland's Okmetic, but said they generated little revenue in comparison[19].

As of 2018, five Chinese firms were making or planning 300 mm wafers[20]. In its October 2025 listing prospectus, Eswin, a Chinese wafer maker, counts seven Chinese firms, itself included, making 300 mm wafers at scale[21]. Their combined capacity was 2.94 million wafers a month at the end of 2024[21].

2 SUMCO, production processes page. SUMCO, accessed 9 September 2026. sumcosi.com

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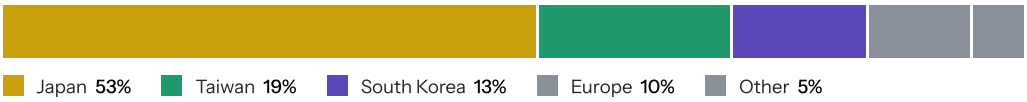
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20 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 57. cset.georgetown.edu

21 西安奕材首次公开发行股票并在科创板上市招股说明书. CNINFO, 22 October 2025, p. 138. static.cninfo.com.cn

Silicon wafer sales by supplier headquarters, 2023. Japan held 53%, and the source shows no separate share for China or the United States %



Source: METI, December 2025, from Fuji Keizai data (2023); Japan's and Europe's shares are printed, the others are read off METI's chart

The chart shows shares of 2023 sales from a survey by Fuji Keizai, a Japanese research firm[4]. Japan's Ministry of Economy, Trade and Industry reproduced the figures in a December 2025 strategy deck[4]. The ministry's chart of the Fuji Keizai survey prints only Japan's 53 percent, so the other shares are read off it[4]. An earlier page of the ministry's deck prints Europe's share in the same Fuji Keizai survey as 10 percent[22]. The chart gives no separate share to Chinese or US companies[4].

Siltronic's April 2026 investor presentation says the top five wafer makers serve around 75 percent of the market, based on revenue in the first three quarters of 2025[3]. It names the five as Shin-Etsu, SUMCO, GlobalWafers, SK Siltron and Siltronic itself[3].

The wafer makers' 2025 results show strong AI demand and weak demand for chips made on older, mature nodes.

- **Shin-Etsu.** Shin-Etsu Chemical's electronics materials segment, which includes its wafer business, sold ¥1,015.7 billion in the year to March 2026, up 9 percent[11]. AI-related demand stayed strong and demand in other sectors began to rise, lifting sales of silicon wafers, photoresist and photomask blanks[11]. The segment also sells rare earth magnets and other materials, and the results release does not break out wafer sales[23]. So the figure cannot be compared with the wafer revenue of the other firms listed.
- **SUMCO.** Sold ¥409,670 million and still lost ¥11,751 million, as operating profit fell from ¥36,924 million to ¥1,342 million[24]. In the same year it added leading-edge 300 mm capacity and reorganized production of 200 mm and smaller wafers, where demand stayed weak[12].
- **Siltronic.** Lost €77.9 million on €1,346.7 million of sales while investing €369.1 million, with its new Singapore fab as the focus[14].
- **GlobalWafers.** NT$60.6 billion of revenue, down 3.24 percent in local currency[13].

3 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 6. siltronic.com

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Worldwide, the wafer area shipped rose 5.8 percent in 2025, but revenue fell 1.2 percent[10]. SEMI attributes the softer revenue mostly to traditional chip applications, where demand and pricing had yet to improve[10].

China made 1.50 million metric tons of polysilicon, the raw material, in 2023, about 93 percent of world output[15]. Solar-grade material was an estimated 98 percent of that output and electronic-grade 2 percent[15]. The Commerce Department found in 2026 that semiconductor-grade polysilicon makes up only 2.4 percent of global production, a split close to China's[8]. These figures alone do not show whether Chinese polysilicon goes into chips.

China's polysilicon output by grade, 2023. China made 1.50 million metric tons that year, about 93% of world output %



Source: USGS Minerals Yearbook, China, 2023

Siltronic put the 2025 market for electronic-grade silicon at $1.4 billion, with five major suppliers, none of them Chinese: Wacker, Hemlock, OCI, Tokuyama and Mitsubishi[6].

Siltronic of Germany describes itself as the only Western-based maker among the top five wafer makers, based on revenue in the first three quarters of 2025[3]. GlobalWafers opened a $3.5 billion plant in Sherman, Texas in May 2025, the first production line of its kind built in the United States in over twenty years[7]. The same day it announced another $4 billion for phases three and four in Sherman[7]. The US government had awarded it up to $406 million under the CHIPS incentives program for wafer plants in Texas and Missouri[16].

Presidential Proclamation 11052, signed 6 August 2026, sets a minimum import price of $100 per kilogram for polysilicon ingots and wafers from 4 December 2026[8]. From the same date it adds an extra 15 percent duty on imported ingots and other listed products made from polysilicon[8]. A September 2026 Commerce rule meant to stop stockpiling before the duties start lists tariff code 3818.00.0040 among the products it covers[25]. That code covers round, doped, single-crystal silicon wafers[26].

Commerce may approve plans to build US polysilicon, ingot, wafer or solar cell plants whose construction starts by 20 January 2029[8]. Firms with approved plans can import equipment and some of those products without the new duties[8].

3 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 6. siltronic.com
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The chokepoint

China makes most of the world's polysilicon, but in 2019 its firms had only a small share of the wafer market. As of 2019 Chinese firms held under 5 percent of the wafer market and under 1 percent of the 300 mm market[19]. As of 2018, Chinese producers made 12 percent of the wafers for Chinese-headquartered 300 mm fabs, against 18 percent for 100 to 200 mm fabs[27].

Eswin, mainland China's largest maker of 300 mm wafers, says it ranked sixth in the world with about 7.7 percent of global 300 mm shipments in the first half of 2026[5]. Its October 2025 Shanghai listing prospectus, drawing on SEMI data, estimates that the five largest makers shipped 79.08 percent of 300 mm wafers in 2024[28]. It puts Eswin at 6.01 percent and all other makers at 14.91 percent[28].

Making the smoothest, high-purity 300 mm wafers takes considerable know-how that is hard to write down[20]. Customers have to qualify a new wafer plant before they buy from it, and Siltronic completed key customer qualifications at its new Singapore fab in July 2025[9].

In SEMI figures from November 2020 cited by CSET, 99.7 percent of world fab capacity at 45 nm and below used 300 mm wafers[20]. SEMI estimated that 300 mm wafers made up about 78.8 percent of all wafer area shipped worldwide in 2025, according to Eswin's 2025 annual report summary[29].

The last increase in wafer size, to 300 mm, happened in 2002[30]. Bigger discs make printing, or photolithography, a bigger share of the processing cost[30]. A 2013 University of Texas lecture cited by CSET put photolithography at half the cost of processing a 300 mm wafer, against a quarter for a 150 mm one[30]. Wafers are still produced in diameters of up to 300 mm[10].

Some GPU dies fill the whole area a lithography tool can print in one exposure. One fatal defect on such a die ruins far more silicon than one on a small chip. Extreme-ultraviolet printing also stays in focus over only a tiny range of heights (see Lithography), so the disc has to be flat at every point. SEMI's wafer group says the move to sub-3nm processes is raising requirements for wafer quality and consistency[10].

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5 西安奕斯伟材料科技股份有限公司2026年半年度报告. CNINFO, 25 August 2026, p. 18. static.cninfo.com.cn

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FOUR THINGS TO REMEMBER

- Every chip starts on a wafer, a polished disc of pure silicon about 30 centimeters across.
- The top five wafer makers, in Japan, Taiwan, Germany and South Korea, served around 75 percent of the market in 2025.
- Customers have to test and approve a new wafer plant before they buy from it.
- China makes most of the world's polysilicon, but none of the five major suppliers of electronic-grade silicon, the grade chips need, is Chinese.

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22. A METI, Semiconductor and digital industry strategy: future direction (半導体・デジタル産業戦略の今後の方向性), 23 December 2025. Ministry of Economy, Trade and Industry (Japan) · 23 December 2025 · p. 4
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CHAPTER
03
STAGE 3 OF 15

Chemicals, Gases and Photoresist

Japanese companies held 78 percent of world photoresist sales in 2023. Supply of chip chemicals, gases and metals has been disrupted three times since 2019.

IN PLAIN TERMS

Photoresist is a coating that changes wherever light hits it. A machine spins a coat of it, thinner than a soap bubble, across the wafer. Light shines through a stencil onto the coat, and a liquid wash then takes away the parts the light reached. What remains is a pattern, and the next machine cuts that pattern into the wafer beneath, a step called etching. Hundreds of other chemicals, gases and materials lay down layers, strip them off and clean the wafer between steps[1]. The cleaning chemicals are purified until metal and other impurities are down to parts per trillion, the Japanese chemical maker Sumitomo Chemical said in 2021[2]. A chipmaker has to test and approve each new resist in its own factory before using it[3]. Even a resist that passes the first time takes seven to 17 months to approve, the Semiconductor Industry Association wrote in 2023[3]. Japanese companies held 78 percent of world photoresist sales in 2023, by a Fuji Keizai survey that Japan's trade ministry reproduced[4].

IN SHORT

In July 2019 Japan began requiring an export license for each contract to sell photoresist, hydrogen fluoride or fluorinated polyimide to South Korea[5]. Japan's hydrogen fluoride exports to South Korea then fell 87.9 percent, but its photoresist exports did not[5]. Production of all three chemicals rose inside South Korea[5].

China banned exports of the metals gallium, germanium and antimony to the United States in December 2024[6]. In Europe, germanium metal averaged an estimated $4,100 a kilogram in 2025, against $1,392 in 2023, on Argus Media prices reported by the US Geological Survey[7]. In November 2025 China suspended the ban until 27 November 2026[8].

1 Testimony of the Semiconductor Industry Association (SIA) before the Senate... U.S. Senate Committee on Environment and Public Works, 4 March 2026, p. 6. epw.senate.gov

2 Sumitomo Chemical to Expand Production Capacity for High-Purity Chemicals for... Sumitomo Chemical, 24 August 2021. sumitomo-chem.co.jp

3 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 22. semiconductors.org

4 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

5 The impact of export controls on international trade: evidence from the... Research Institute of Economy, Trade and Industry, 8 May 2023. rieti.go.jp

6 商务部公告2024年第46号 关于加强相关两用物项对美国出口管制的公告. Ministry of Commerce of the People's Republic of China, 3 December 2024. mofcom.gov.cn

7 Mineral Commodity Summaries 2026: Germanium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

8 商务部公告2025年第72号 公布调整实施商务部公告2024年第46号的决定. Ministry of Commerce of the People's Republic of China, 9 November 2025. mofcom.gov.cn

CHOKEPOINT CARD | HIGH CONCENTRATION | MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
High	Moderate Rated on photoresist, the hardest input to replace at this stage[9].	$45.8bn market for wafer fabrication materials in 2025[10]. SEMI's count includes silicon wafers and photomasks[11]

WHO LEADS

- JP **JSR** A vehicle of the state-backed Japan Investment Corporation bought 84.36% of JSR in April 2024[12]
- JP **Tokyo Ohka Kogyo** Resist maker with ¥139,668m of sales in the first half of fiscal 2026[13]
- FR **Air Liquide** €2,465m of electronics revenue in 2025, 9.1% of group sales[14]
- US **Entegris** Bought the polishing-slurry maker CMC Materials in July 2022[15]
- JP **JX Advanced Metals** Says it holds about 65% of the world market for semiconductor sputtering targets as of fiscal 2026[16]

WHERE IT IS MADE

- JP **Japan** Photoresist, ultra-pure hydrofluoric acid, sputtering targets, polishing slurries
- US **United States** Entegris polishing slurries, Qnity Electronics resist, Air Products gases
- FR **France** Air Liquide electronics gases and precursors
- KR **South Korea** Dongjin resist, with output of resist and hydrogen fluoride rising after 2019
- CN **China** Gallium, germanium and antimony refining

WHY SUBSTITUTION IS POSSIBLE

Each formula is tuned to the layers it prints and has to pass qualification in the fab[9]. So a new supplier has to develop its version together with the fab, one layer at a time[3]. When Japan required individual export licenses for resist and two other chemicals sold to South Korea in 2019, South Korea imported more from the United States, Taiwan and Belgium[5].

WHERE CHINA STANDS

In March 2026 the Chinese resist maker Hubei Dinglong said China had yet to reach large-scale production of advanced resist for 248 nm, 193 nm or extreme-ultraviolet printing[17]. Since August 2023 China has required licenses to export gallium and germanium[18].

WHERE THE US STANDS

In photoresist, American companies ranked a distant second to Japanese companies by sales in 2023[4]. In 2021 JSR of Japan agreed to buy Inpria, an Oregon maker of resist for extreme ultraviolet[19].

3 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 22. semiconductors.org

4 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

5 The impact of export controls on international trade: evidence from the... Research Institute of Economy, Trade and Industry, 8 May 2023. rieti.go.jp

9 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 21. semiconductors.org

10 Global Semiconductor Materials Market Revenue Reaches Record $73.2 Billion in... PR Newswire, 12 May 2026. prnewswire.com

11 Materials Market Data Subscription (MMDS). SEMI, accessed 20 October 2026. semi.org

12 Announcement Regarding Result of Tender Offer for JSR Corporation (Securities... Japan Investment Corporation, 17 April 2024. jiccapital.co.jp

13 Consolidated Financial Results for the Second Quarter (Interim) of the Fiscal... Tokyo Ohka Kogyo, 6 August 2026, p. 4. tok.co.jp

14 Air Liquide, electronics page. Air Liquide, accessed 9 September 2026. airliquide.com

15 ENTEGRIS INC, Form 8-K current report for the period ended 2022-07-06. U.S. Securities and Exchange Commission, 6 July 2022. sec.gov

16 Quick Guide to JX Advanced Metals. JX Advanced Metals, accessed 9 September 2026. jx-nmm.com

17 湖北鼎龙控股股份有限公司2025年年度报告摘要. CNINFO, 27 March 2026, p. 2. static.cninfo.com.cn

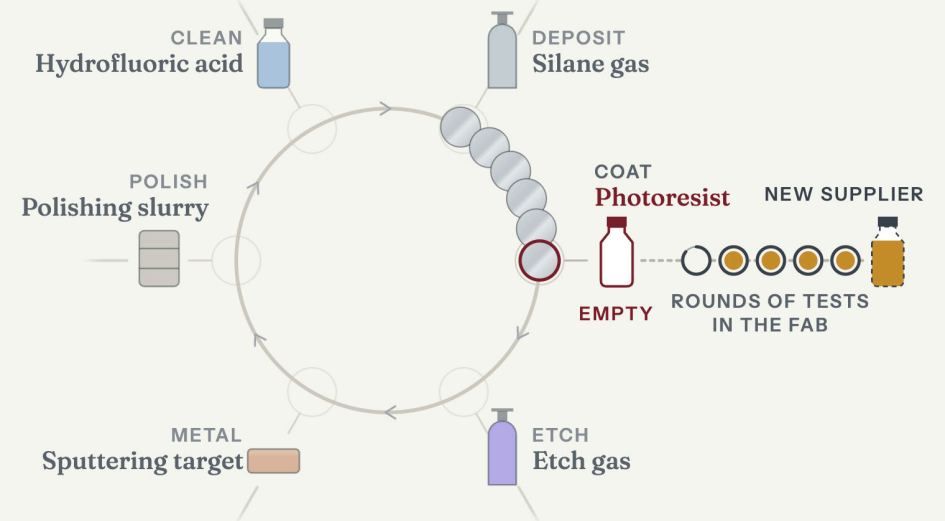
18 商务部 海关总署公告2023年第23号 关于对镓、锗相关物项实施出口管制的公告. Ministry of Commerce of the People's Republic of China, 3 July 2023. mofcom.gov.cn

19 JSR Agrees to Acquire EUV Pioneer Inpria Corporation. JSR Corporation, 17 September 2021. jsr.co.jp

A chip fab draws on hundreds of chemicals, gases and other materials[1]. A shortage of any one of them can stop it. The French gas company Air Liquide alone supplies more than 200 molecules and roughly 50,000 cylinders a year[14].

How it works

Why one missing material can stop a fab



1. **One layer.** A chip is built up in layers. For each layer the wafer goes around a loop of steps like this one, and each step uses its own chemical, gas or metal.
2. **Many wafers.** A fab runs many wafers at once, so every material is always in use. Deliveries keep each stock topped up.
3. **One runs out.** If one material stops arriving, the fab works through its stock.
4. **The fab stops.** When the stock runs out, wafers pile up at that step, and the rest of the fab stands idle.
5. **A new supplier.** A new supplier's resist goes through round after round of tests in the fab. Even one that passes the first time takes seven to 17 months to approve.

Simplified diagram. Six steps and their materials stand for the many a fab uses, and each real layer uses its own mix.

The resist has to change only where the light lands. It also has to work with very little light, so the scanner, the machine that projects the pattern, can expose each patch of the wafer in milliseconds.

Since the 1990s the standard has been chemically amplified resist. In resists for 248 nm and 193 nm light, a photon absorbed in the film can free an acid molecule, and little more happens until the wafer is baked. In the bake, each acid molecule sets off hundreds of reactions around itself that turn the film soluble, so a little light does a lot of chemistry[20]. The wash then takes away the exposed film and leaves the rest.

The 248 nm light comes from a krypton fluoride (KrF) laser. The 193 nm light comes from an argon fluoride (ArF) laser. Immersion scanners also pass it

1 Testimony of the Semiconductor Industry Association (SIA) before the Senate... U.S. Senate Committee on Environment and Public Works, 4 March 2026, p. 6. epw.senate.gov

14 Air Liquide, electronics page. Air Liquide, accessed 9 September 2026. airliquide.com

20 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 6. semiconductors.org

through a layer of water, which sharpens the image.

That chemistry works less well with extreme ultraviolet (EUV) light (see Lithography). An absorbed EUV photon instead knocks electrons loose, and a cascade of lower-energy secondary electrons makes the chemical changes in the resist[21]. A 13.5 nm EUV photon carries about fourteen times the energy of a 193 nm one, so the same dose of light arrives as far fewer photons. With fewer photons, random variation in where each one lands matters more. It shows up as ragged line edges and missing contact holes, the tiny holes that link one layer of a chip to the next.

Resist makers responded with metal-oxide resists, which are built on clusters with a tin-oxide core and absorb more of the EUV light than chemically amplified resists[22]. In 2021 the Japanese resist maker JSR agreed to buy Inpria, an Oregon maker of these resists[19].

The other chemicals and gases put material on the wafer, take it off or keep it clean. Bulk gases such as nitrogen come by pipeline or from plants built at the fab, and specialty gases come in cylinders[14]. Silane gas carries the silicon for new layers into the chamber that holds the wafer. A plasma splits nitrogen trifluoride, and the fluorine it frees scrubs the chamber walls clean between wafers. Hydrofluoric acid cleans the wafer itself. Polishing slurries grind each new layer flat before the next goes on.

Sputtering targets are slabs of pure metal. A plasma knocks atoms off them, and the atoms settle on the wafer as thin metal films, including the barrier and seed layers for copper wiring[23]. Electroplating then fills in the copper wiring itself[24]. Filters and wafer carriers keep contamination out.

Variants and trade-offs

RESIST CLASS	WAVELENGTH	WHERE IT IS USED	LIMIT
i-line novolak	365 nm	Packaging, mature nodes	Resolution
KrF chemically amplified	248 nm	Implant layers, thick films	Resolution
ArF immersion chemically amplified	193 nm	Most critical layers below 28 nm	Needs several exposures per layer
EUV chemically amplified	13.5 nm	Leading-edge single-exposure layers	Shot noise from too few photons
EUV metal-oxide	13.5 nm	High-NA tools, tightest pitches	Cost, defects, outgassing

14 Air Liquide, electronics page. Air Liquide, accessed 9 September 2026. airliquide.com

19 JSR Agrees to Acquire EUV Pioneer Inpria Corporation. JSR Corporation, 17 September 2021. jsr.co.jp

21 How Scientists Are Accelerating the Development of Next-Gen Microelectronics. Lawrence Berkeley National Laboratory, 21 August 2023. newscenter.lbl.gov

22 Inpria, company website. Inpria, accessed 9 September 2026. inpria.com

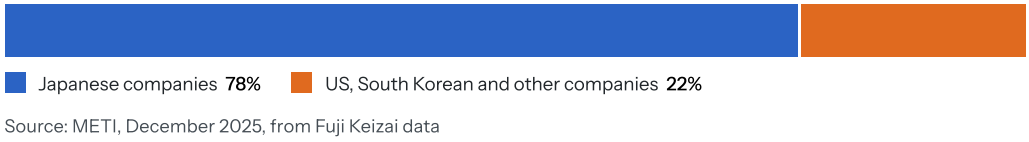
23 Applied Materials, PVD technology page. Applied Materials, accessed 9 September 2026. appliedmaterials.com

24 SABRE Product Family. Lam Research, 11 August 2017. lamresearch.com

Each resist class is made for one kind of lithography, such as EUV, ArF immersion, ArF, KrF or i-line[25].

Who makes it

In a January 2021 brief, Georgetown University's Center for Security and Emerging Technology (CSET) put Japan's share of the semiconductor photoresist market at 90 percent, citing a 2019 estimate[25]. The brief put most of the rest with US and South Korean firms[25]. In a December 2025 strategy presentation, Japan's Ministry of Economy, Trade and Industry (METI) published figures from a separate survey by Fuji Keizai, a Japanese market research firm[4]. That survey counts sales by company nationality and puts Japanese companies at 77 percent in 2021 and 78 percent in 2023, with American and South Korean companies next[4].



Source: METI, December 2025, from Fuji Keizai data

CSET's January 2021 brief lists five Japanese resist makers: JSR, Tokyo Ohka Kogyo, Shin-Etsu Chemical, Fujifilm and Sumitomo Chemical[26]. It also lists Dongjin of South Korea, four Chinese firms and three firms it counts as American: DuPont, EMD Performance Materials and Inpria[26]. Qnity Electronics, the former DuPont electronics business, says in its 2025 annual report that it offers lithography materials for every wavelength, including EUV[27]. In its 2025 annual report summary, the Chinese firm Hubei Dinglong said China had yet to reach large-scale production of advanced resist for 248 nm, 193 nm or EUV printing[17]. The firm also said three of its own resists for 248 nm and 193 nm printing had moved into stable volume supply[28].

A vehicle of Japan Investment Corporation, a state-backed fund, bought 84.36 percent of JSR's shares at ¥4,350 each in a tender offer that closed on 16 April 2024[12]. JSR's shares were delisted from the Tokyo Stock Exchange on 25 June 2024[29].

Tokyo Ohka Kogyo sold ¥139,668 million in the first half of fiscal 2026, up 25.1 percent, as strong demand for AI servers in data centers raised demand for

4 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

12 Announcement Regarding Result of Tender Offer for JSR Corporation (Securities... Japan Investment Corporation, 17 April 2024. jiccapital.co.jp

17 湖北鼎龙控股股份有限公司2025年年度报告摘要. CNINFO, 27 March 2026, p. 2. static.cninfo.com.cn

25 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 59. cset.georgetown.edu

26 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 56. cset.georgetown.edu

27 Qnity Electronics, Inc., Form 10-K annual report for the period ended... U.S. Securities and Exchange Commission, 26 February 2026. sec.gov

28 湖北鼎龙控股股份有限公司2025年年度报告摘要. CNINFO, 27 March 2026, p. 8. static.cninfo.com.cn

29 Announcement of Delisting of Shares of JSR Corporation. JSR Corporation, 24 June 2024. jsr.co.jp

chips[13]. Fujifilm planned to raise its polishing-slurry capacity at Kumamoto by about 30 percent from January 2025[30]. It also scheduled new production and quality-evaluation equipment for EUV resist at Shizuoka and Pyeongtaek to start in October 2025[31].

World sales of wafer fabrication materials reached $45.8 billion in 2025, and sales of packaging materials reached $27.4 billion[10]. SEMI counts silicon wafers and photomasks among wafer fabrication materials, along with resists, gases, chemicals and polishing materials[11].

In electronic gases, CSET named Merck, Air Products and Air Liquide as the market leaders in January 2021[32]. Air Liquide's electronics business sold €2,465 million in 2025, 9.1 percent of group revenue[14]. SK Specialty has made nitrogen trifluoride since 2001, when it became the first company to produce the gas in South Korea[33]. Kanto Denka Kogyo refines its own fluorine gas and turns it into specialty gases for deposition, etching and cleaning[34]. Stella Chemifa makes ultra-high-purity hydrofluoric acid for chipmaking[35]. Morita Chemical Industries sells an electronic-grade hydrofluoric acid[36].

CSET's January 2021 brief, drawing on a 2019 report by the market research firm Mordor Intelligence, put five firms at more than 60 percent of the wet chemicals market[32]. Citing a 2019 report by the market research firm Market Reports World, it put DuPont and Cabot Microelectronics at about 56 percent of a $790 million market for polishing slurries[25]. Counting by company nationality, the same Fuji Keizai survey, as published by METI, puts Japanese companies at 51 percent of polishing-slurry sales and 59 percent of high-purity cleaning-liquid sales in 2023[4]. Both shares were 49 percent in 2021[4].

Cabot Microelectronics renamed itself CMC Materials in October 2020[37]. The American firm Entegris completed its purchase of CMC Materials on 6 July 2022[15]. DuPont separated its electronics business into an independent company, Qnity Electronics, on 1 November 2025[38]. Qnity sells polishing slurries and pads[39].

4 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

10 Global Semiconductor Materials Market Revenue Reaches Record $73.2 Billion in... PR Newswire, 12 May 2026. prnewswire.com

11 Materials Market Data Subscription (MMDS). SEMI, accessed 20 October 2026. semi.org

13 Consolidated Financial Results for the Second Quarter (Interim) of the Fiscal... Tokyo Ohka Kogyo, 6 August 2026, p. 4. tok.co.jp

14 Air Liquide, electronics page. Air Liquide, accessed 9 September 2026. airliquide.com

15 ENTEGRIS INC, Form 8-K current report for the period ended 2022-07-06. U.S. Securities and Exchange Commission, 6 July 2022. sec.gov

25 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 59. cset.georgetown.edu

30 Fujifilm to Enhance its Production Capacity of Advanced Semiconductor Material... Fujifilm, 5 December 2024. fujifilm.com

31 Fujifilm Launches EUV Resist and EUV Developer. Fujifilm, 29 October 2024. fujifilm.com

32 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 60. cset.georgetown.edu

33 SK specialty, company website. SK specialty, accessed 18 October 2026. skspecialty.com

34 Specialty Gases. Kanto Denka Kogyo, accessed 18 October 2026. kantodenka.co.jp

35 High-purity Chemical Business. Stella Chemifa, accessed 18 October 2026. stella-chemifa.co.jp

36 Electronic grade Hydrofluoric acid. Morita Chemical Industries, accessed 18 October 2026. morita-kagaku.co.jp

37 CMC Materials, Inc., Form 8-K current report for the period ended 2020-09-28... U.S. Securities and Exchange Commission, 1 October 2020. sec.gov

38 DuPont Completes Separation of Qnity Electronics. DuPont, 3 November 2025. investors.dupont.com

39 Chemical Mechanical Planarization (CMP) Materials for Semiconductor. Qnity Electronics, accessed 9 September 2026. qnityelectronics.com

In sputtering targets for semiconductors, JX Advanced Metals puts its own share of the world market at about 65 percent as of fiscal 2026[16].

The chokepoint

Supply at this stage has been disrupted three times since 2019, twice by governments and once by a war.

In July 2019 Japan put photoresist, hydrogen fluoride and fluorinated polyimide under individual export licenses for South Korea, so each export contract needed its own license[5]. Japan had supplied more than 90 percent of South Korea's imports of two of the three[5]. Its hydrogen fluoride exports to South Korea then fell 87.9 percent[5]. Japanese photoresist exports to South Korea did not fall, possibly in part because Japan allowed three-year bulk licenses for some resist sales in December 2019[5].

South Korea bought more hydrogen fluoride from the United States and Taiwan and more photoresist from Belgium[5]. Production of all three chemicals rose inside South Korea[5]. The rise came both from South Korean firms and from Japanese companies' affiliates in South Korea[5].

Researchers at Japan's Research Institute of Economy, Trade and Industry (RIETI) found that Japanese hydrogen fluoride exports to the United States rose[5]. They say some Japanese supply may have reached South Korea by way of the United States[5]. Japan ended the individual licenses on 23 March 2023, when it put exports of the three chemicals to South Korea under its bulk export license system[40].

In 2022 Russia invaded Ukraine, which supplied about 70 percent of the world's neon and 90 percent of the semiconductor-grade neon used by US industry[41]. Gas mixtures containing neon fill the lasers used in lithography, but EUV scanners use a tin light source instead[42]. Many chipmakers had added to their gas stockpiles after Russia annexed Crimea in 2014[41]. Samsung, SK hynix, Renesas and Rohm reported adequate neon supplies from China[41].

At this stage, China's export controls cover metals. On 3 July 2023 China announced that exports of gallium and germanium would need licenses from 1 August[18]. On 3 December 2024 it said it would in principle grant no licenses to export gallium, germanium, antimony or superhard materials to the United States[6]. On 9 November 2025 it suspended that ban until 27 November 2026[8].

5 The impact of export controls on international trade: evidence from the... Research Institute of Economy, Trade and Industry, 8 May 2023. rieti.go.jp

6 商务部公告2024年第46号 关于加强相关两用物项对美国出口管制的公告. Ministry of Commerce of the People's Republic of China, 3 December 2024. mofcom.gov.cn

8 商务部公告2025年第72号 公布调整实施商务部公告2024年第46号的决定. Ministry of Commerce of the People's Republic of China, 9 November 2025. mofcom.gov.cn

16 Quick Guide to JX Advanced Metals. JX Advanced Metals, accessed 9 September 2026. jx-nmm.com

18 商务部 海关总署公告2023年第23号 关于对镓、锗相关物项实施出口管制的公告. Ministry of Commerce of the People's Republic of China, 3 July 2023. mofcom.gov.cn

40 Circular Notice on Partial Amendment to the Circular Notice on the Operation of... Ministry of Economy, Trade and Industry (Japan), 23 March 2023. web.archive.org

41 Russia's Invasion of Ukraine Impacts Gas Markets Critical to Chip Production. Center for Strategic and International Studies, 14 March 2022. csis.org

42 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 97. cset.georgetown.edu

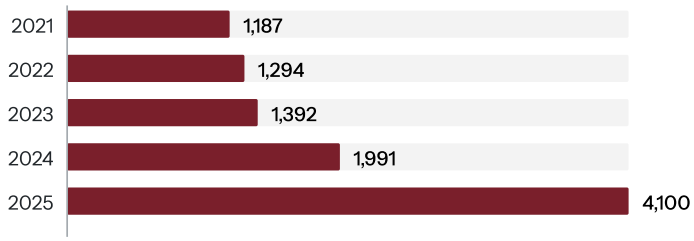
In February 2026 the US Geological Survey put China's share of world production of primary low-purity gallium at 99 percent[43].

The White House set out a trade deal with China on 1 November 2025[44]. It said that under the deal China would also issue general licenses for exports of gallium, germanium and antimony for the benefit of US end users and their suppliers[44].

In the United States, germanium goes mainly into fiber optics, infrared optics, semiconductors and solar cells, and radiation detectors[7]. Of the gallium used in the United States in 2025, 73 percent went into integrated circuits built on gallium arsenide or gallium nitride[43]. Another 26 percent went into optoelectronic devices such as LEDs and laser diodes[43]. Gallium is recovered mostly as a byproduct of processing bauxite, the ore that aluminum is made from[43]. Germanium comes from certain zinc ores and coal deposits, so the supply of each metal depends on another industry's economics[45].

In Europe, germanium metal averaged an estimated $4,100 a kilogram in 2025, against $1,392 in 2023, on Argus Media prices reported by the US Geological Survey[7]. US imports of the metal fell an estimated 67 percent in 2025 because of China's ban[7]. In October 2024 the US Geological Survey estimated that a total Chinese ban on gallium and germanium exports could cut US GDP by $3.4 billion[46].

Germanium metal, annual average price in Europe ($/kg). China required export licenses from August 2023, banned shipments to the US in December 2024 and suspended the ban from November 2025 to 27 November 2026 $/KG



Source: USGS Mineral Commodity Summaries, February 2026 (Argus Media European prices, 2025 estimated)

A slower risk is the European Union's proposed restriction on PFAS, the long-lived fluorinated chemicals. These chemicals are used throughout a fab, from some etch gases to seals, filters and the plastic linings of chemical tanks and pipes. Most of the photoacid generators that free the acid in resist contain PFAS too[47]. The Semiconductor Industry Association wrote in 2023 that photoresist needs the most rigorous qualification of any chemical a chipmaker uses, because each resist is tuned to the layers it prints[9]. The resist

7 Mineral Commodity Summaries 2026: Germanium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

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45 Quantifying Potential Effects of China's Gallium and Germanium Export... U.S. Geological Survey, 15 October 2024, p. 11. pubs.usgs.gov

46 Quantifying Potential Effects of China's Gallium and Germanium Export... U.S. Geological Survey, 15 October 2024, p. 9. pubs.usgs.gov

47 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 4. semiconductors.org

maker adjusts its formula over rounds of tests in the fab, and some resists have to be qualified one after another[3].

The national authorities of Denmark, Germany, the Netherlands, Norway and Sweden prepared the proposal and updated it after evaluating more than 5,600 comments[48]. ECHA, the EU chemicals agency, published the updated proposal on 20 August 2025[48]. The updated proposal assessed options that would let electronics and semiconductor makers keep using PFAS where the risks can be controlled[48].

In March 2026 ECHA's two scientific committees backed an EU-wide restriction with targeted derogations, or exemptions[49]. One committee gave its backing in a final opinion and the other in a draft, and their evaluation is due to finish by the end of 2026[49]. The European Commission will decide in consultation with the member states[48].

FOUR THINGS TO REMEMBER

- Photoresist holds the pattern printed by light so the next step can etch it into the wafer.
- Japanese companies held 77 percent of world photoresist sales in 2021 and 78 percent in 2023.
- A chipmaker has to approve each new resist in its own fab, and even one that passes the first time takes seven to 17 months.
- China has required licenses to export gallium and germanium since August 2023.

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CHAPTER
04
STAGE 4 OF 15

Lithography

ASML, a Dutch firm, is the only maker of extreme ultraviolet scanners, the machines that print the finest layers of a chip. It sold 48 of them in 2025.

IN PLAIN TERMS

Lithography prints the pattern of a circuit onto the wafer. A machine holds a stencil of one layer of the circuit up to a light source and projects the image, four times smaller, onto the silicon. The shorter the wavelength of the light, the finer the lines a machine can print. A machine can still print lines narrower than its light's wavelength. Immersion scanners shine the light through a layer of water between lens and wafer, which lets them print smaller features with the same light[1]. ASML rates its NXT:2150i immersion scanner at features of 38 nanometers or less with 193-nanometer argon fluoride laser light[2]. Air soaks up extreme ultraviolet light, the shortest in use, so machines that print with it work in a vacuum. Only ASML, a Dutch company, makes them. Export controls are government rules on who a company may sell to. Every advanced chip passes through ASML's machines, so export controls on ASML affect all advanced chipmaking.

IN SHORT

ASML also holds above 80 percent of the market for deep ultraviolet (DUV) lithography, the kind that uses longer-wavelength light[3]. ASML says the production capacity of Zeiss, its sole supplier of optics, limits how many lithography systems it can build[4].

Export controls bar extreme ultraviolet (EUV) scanners from Chinese chip foundries[5]. In December 2025 Reuters reported a Chinese prototype that had made EUV light but had printed no chip[6]. In July 2026 Reuters reported, citing one unnamed source, that a Shanghai firm with two state shareholders has started producing immersion DUV tools, which shine their light through water[7].

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CHOKEPOINT CARD	EXTREME CONCENTRATION	VERY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	Very hard ASML is the only maker of EUV scanners, and Zeiss is its sole supplier of their mirrors[4].	About $200M for a standard EUV scanner, an estimated $350-400M for the newer High-NA version and roughly $60M for a tool in the DUV segment, according to Reuters in July 2026[3]

WHO LEADS	NL **ASML** 100% of EUV (extreme ultraviolet) systems sold in 2025[8] JP **Nikon** 22 new chip lithography systems in the year to March 2026[9] JP **Canon** 234 chip lithography systems in 2025, 188 of them i-line and 46 KrF[10]
WHERE IT IS MADE	NL **Netherlands** ASML, the Dutch maker of every EUV scanner DE **Germany** Zeiss SMT optics, Oberkochen, and Trumpf carbon dioxide drive lasers, Ditzingen US **United States** ASML light-source research and manufacturing, San Diego JP **Japan** Nikon and Canon lithography scanners and steppers
WHY SUBSTITUTION IS SLOW	A newcomer would first have to make mirrors a meter wide, polished smooth to within tens of picometers, a picometer being a trillionth of a meter[1]. It would also need a light source that hits tin droplets with a laser 50,000 times a second[11].

WHERE CHINA STANDS	WHERE THE US STANDS
In September 2025 CSIS judged that EUV remained firmly out of reach for China despite massive state investment[5]. SMEE, the older Chinese supplier, had its largest share in i-line tools, the coarsest kind, and even there held 4 percent of the world market as of 2025[5].	No American firm makes scanners, CSET found in 2021[12]. But a US rule published in October 2023 covers certain immersion scanners bound for advanced chipmaking if they hold any American-made parts[13]. The US entity list also requires a license to supply the firms it names[14].

ASML sold 327 lithography systems in 2025, 48 of them extreme ultraviolet (EUV)[15]. Nobody else makes EUV systems[8].

How it works

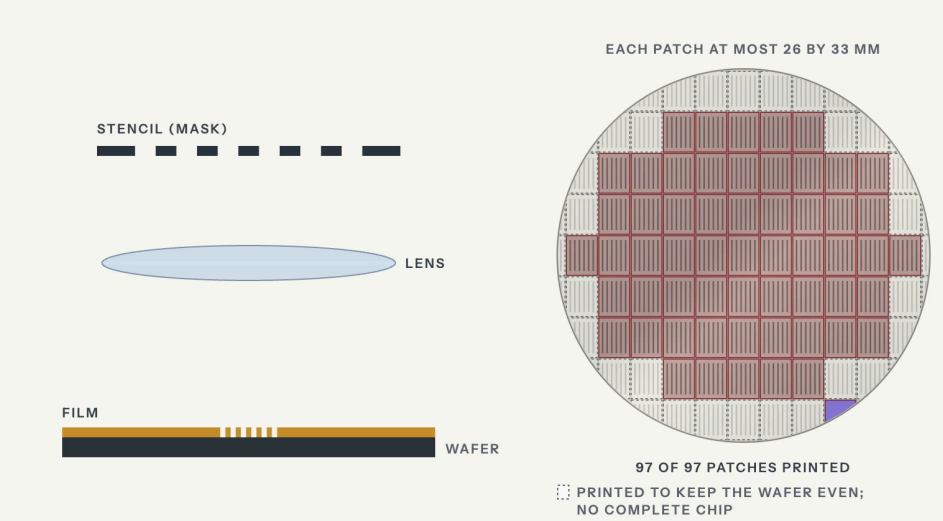
The stencil is called a photomask, and the machine a scanner (see Photomasks and pellicles).

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The wafer is coated with photoresist, a film that changes wherever light touches it. The mask holds one layer of the circuit as a pattern of clear and dark areas, drawn at four times its size on the chip. The machine shines light through the mask, shrinks the image four times with a lens, and lands it on the film.

A wash then removes the film where the light hit, or, with some films, where it did not, leaving the pattern standing on the wafer[16]. The film then serves as a stencil for the next step, which either etches the pattern into the layer beneath or fires ions into the open areas to change how the silicon conducts[16]. Then the film is stripped and the next mask goes in. Chips are made up of dozens of layers, ASML says[16]. Photronics, a US mask maker, says masks are typically sold in sets, with each layer's pattern on a different mask[17].

How lithography prints a chip



1. **Coat.** The wafer is coated with a thin film that changes wherever light touches it.
2. **Stencil.** A stencil, called a mask, holds the pattern for one layer of the chip, drawn four times larger than life.
3. **Shrink.** Light shines through the stencil, and a lens shrinks the image four times and lands it on the film.
4. **Wash.** A wash removes the film wherever the light landed, and the pattern is left on the wafer.
5. **Repeat.** The machine steps across the wafer, printing one patch at a time, each about the size of a postage stamp.

Simplified. The newest machines, which use extreme ultraviolet light, carry the image with mirrors in place of a lens.

The image from one mask covers a patch of up to 26 by 33 mm[18]. The wafer moves under the lens one patch at a time, close to a hundred patches per wafer. ASML's own wafers-per-hour rating for an EUV scanner assumes 96 of them[18]. Before the wafer is exposed, the machine measures alignment marks printed in earlier layers so the new layer lines up with them[18]. ASML rates the

16 6 crucial steps in semiconductor manufacturing. ASML, 4 October 2023. asml.com

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18 TWINSCAN NXE:3400B: EUV lithography systems. ASML, accessed 27 September 2026. asml.com

NXE:3400B's matched-machine overlay, a measure of that alignment, at 2 nm[18]. That model first shipped in 2017[19].

Finer lines need shorter light. Glass and air both absorb the shortest light in use, at 13.5 nm[20]. So an EUV scanner works in a vacuum and uses mirrors instead of lenses, and its masks are mirrors too[20]. Deep ultraviolet (DUV) machines use light with a wavelength more than fourteen times longer[11]. ASML calls its immersion DUV scanners the workhorses of the industry, used in high-volume manufacturing of the most advanced logic and memory chips[21].

When one exposure cannot draw a pattern finely enough, chipmakers use multipatterning. In the litho-etch kind, the pattern is split between two or more masks printed in turn[22]. Each added mask brings more process steps, more cost and a risk that the parts of the pattern misalign[23]. Self-aligned double and quadruple patterning (SADP and SAQP) print the lines with one exposure and then multiply them with deposition and etch steps[22]. In SAQP each printed line ends up as four[22]. Because one exposure places all those lines, they cannot misalign with each other, though a later exposure still prints blocks that break them where the circuit needs gaps[22]. ASML says one EUV exposure per layer cuts masks and process steps and lowers cost[8].

Variants and trade-offs

Deep ultraviolet

Deep ultraviolet (DUV) light comes from gas lasers called excimer lasers: 248 nanometers from krypton fluoride (KrF), 193 from argon fluoride (ArF), shone through air or water[11]. Cymer, an independently run ASML business in San Diego, makes these excimer lasers, which generate the light that scanners use to print patterns on wafers[24]. Gigaphoton, a Komatsu company in Japan, also makes excimer lasers for lithography tools[25]. Most of the machines sold are still DUV tools. Of the systems ASML sold in 2025, 131 were ArF immersion, 78 KrF and 16 ArF dry, all deep ultraviolet, and another 54 were older i-line tools[26]. The i-line tools use 365 nm light from a mercury lamp, and ASML says they print features as small as 220 nm[11].

Reuters gives one price for the whole DUV segment, with tools costing roughly $60 million each[3]. In 2025, those 131 ArF immersion tools made up 42 percent of ASML's system sales and the 78 KrF tools only 4 percent[26]. In DUV, ASML competes with Nikon and Canon, and it still holds above 80 percent of that market[3]. Without EUV, chipmakers build their finest layers on immersion DUV

3 Explainer: The $400 million ASML 'printers' key for the AI chip boom. Reuters, 28 July 2026. reuters.com
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26 ASML 2025 fourth-quarter and full-year results (investor presentation). ASML, 28 January 2026, p. 9. ourbrand.asml.com

scanners, in several passes per layer[8]. From 1 September 2023, Dutch export licensing covered the most advanced of these scanners, ASML's NXT:2000i and later models[27].

Extreme ultraviolet

The 13.5 nm light comes from molten tin droplets shot from a generator[11]. A laser hits each droplet twice, first with a low-intensity pulse that flattens it and then with a more powerful one that vaporizes it into plasma[11]. The process repeats 50,000 times a second[11]. Mirrors made by Zeiss, a German optics company, carry the light to the wafer[1]. The largest are a meter across and smooth to within tens of picometers, a picometer being a thousandth of a nanometer[1].

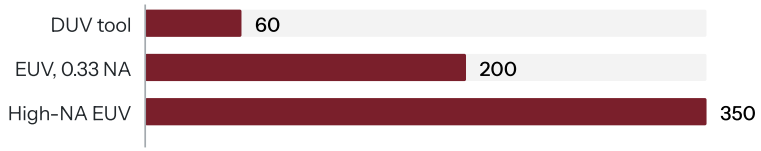
Brighter light means more wafers an hour, and ASML demonstrated the first 1,000-watt EUV source in April 2025[28]. Its TWINSCAN NXE:3800E ships at its full 220 wafers an hour, 37 percent more than the NXE:3600D[29].

High-NA EUV

High-NA scanners raise the numerical aperture (NA), a measure of how well the optics collect and focus light, from 0.33 to 0.55[30]. That cuts the smallest printable feature from 13 nm to 8 nm[30]. Their optics shrink the mask image four times in one direction and eight in the other, halving the patch of wafer one exposure covers and doubling the exposures per wafer[30]. The second-generation TWINSCAN EXE:5200B runs at 175 wafers an hour, 60 percent above the EXE:5000[8].

Reuters puts a standard EUV tool at around $200 million and a High-NA machine at an estimated $350 to $400 million[3].

What one lithography system costs. The DUV bar is one figure for the whole segment, and the High-NA bar shows the low end of an estimated $350-400M range $M



Source: Reuters explainer on ASML's lithography machines, first published 28 January 2026 and updated 28 July 2026

Intel and TSMC have chosen differently on High-NA:

- **Intel Foundry** says it uses High-NA in volume production for select layers of some Intel 18A products[31]. It installed the first EXE:5200B[32]. By September

1 Lenses & mirrors - Lithography principles. ASML, accessed 9 September 2026. asml.com
3 Explainer: The $400 million ASML 'printers' key for the AI chip boom. Reuters, 28 July 2026. reuters.com
8 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 16. ourbrand.asml.com
11 All about light and lasers in lithography. ASML, accessed 9 September 2026. asml.com
27 Statement regarding export control regulations Dutch government. ASML, 30 June 2023. asml.com
28 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 30. ourbrand.asml.com
29 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 17. ourbrand.asml.com
30 5 things you should know about High NA in EUV. ASML, 25 January 2024. asml.com
31 Intel Foundry and ASML Accelerate Industry Readiness for High-NA EUV. Intel, 7 September 2026. intel.com
32 High NA EUV reaches new readiness milestone with first high-volume Logic product. ASML, 15 July 2026. asml.com

2026 it had run more than one million wafers on High-NA tools, a count that includes tool testing and research as well as volume production[31].

- **TSMC** waited, and in September 2026 said it will use High-NA in high-volume manufacturing from 2030[33].

ASML sold four High-NA (EXE) systems in 2025, against two in 2024[34]. In its 2025 annual report, filed in February 2026, ASML said it expected High-NA to start supporting high-volume manufacturing in 2027[4]. On 8 September 2026 ASML and TSMC announced a move from 6-inch photomasks to 12-inch ones, to remove stitching constraints[33]. Stitching constraints are the design limits that come from printing a large chip in two halves. They aim for a pilot line for the new masks by 2031 and for High-NA systems that use them to enter production by 2033[33].

EUV systems ASML sold each year TOOLS



Source: ASML 2025 annual report (2024 and 2025); 2023 from ASML's Q4 2023 investor presentation

Who makes it

ASML's net sales were EUR 32.7 billion in 2025, at a 52.8 percent gross margin[15]. In October 2026 it expected 2026 net sales of EUR 44 to 45 billion, at a gross margin of around 56 percent[35]. Its 2025 annual report lists 5,100 suppliers[4]. Three of them and ASML's own San Diego site make the main parts of the EUV machine:

- **Zeiss SMT**, Oberkochen, makes the illumination system and the six-mirror projection optics[36].
- **Trumpf**, Ditzingen, makes the carbon dioxide drive laser, which amplifies a few watts to 40 kilowatts[37].
- **ASML San Diego**, which includes the light-source maker Cymer that ASML bought in 2013, designs the source where the laser turns tin into plasma, and builds the droplet generator[24].
- **VDL ETG** builds the frames that suspend and position the mirrors[38].

ASML says the production capacity of Zeiss, its sole supplier of optics, limits how many lithography systems it can build[4].

4 ASML HOLDING NV, Form 6-K report of foreign private issuer for the period ended... U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

15 Financial performance: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 5. ourbrand.asml.com

24 Explore ASML San Diego. ASML, accessed 9 September 2026. asml.com

31 Intel Foundry and ASML Accelerate Industry Readiness for High-NA EUV. Intel, 7 September 2026. intel.com

33 ASML and TSMC Announce Initiative to Pioneer Industry Transition to... TSMC, 8 September 2026. pr.tsmc.com

34 Financial performance: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 7. ourbrand.asml.com

35 ASML reports €11.7 billion total net sales and €5.0 billion net income in Q3... U.S. Securities and Exchange Commission, 14 October 2026. sec.gov

36 EUV lithography and technology. Carl Zeiss, accessed 9 September 2026. zeiss.com

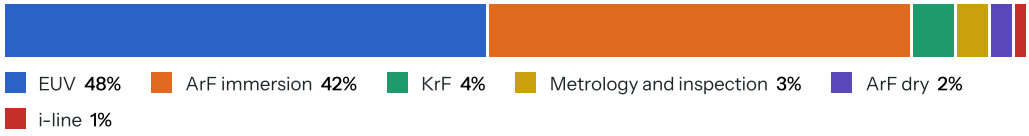
37 Good things come in ever-smaller packages. TRUMPF, accessed 9 September 2026. trumpf.com

38 VDL ETG builds complex frames for Zeiss. VDL Groep, 22 March 2019. web.archive.org

In Japan, Nikon and Canon also make lithography tools. Nikon sold 22 new chip lithography systems in the year to March 2026, in a market it sizes at 570 units for 2025[9]. Its Precision Equipment business made an operating loss in the year to March 2026[39]. Canon sold 234 chip lithography systems in 2025, 188 of them i-line and 46 KrF[10]. Its range runs from KrF scanners to i-line steppers, including some built for advanced packaging[40]. Canon also makes a nanoimprint system, the FPA-1200NZ2C, which stamps the pattern instead of projecting it, down to a 14 nm linewidth[41]. Canon delivered one in 2024 to the Texas Institute for Electronics, to be used for research and prototypes[41].

In January 2021 CSET, a research center at Georgetown University, found that the United States lacked firms in a few key parts of the chip supply chain, above all lithography tools[12].

ASML net system sales by technology, 2025 %



EUV 48% ArF immersion 42% KrF 4% Metrology and inspection 3% ArF dry 2% i-line 1%

Source: ASML Q4 2025 investor presentation

Controls and China's answer

Export controls bar EUV from Chinese foundries[5]. In September 2025 CSIS judged that EUV remained "firmly out of reach" for China despite massive state investment[5]. For DUV scanners and other tools, the restrictions came in this order:

- **Dutch national licensing** of advanced DUV took effect on 1 September 2023[42].
- **A US rule** published on 25 October 2023 put certain immersion scanners bound for advanced chipmaking under US export rules if they held any American content at all[13]. The rule did not apply if the country the scanners were first exported from already controlled them[13].
- **A partial license revocation** by the Dutch government, disclosed by ASML on 1 January 2024, stopped shipments of NXT:2050i and NXT:2100i immersion scanners to a few Chinese customers[43].
- **The Netherlands widened its licensing** from 7 September 2024[42]. On 15 January 2025 it announced that from 1 April 2025 the license requirement

5 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

9 Financial Results: The Year Ended March 31, 2026. Nikon, 8 May 2026, p. 8. nikon.com

10 Canon Inc. FY 2025 Results. Canon, 29 January 2026, p. 22. global.canon

12 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 4. cset.georgetown.edu

13 Export Controls on Semiconductor Manufacturing Items. Federal Register, 25 October 2023. federalregister.gov

39 Financial Results: The Year Ended March 31, 2026. Nikon, 8 May 2026, p. 6. nikon.com

40 Semiconductor Lithography Equipment. Canon, accessed 18 October 2026. global.canon

41 Canon delivers FPA -1200NZ2C nanoimprint lithography system for semiconductor... Canon, 26 September 2024. global.canon

42 The Netherlands expands export control measure for advanced semiconductor... Government of the Netherlands, 6 September 2024. government.nl

43 Statement regarding partial revocation export license. ASML, 1 January 2024. asml.com

would also cover a very limited number of further goods, among them specific measuring and inspection equipment[44].

- **The BIS Affiliates Rule**, issued by the US Bureau of Industry and Security, extended entity-list controls to affiliates owned 50 percent or more by listed entities from 29 September 2025[14]. BIS then suspended it until 9 November 2026[45]. Exporting goods covered by US export rules to a firm on the entity list needs a US license[14].
- **The EU** added certain Dutch controls to the EU Control List in November 2025[4]. ASML says this changed nothing for its immersion tools, because they already needed a Dutch license[4].

China's share of ASML's net system sales fell from 41 percent in 2024 to 33 percent in 2025[26].

On DUV, Reuters reported in July 2026, citing one unnamed source, that Shanghai Aishengna Electronic Technology Group has started producing home-grown immersion DUV tools[7]. The firm was registered in August 2023 with 7 billion yuan of capital and two state shareholders, Shanghai Electric Holding and a Shanghai International Trust subsidiary[7]. It has no website and has taken in teams from SMEE, China's older scanner maker, and the startup Yuliangsheng[7]. SMEE is strongest in i-line, the coarsest kind of lithography tool, and even there holds only four percent of the world market[5].

On EUV there is less evidence. Reuters reported in December 2025 that a Shenzhen team of former ASML engineers had built a prototype that made extreme ultraviolet light but had printed no chip[6]. To print a chip, a machine also needs mirrors that carry that light to the wafer, and ASML's sole supplier of those mirrors is Zeiss[4].

ASML net system sales by the region tools shipped to, 2025 %



China 33% South Korea 25% Taiwan 22% United States 12% Japan 5%
Rest of Asia 2% EMEA 1%

Source: ASML Q4 2025 investor presentation

The chokepoint

If ASML stopped shipping, no fab anywhere could add EUV scanners. Tools already in fabs would keep printing for as long as they were serviced. In its 2025 annual report ASML said about 95 percent of the systems it had sold in

4 ASML HOLDING NV, Form 6-K report of foreign private issuer for the period ended... U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

5 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

6 How China built its 'Manhattan Project' to rival the West in AI chips. Reuters, 17 December 2025. reuters.com

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14 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. Federal Register, 30 September 2025. federalregister.gov

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44 Klever: export controls on advanced semiconductor manufacturing equipment to be... Government of the Netherlands, 15 January 2025. government.nl

45 One Year Suspension of Expansion of End-User Controls for Affiliates of Certain... Federal Register, 12 November 2025. federalregister.gov

the previous 30 years were still in use[4]. It guarantees service and spare parts for each platform until at least a committed date, subject to export controls[4]. ASML itself could not quickly rebuild its production, because Zeiss makes its optics and Trumpf its drive laser.

In July 2026 ASML put its 2026 capacity at around 65 standard (low-NA) EUV scanners and said it plans to add 30 percent for 2027[46].

For AI accelerators, EUV is the point of control, because it prints the finest layers, the ones that set how densely transistors and wires pack[20]. TSMC expects the number of layers needing High-NA to rise, driven mainly by the more complex transistors that AI chips need[33].

FOUR THINGS TO REMEMBER

- Lithography prints each layer of a chip's circuit onto the wafer.
- Extreme-ultraviolet (EUV) light prints the finest layers in fewer steps than deep ultraviolet, but air and glass absorb it, so EUV machines work in a vacuum and use mirrors.
- ASML of the Netherlands is the only maker of EUV machines, and Zeiss alone supplies their mirrors.
- Export controls block EUV sales to Chinese chipmakers, and a Chinese prototype reported in December 2025 had printed no chip.

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CHAPTER
05
STAGE 5 OF 15

Photomasks and Pellicles

Only three firms in the world make the blank plates for the masks used to print the finest layers of leading-edge chips. Japan's AGC and Hoya are two of them.

IN PLAIN TERMS

A photomask is the stencil that lithography prints from, at least one for each layer of a chip[1]. The same mask prints its layer on wafer after wafer, so one mask shapes millions of chips. The most complicated layers are printed with extreme ultraviolet (EUV) light, and the rest can often be printed with deep ultraviolet light shone through a quartz plate. No glass lets EUV light through, so the mask for those layers is a mirror. The mirror is built up from about forty pairs of thin layers, and the light bounces off it. A dark pattern drawn on top of the mirror soaks up the light wherever no line should print. Any speck of dust on the mask repeats on every wafer it prints. So a thin membrane on a frame, called a pellicle, sits just above the mask and catches dust where it is too far out of focus to print. Japan's AGC and Hoya are two of the three firms that make those mirrors.

IN SHORT

On CSET's count from TechInsights data, Austria's IMS Nanofabrication held 46.6 percent of the 2025 world market for electron-beam mask writers, which draw the pattern onto a mask[2]. Hoya says its share of the mask blank market is exceptionally high[3].

SemiAnalysis put a full set of masks for a leading-edge chip at $40 million in May 2026[4]. In July 2022 it wrote that more companies will lack the volume to spread the fixed cost of mask sets[5]. On 8 September 2026 ASML and TSMC announced a plan to move from 6-inch to 12-inch masks[6]. They expect the change to raise fab productivity and lower chipmaking costs[6].

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CHOKEPOINT CARD EXTREME CONCENTRATION | HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	**Hard** AGC said in December 2024 that only three firms supply the blank plates for EUV masks[7].	**$40M** for a leading-edge mask set, SemiAnalysis's May 2026 figure[4]. In July 2022 it put a 7 nm set above $10M[5]

WHO LEADS

- JP **AGC** Ranks itself No.2 worldwide in EUV mask blanks on its own fiscal 2026 estimates[8]
- JP **Hoya** Said in its 2026 report that its share of mask blanks is exceptionally high[3]
- JP **Lasertec** Launched the world's first tool that inspects patterned EUV masks at the scanner's own wavelength in 2019[9]
- AT **IMS Nanofabrication** CSET puts its 2025 share of the market for all electron-beam mask writers at 46.6%, using TechInsights data[2]
- US **Photronics** $849.3M of revenue in fiscal 2025[10]

WHERE IT IS MADE

- JP **Japan** Mask blanks (AGC, Hoya), DUV blanks (Shin-Etsu), inspection (Lasertec), writers (NuFlare), pellicles (Mitsui Chemicals)
- AT **Austria** IMS Nanofabrication multi-beam mask writers
- DE **Germany** Zeiss mask checking in EUV light (AIMS EUV) and mask repair
- US **United States** Photronics merchant mask shops, KLA inspection
- TW **Taiwan** TSMC, which held 10% of IMS at the end of 2024

WHY SUBSTITUTION IS SLOW

AGC says it spent more than a decade meeting the blank's requirements, and mass production began only in 2017[7]. Germany's Zeiss calls its AIMS the only system that tests how a mask will print under conditions equivalent to the scanner's[11].

WHERE CHINA STANDS

As of 2026 SMIC runs two mask shops of its own in Shanghai[12]. No Chinese firm appears in CSET's 2025 market shares for electron-beam mask writers, the machines that draw the pattern onto a mask[2].

WHERE THE US STANDS

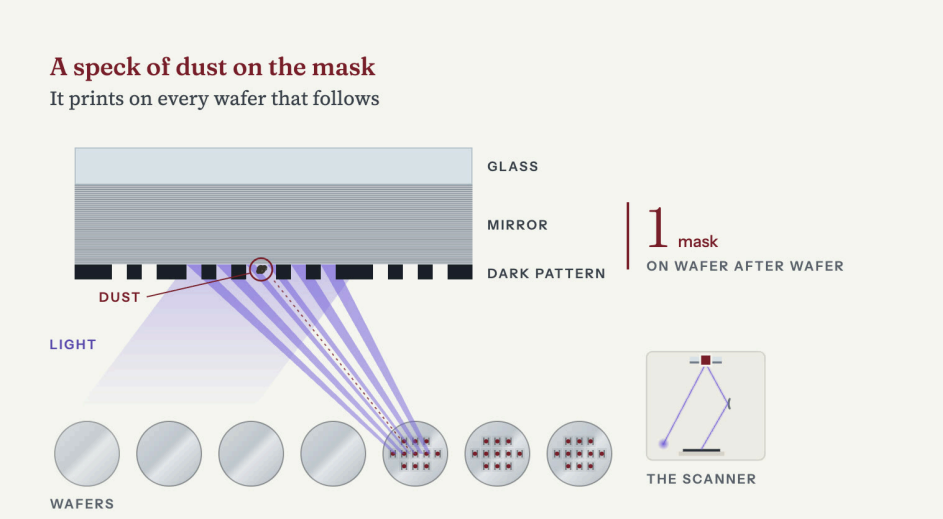
In December 2025 Photronics called itself a leading worldwide maker of photomasks[10]. As of 2026 KLA sells tools that inspect EUV and optical masks[13].

Each EUV layer of an AI accelerator is printed from a mask built on a blank plate. Only three companies worldwide supply those plates, and Japan's AGC is one of them[7]. Hoya, also Japanese, is another[3]. AGC says it is the only company in the world that takes an EUV blank all the way from the glass to the finished coating[14].

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
3 HOYA, Information Technology business, review of operations, Integrated Report... HOYA, accessed 9 September 2026. hoya.com
4 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com
5 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due... SemiAnalysis, 24 July 2022. newsletter.semianalysis.com
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8 AGC Data Book. AGC Inc., 29 May 2026, p. 8. agc.com
9 Lasertec Releases ACTIS A200HiT Series Actinic EUV Patterned Mask Inspection... Lasertec, 31 October 2025. lasertec.co.jp
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11 Mask Qualification Solutions by ZEISS SMT. Carl Zeiss, accessed 9 September 2026. zeiss.com
12 Mask Service. SMIC, accessed 9 September 2026. smics.com
13 Reticle Qualification. KLA Corporation, accessed 20 October 2026. kla.com
14 AGC to Boost Production Capacity of EUVL Photomask Blanks. AGC, 27 April 2023. agc.com

How it works

How a photomask prints its pattern



1. **The scanner.** A photomask is the stencil that lithography prints from, at least one for each layer of a chip. Inside the scanner, light bounces off the mask and carries its pattern down to the wafer.
2. **A mirror.** No glass lets extreme ultraviolet (EUV) light through. So the mask for EUV layers is a mirror, built up from about forty pairs of very thin layers laid on glass, and the light bounces off it.
3. **The pattern.** A dark pattern drawn on the mirror soaks up the light wherever no line should print. The light that bounces back carries the pattern of one layer.
4. **Wafer after wafer.** The same mask prints its layer on wafer after wafer, so one plate shapes millions of chips.
5. **One speck.** A speck of dust on the mask prints on every wafer, until someone finds it. So a thin membrane on a frame, called a pellicle, covers the mask's patterned side and catches dust where it is too far out of focus to print.

Simplified. The mask is drawn face down, as it sits in the scanner. The mirror layers and the dark pattern are drawn far thicker than they are, and the light's path through the scanner's mirrors is cut to one bounce.

ASML says a chip's most complicated layers are made on EUV machines, while the rest can often be printed on older deep ultraviolet machines[15]. A mask starts as a blank, a polished plate with no pattern on it yet. For deep ultraviolet the blank is a window of quartz carrying a light-blocking film such as molybdenum silicide, and Shin-Etsu is one maker of those blanks[16]. For EUV the blank is a mirror of about 40 molybdenum and silicon layer pairs, 7 nm to a pair, laid on glass[17]. A dark tantalum film on top, called the absorber, will carry the pattern[18].

The blank is coated with a sensitive film called resist. A mask writer then draws the design into the resist with electrons. A single-beam writer flashes one shape at a time, while NuFlare's multi-beam writer for the most advanced masks controls 260,000 beams at once[19].

15 See ASML's DUV lithography systems. ASML, accessed 9 September 2026. asml.com
16 Photomask blanks. Shin-Etsu Chemical, 3 April 2019. shinetsu.co.jp
17 The refined EUV mask model. ArXiv, 19 December 2019, p. 2. arxiv.org
18 The refined EUV mask model. ArXiv, 19 December 2019, p. 1. arxiv.org
19 EB Mask Writer. NuFlare Technology, accessed 9 September 2026. nuflare.co.jp

A wash removes the drawn resist. An etch then removes the blocking film or absorber wherever the resist is gone, and the rest of the resist is stripped. The plate now passes light, or reflects it, in the shape of one layer of the chip. On standard EUV scanners the pattern is drawn four times larger than it will print[20]. An inspection tool compares the plate to the design, and a repair tool fixes the stray defects it finds.

EUV masks bring problems of their own:

- **Mask 3D effects.** Light strikes the mirror at a slant, and the absorber has height, so its edges cast small shadows that distort the printed shape. Software has to correct for the shadows in advance.
- **Write precision.** Every edge of the absorber has to sit within a few nanometers of where the design says.
- **Particles.** Anything on the mirror prints on every wafer until someone finds it.

High-NA scanners, the newest EUV machines, print finer lines with larger mirrors. To keep masks at their usual size, their mirrors shrink the pattern four times in one direction and eight times in the other[20]. So each exposure covers half the area of a standard exposure, and a wafer takes twice as many exposures[20].

On 8 September 2026 ASML and TSMC announced a plan to move from the current 6-inch masks to 12-inch masks[6]. They expect the change to raise fab productivity, lower chipmaking costs and remove stitching constraints, the design limits that come from printing a large chip in two halves[6]. They aim to set up a 12-inch mask pilot line by 2031, so that High-NA scanners using 12-inch masks can enter advanced chip production by 2033[6].

Variants and trade-offs

Blanks

AGC says it spent more than a decade meeting the blank's requirements, which include near-zero thermal expansion, perfectly smooth and flat surfaces and no particle contamination[7]. It says mass production began only in 2017[7].

AGC publishes no market share figure for EUV blanks. In its data book, AGC ranks itself No.2 worldwide in EUV blanks, based on its own fiscal 2026 estimates[8]. Hoya says its share of the mask blank market is exceptionally high, and that customers may gradually start buying EUV blanks from more than one supplier[3].

In April 2023 AGC said its EUV blank capacity would grow by about 30 percent in 2025[14]. The growth would come from a line in Fukushima, a project that METI, Japan's industry ministry, selected under a program to strengthen

3 HOYA, Information Technology business, review of operations, Integrated Report... HOYA, accessed 9 September 2026. hoya.com
6 ASML and TSMC Announce Initiative to Pioneer Industry Transition to... TSMC, 8 September 2026. pr.tsmc.com
7 The Japanese glass behind next-generation chips. AGC, 13 December 2024. agc.com
8 AGC Data Book. AGC Inc., 29 May 2026, p. 8. agc.com
14 AGC to Boost Production Capacity of EUVL Photomask Blanks. AGC, 27 April 2023. agc.com
20 5 things you should know about High NA in EUV. ASML, 25 January 2024. asml.com

supply chains[14]. AGC also said it was aiming for more than ¥40 billion of sales in the business by 2025[14]. Its EUV blank sales grew 50 percent in 2023, AGC said in December 2024[7]. In February 2025 AGC reported that the sales reached ¥40 billion in 2024, a year ahead of the 2025 target[21].

The sales then declined in 2025 and were recovering in 2026, AGC said in June 2026[22]. In August 2026 AGC set out capacity expansions of 20 to 50 percent across EUV blanks, polishing slurry and lens materials for lithography tools[23]. It gave no separate figure for the blanks, and mass production on the added capacity is to begin in 2028[23].

Writers

An electron-beam writer draws the pattern into the resist on the blank. Multi-beam writers fire many beams at once, and Japan's NuFlare sells them for the 3 nm node[19].

In September 2023 Intel called Austria's IMS Nanofabrication the established industry leader in the multi-beam mask writing tools that advanced EUV requires[24]. Counting all electron-beam mask writers, CSET, a research center at Georgetown University, put IMS's share of a $994 million world market at 46.6 percent in 2025, using TechInsights data[2]. NuFlare had 39.2 percent and JEOL 14.2 percent, in the same CSET analysis of TechInsights data[2].

Intel agreed in September 2023 to sell roughly 10 percent of IMS to TSMC at a valuation of about $4.3 billion[24]. TSMC's subsidiary TSMC Development held that 10 percent at the end of 2024, at a fair value of $432.8 million in TSMC's accounts[25]. Intel sold a combined 32 percent of IMS in 2023 and still held the other 68 percent on 27 December 2025[26].

Inspection and repair

A mask defect cuts the number of good chips on every wafer that mask prints, so masks are checked as blanks, after patterning and again at the fab. Actinic inspection uses 13.5 nm light, the wavelength of EUV lithography[9]. In 2019 Lasertec launched the world's first actinic inspection system for patterned EUV masks, the ACTIS A150[9]. In October 2025 it launched the ACTIS A200HiT series, which it says triples the A150's inspection speed[9]. Lasertec reported net sales of JPY 230,485 million in the year to June 2026[27].

As of 2026 KLA's Teron systems for mask shops detect pattern and particle defects on both EUV and optical masks[13].

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
7 The Japanese glass behind next-generation chips. AGC, 13 December 2024. agc.com
9 Lasertec Releases ACTIS A200HiT Series Actinic EUV Patterned Mask Inspection... Lasertec, 31 October 2025. lasertec.co.jp
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19 EB Mask Writer. NuFlare Technology, accessed 9 September 2026. nuflare.co.jp
21 Financial Results for FY2024. AGC Inc., 7 February 2025, p. 45. agc.com
22 Semiconductor-related Business Briefing: Event Summary. AGC Inc., 2 June 2026, p. 17. agc.com
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24 Intel to Sell Minority Stake in IMS Nanofabrication Business to TSMC. Intel, 12 September 2023. intc.com
25 TAIWAN SEMICONDUCTOR MANUFACTURING CO LTD, Form 6-K report of foreign private... U.S. Securities and Exchange Commission, 27 February 2025. sec.gov
26 INTEL CORP, Form 10-K annual report for the period ended 2025-12-27 (10-K). U.S. Securities and Exchange Commission, 23 January 2026. sec.gov
27 Lasertec, business report. Lasertec, September 2026. lasertec.co.jp

Zeiss makes AIMS EUV, which checks a mask in EUV light under the scanner's own imaging conditions to judge how a defect will affect printing[11]. Zeiss calls AIMS the only system that tests how a mask will print under conditions equivalent to the scanner's[11]. Zeiss says the tool can be extended to ASML's High-NA scanner, the EXE:5000[11]. Zeiss also sells electron-beam repair tools that work on EUV masks[28].

Pellicles

A pellicle is a thin membrane on a frame above the mask. Particles land on it, too far out of focus to print, and the pattern below stays clean.

In 2022 ASML described its most advanced EUV pellicle as a membrane 13 nanometers thick that can withstand up to 500 degrees Celsius in a high vacuum[29]. Every photon it absorbs is light that never reaches the wafer[29]. The light crosses it twice, on the way in and on the way back off the mirror, so absorption counts double. ASML released its first EUV pellicles in 2016 and calls the pellicle one of the breakthroughs that brought EUV lithography to high-volume manufacturing[29].

Carbon nanotube membranes are the next step, aimed at next-generation High-NA, high-output EUV scanners[30]. In 2020 imec, a Belgian chip research center, mounted them on masks and exposed them in an ASML NXE:3300 scanner[31]. The membranes passed up to 97 percent of the light in a single crossing, and imec found their effect on imaging low and correctable[31].

Mitsui Chemicals began commercial production of silicon-based EUV pellicles under an ASML license in 2021[30]. In 2024 it said it would build a carbon nanotube membrane plant at Iwakuni-Ohtake, tentatively due to finish in December 2025[30]. In December 2025 Mitsui Chemicals said the plant was scheduled for completion in its fiscal 2025, in line with its initial plan[32]. The membranes are designed to pass 92 percent or more of the light and to withstand exposure power above 1 kW[30].

Cost and design economics

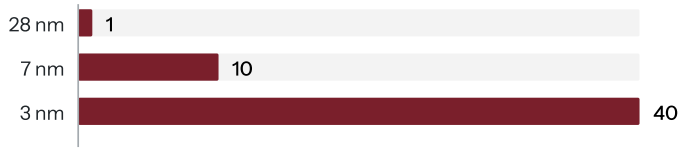
In July 2022 SemiAnalysis put mask sets above $1 million at 28 nm and above $10 million at 7 nm[5]. It expected 3 nm mask sets to reach the $40 million range[5]. Its May 2026 primer still puts a leading-edge mask set at $40 million[4]. The same primer puts a single respin, remaking a chip after a design flaw, at $50 million to $100 million at leading-edge nodes[4].

Mask costs restrict which firms can use a node. SemiAnalysis wrote that at 5 nm and 3 nm, only firms with large wafer volume see the cost per transistor

4 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com
5 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due... SemiAnalysis, 24 July 2022. newsletter.semianalysis.com
11 Mask Qualification Solutions by ZEISS SMT. Carl Zeiss, accessed 9 September 2026. zeiss.com
28 Mask Repair Solutions by ZEISS SMT. Carl Zeiss, accessed 9 September 2026. zeiss.com
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30 Mitsui Chemicals Sets up Production Facilities for CNT Pellicles to Be Used in... Mitsui Chemicals, 28 May 2024. jp.mitsuichemicals.com
31 Imec demonstrates CNT pellicle utilization on EUV scanner. Imec, 6 October 2020. imec-int.com
32 Business Strategy Presentation (L&HC, Mobility, ICT) in December 2025. Mitsui Chemicals, 17 December 2025, p. 31. jp.mitsuichemicals.com

fall[5]. It wrote that more companies will lack the volume to spread the fixed cost of mask sets[5]. It expected failed chip designs to show up first among AI chip startups, which work on much tighter budgets[5].

What one mask set costs, by node, 2022 ($M). The 28 nm and 7 nm figures are minimums and 3 nm was a projection $M



Source: SemiAnalysis, The Dark Side of the Semiconductor Design Renaissance, July 2022. SemiAnalysis's May 2026 EDA Market Primer puts a leading-edge mask set at $40M

Who makes it

Mask making splits between chipmakers that make masks for their own use and independent merchant firms that sell to them and others[1]. Photronics says that chipmakers' own shops once supplied most of the market, and that production has more recently tended to move toward the independents[1]. TSMC makes masks too, and lists mask making among the services it offers besides making wafers[33].

Photronics calls itself a leading worldwide maker of masks for chips and flat panel displays[10]. It had $849.3 million of revenue in fiscal 2025, $615.1 million of it from chip masks[10]. The competitors it names in chip and display masks include Dai Nippon Printing, Hoya, Tekscend Photomask and two Chinese firms, Shenzhen Newway Photomask and Shenzhen Qingyi Photomask[1]. It also competes with chipmakers' own mask shops[1].

SMIC, a Chinese foundry, runs two in-house mask shops in Shanghai that make masks for deep ultraviolet machines and older ones[12]. Photronics began operating two mask plants in China, in Hefei and Xiamen, in 2019[1]. Its chip mask business in Xiamen is a joint venture in which Dai Nippon Printing holds 49.99 percent[1]. No Chinese firm appears in CSET's 2025 market shares for electron-beam mask writers[2].

The chokepoint

Mask blanks are a small, highly concentrated business. In 2019 Lasertec launched the first tool that inspects patterned EUV masks in EUV light[9].

Pellicles already work in high-volume EUV manufacturing[29]. But ASML says no material is completely transparent to EUV light, so any pellicle in the light's

1 PHOTRONICS INC, Form 10-K annual report for the period ended 2025-10-31 (10-K). U.S. Securities and Exchange Commission, 17 December 2025. sec.gov

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

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33 TAIWAN SEMICONDUCTOR MANUFACTURING CO LTD, Form 20-F annual report for the... U.S. Securities and Exchange Commission, 16 April 2026. sec.gov

path costs customers productivity[29]. In 2024 Mitsui Chemicals said it would set up production of carbon nanotube membranes for next-generation High-NA, high-output EUV scanners[30].

FOUR THINGS TO REMEMBER

- A photomask is the stencil for one layer of a chip, and any flaw on it repeats on every wafer.
- Three firms make EUV mask blanks, and Lasertec launched the first tool that checks finished masks in EUV light.
- AGC spent more than a decade learning to make EUV blanks before mass production began in 2017.
- China makes masks for deep ultraviolet machines and older ones, in SMIC's own mask shops and at independent firms such as Shenzhen Qingyi and Shenzhen Newway.

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CHAPTER
06
STAGE 6 OF 15

Deposition and Etch

Deposition and etch tools build a chip layer by layer. They are among the chipmaking tools where Chinese toolmakers gained the most share from 2019 to 2024.

IN PLAIN TERMS

Deposition and etch build the chip itself, one layer at a time. Deposition lays down a thin film, sometimes only a few atoms thick. Lithography prints a pattern on it. Etch then eats away everything the pattern does not protect, and the next film goes down on top. Making one chip takes 1,000 to 1,500 steps, the Semiconductor Industry Association estimated in 2025[1]. Some of the holes the etch cuts in memory chips are more than 50 times deeper than they are wide, like a finger-wide shaft about a meter deep[2]. The film has to coat them right to the bottom. The etch has to eat one material and leave the one beside it untouched. Applied Materials and Lam Research in America, Tokyo Electron in Japan and ASM International in the Netherlands make most of these machines. Chinese toolmakers have gained share in both kinds of tool since 2019.

IN SHORT

American companies held 60.3 percent of the world market for deposition tools in 2025, Japanese companies 14.0 percent and China-based companies 9.8 percent, by CSET's count from TechInsights data[3]. In etch and cleaning tools that year, American companies held 52.6 percent, Japanese companies 32.6 percent and China-based companies 10.1 percent[4].

Atomic layer deposition tools coat a hole's walls and floor as evenly as its rim[5]. China-based companies held only 2.6 percent of the world market for them in 2025, by CSET's count from TechInsights data[6]. The December 2024 US export rule targets tools for advanced atomic layer deposition and for coating holes more than 200 times deeper than they are wide[7].

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7 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 33. govinfo.gov

CHOKEPOINT CARD	HIGH CONCENTRATION	MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
High	Moderate In 2025 Lam Research held 41.1 percent of dry etch tools, Tokyo Electron 23.4 percent and Applied Materials 17.4 percent, by CSET's count from TechInsights data[8].	$26.2B deposition tool market in 2025, by CSET's count from TechInsights data[3]. Dry etch tools were a $21.9B market in 2025[8]

WHO LEADS	US **Applied Materials** 82.4% of physical vapor deposition tools in 2025, by CSET's count from TechInsights data[9] US **Lam Research** 51.2% of conductor etch tools in 2025[10] JP **Tokyo Electron** 55.9% of dielectric etch tools in 2025[11] NL **ASM International** 54.5% of atomic layer deposition tools in 2025[6] CN **Naura** 14.7% of physical vapor deposition tools in 2025[9]
WHERE IT IS MADE	US **United States** Applied Materials, Lam Research JP **Japan** Tokyo Electron, Kokusai Electric NL **Netherlands** ASM International, atomic layer deposition KR **South Korea** Large equipment buyer CN **China** Naura, AMEC and Piotech, in the largest equipment market ($49.3B in 2025)
WHY SUBSTITUTION IS POSSIBLE	Atomic layer deposition lays down a film a fraction of a layer of atoms at a time[5]. At least five makers sell it[6].

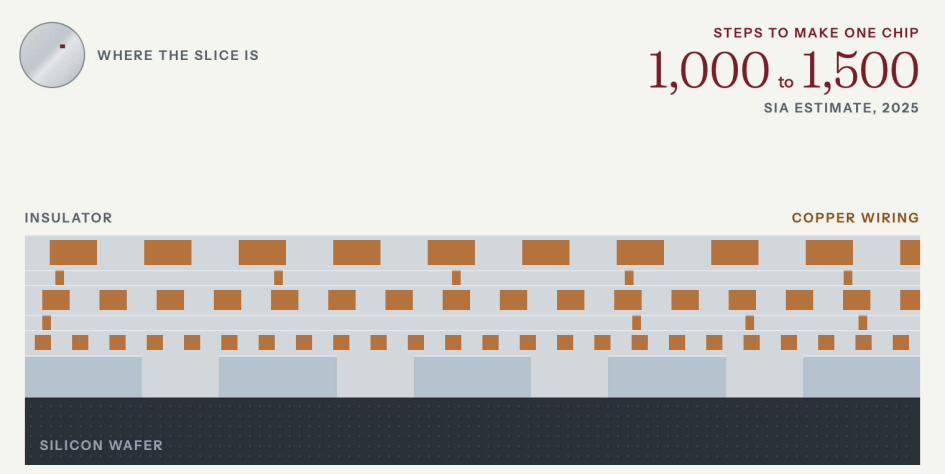
WHERE CHINA STANDS	WHERE THE US STANDS
Chinese toolmakers held 12.4 percent of dry etch in 2025, according to CSET's analysis of TechInsights data[8]. CSET's earlier analysis of TechInsights data had them under 1 percent of atomic layer deposition in 2024[12].	A December 2024 US rule lists some dry etch, atomic layer deposition and deep-hole deposition tools as controlled exports[13]. Its foreign direct product rule reaches some tools built outside the United States[14].

Applied Materials and Lam Research are the two largest deposition and etch suppliers. Applied Materials booked $28.368 billion of revenue in fiscal 2025[15]. Lam Research booked $23.233 billion in fiscal 2026[16].

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
5 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 16. govinfo.gov
6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
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12 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu
13 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 32. govinfo.gov
14 Commerce Strengthens Export Controls to Restrict China's Capability to Produce... U.S. Bureau of Industry and Security, 2 December 2024. bis.gov
15 Revenue from Contract with Customer, Excluding Assessed Tax... U.S. Securities and Exchange Commission, accessed 9 September 2026. data.sec.gov
16 LAM RESEARCH CORP, Form 8-K current report for the period ended 2026-07-29 (8-K). U.S. Securities and Exchange Commission, 29 July 2026. sec.gov

How it works

One round of deposition and etch



1. **The wafer.** A slice through one spot on a wafer, magnified until its layers show.
2. **Deposit.** A thin film is laid over the whole wafer.
3. **Coat.** A light-sensitive coating, the resist, is spun on top.
4. **Expose.** Light shines through a stencil and changes the resist wherever it lands.
5. **Develop.** A wash removes the resist the light reached, leaving the stencil's pattern.
6. **Etch.** A plasma cuts straight down through the film wherever the resist is gone.
7. **Strip.** The leftover resist is removed. The pattern now sits in the film.
8. **Repeat.** Insulator fills the gaps and the next layer goes on. Making one chip takes 1,000 to 1,500 steps, the Semiconductor Industry Association estimated in 2025.

Simplified cross-section. The layers are drawn much thicker than they are.

Most etch steps, and many deposition steps, happen in a vacuum chamber that holds one wafer at a time. A pump empties the chamber of air, and a measured flow of gas comes in. A radio-frequency field, an electric field that reverses millions of times a second, turns the gas into a plasma. The field splits the gas into charged ions and into neutral fragments called radicals, which are more reactive than whole molecules[17]. Batch furnaces such as Tokyo Electron's TELINDY PLUS instead heat as many as 125 wafers at once, usually without plasma, for oxidation, low-pressure chemical vapor deposition and some atomic layer deposition[18].

In most deposition steps, the gas carries the material the film will be made of. The molecules, or the radicals a plasma splits from them, land on the wafer and stick. The film builds up a few atoms at a time.

To coat the floor of a deep hole as thickly as its rim, the machine can work in cycles, a method called atomic layer deposition. First one gas settles in a single layer on every surface, and the pump removes the rest. Then a second gas reacts with that layer and nothing else, and the pump empties the

17 Etch Essentials: The Building Blocks of AI Era Microchips (Semi 101). Lam Research, 12 June 2024. newsroom.lamresearch.com
18 TELINDY™ Series. Tokyo Electron, accessed 9 September 2026. tel.com

chamber again. Each cycle adds a fraction of a layer of atoms wherever the gas reached, so the film grows as evenly on a hole's walls and floor as on its rim[5].

For etch, the gas attacks one material and leaves the one beside it. Wherever the photoresist covers the wafer, it shields the film beneath. Wherever it does not, the radicals react with the surface and turn it into a gas, which the pump carries away. A second field pulls the ions straight down at the wafer, so the etch cuts vertically and a narrow hole keeps its width as it deepens[17]. When the etch reaches the material underneath, which the gas was chosen not to touch, it slows almost to a stop, and the tool ends the step.

The US export controls describe each deposition or etch tool by the process it is designed to run[19].

Accenture and the Global Semiconductor Alliance estimated in 2020 that a part-finished chip can cross international borders 70 or more times on its way to the end customer[1]. The Semiconductor Industry Association repeated the figure in 2025[1].

Variants and trade-offs

Deposition families

The market sizes and leading shares below are 2025 figures from Georgetown University's Center for Security and Emerging Technology (CSET), based on data from the research firm TechInsights. CSET puts the deposition tool market at $26.2 billion[3]. It puts the dry etch market at $21.9 billion[8].

- **Physical vapor deposition** (PVD), which knocks atoms off a solid target and lands them on the wafer, was a $4.9 billion market in 2025[9]. Applied Materials held 82.4 percent of it and China's Naura 14.7 percent[9].
- **Chemical vapor deposition** (CVD), which reacts gases at the wafer surface to grow a film, was an $11.2 billion market in 2025, led by Applied Materials at 47 percent[20]. The plasma-enhanced version (PECVD) runs cooler and made up most of it, $7.5 billion[21].
- **Atomic layer deposition** (ALD) lays down less than one atomic layer at a time. Each step stops once every surface is covered, so it coats vertical walls as evenly as horizontal floors. US regulators say that even coating makes it important for building 3D structures, such as 3D DRAM, 3D NAND, and the FinFET and gate-all-around transistors in logic chips[5]. ASM International leads it with 54.5 percent of a $4.1 billion market in 2025[6].
- **Epitaxy** grows layers that line up with the crystal beneath them, using high-temperature CVD tools[22]. **Electroplating** deposits the copper used for

1 Testimony of David Isaacs, Semiconductor Industry Association, hearing on... U.S. Senate Committee on Finance, 14 May 2025, p. 2. finance.senate.gov
3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
5 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 16. govinfo.gov
6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
8 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
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17 Etch Essentials: The Building Blocks of AI Era Microchips (Semi 101). Lam Research, 12 June 2024. newsroom.lamresearch.com
19 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 28. govinfo.gov
20 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
21 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
22 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

wiring across the wafer[23].

Etch families

- **Conductor etch**, which cuts gates, metals and silicon, was a $12.2 billion market in 2025, led by Lam at 51.2 percent, by CSET's count from TechInsights data[10].
- **Dielectric etch**, which cuts the insulating oxides and nitrides between them, was an $8.7 billion market in 2025, led by Tokyo Electron at 55.9 percent, by CSET's count from TechInsights data[11].
- **High-aspect-ratio etch** cuts the deep, narrow channels that let 3D NAND stack layers of memory cells[24]. Lam says slight departures from a channel's intended shape can hurt yield[24]. Lam runs this etch well below 0 degrees Celsius[24]. It says its Cryo 3.0 technology can etch channels up to 10 microns deep whose width varies from top to bottom by less than 0.1 percent of their depth[24].

AI hardware needs sideways cuts, deeper holes and a second stack of metal wiring on the back of the wafer.

- **Gate-all-around** transistors need highly selective isotropic etch, which cuts sideways[25]. The US Bureau of Industry and Security (BIS) controls isotropic dry etch tools that remove silicon-germanium at least 100 times faster than silicon, a selectivity of 100:1 or more[19].
- **HBM** (high-bandwidth memory) needs holes etched straight through the silicon, called through-silicon vias (TSVs), so chips can be stacked and wired[26]. BIS controls the tools that etch these holes at aspect ratios (depth to width) of 10:1 or more and at rates above 7 microns a minute[13]. The control covers only tools that also keep the holes' depth even to within 2 percent across the wafer[13].
- **Backside power delivery** moves the wires that carry power to the back of the wafer[27]. In July 2024 Applied Materials said it expected backside power to add $1 billion to the market for its wiring tools, for every 100,000 wafers a month of new fab capacity[28].

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

11 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

13 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 32. govinfo.gov

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23 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 39. cset.georgetown.edu

24 Lam Research Introduces Lam Cryo™ 3.0 Cryogenic Etch Technology to Accelerate... Lam Research, 31 July 2024. newsroom.lamresearch.com

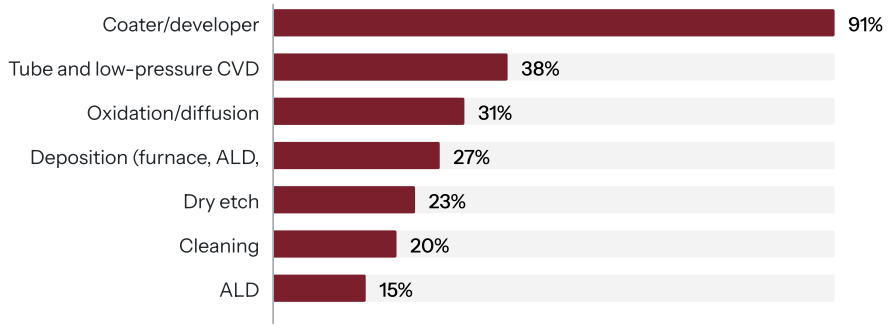
25 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 15. govinfo.gov

26 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 15. govinfo.gov

27 PowerVia Test Shows Industry-Leading Performance. Intel, 5 June 2023. download.intel.com

28 Applied Materials Unveils Chip Wiring Innovations for More Energy-Efficient... GlobeNewswire (release by Applied Materials), 8 July 2024. globenewswire.com

Tokyo Electron world market share by tool type, 2025 %



Source: Gartner, as reported in Tokyo Electron's FY2026 Q4 results presentation

Who makes it

The industry association SEMI put world sales of chipmaking equipment at $135.1 billion in 2025, up 15 percent from $117.1 billion[30]. Sales of wafer processing equipment, the category that includes deposition and etch tools, rose 12 percent[30].

No one firm dominates deposition and etch the way ASML dominates lithography. In dry etch, CSET's count for 2025 from TechInsights data puts Lam Research at 41.1 percent, Tokyo Electron at 23.4 percent and Applied Materials at 17.4 percent[8]. Tokyo Electron's own results presentation, citing Gartner, gives it 23 percent of dry etch, 27 percent of deposition systems (furnace, ALD and low-pressure CVD tools) and 15 percent of ALD in 2025[29]. The same presentation, again citing Gartner, gives it 91 percent of coater/developers in 2025[29]. These machines coat wafers with photoresist and develop the pattern after lithography.

ASM International, the atomic layer deposition specialist, reached a record EUR 3.2 billion of revenue in 2025 as customers built 2 nm gate-all-around capacity[31]. Molybdenum ALD, a new use of its tools that deposits the metal molybdenum, entered volume production that year[31].

What the world spent on chipmaking equipment $B



Source: SEMI, Worldwide Semiconductor Equipment Market Statistics

China and the controls

China is the tool vendors' largest market, buying $49.3 billion of equipment in 2025, half a percent less than in 2024[30]. Taiwan's spending rose 90 percent to $31.5 billion[30]. South Korea's rose 26 percent to $25.8 billion[30]. China still takes a large share of every major vendor's sales.

8 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
29 FY2026 (April 2025-March 2026) Financial Announcement, transcript. Tokyo Electron, 30 April 2026, p. 28. tel.com
30 SEMI Reports Global Semiconductor Equipment Billings Reached $135 Billion in... SEMI, 7 April 2026. semi.org
31 ASM reports Q4 and full-year 2025 results. ASM International, 3 March 2026, p. 2. asm.com

- **Tokyo Electron**: 34.1 percent of sales in the year to March 2026, down from 41.7 percent[32]. It was 30.4 percent in the quarter to June 2026[33].
- **Lam Research**: about 34 percent of revenue in the year to June 2026, the same as a year earlier[34].
- **Applied Materials**: 28 percent in the third fiscal quarter of 2026, down from 35 percent a year earlier[35].
- **ASM International**: more than 30 percent of 2025 revenue[31].

Between 2019 and 2024, by CSET's analysis of TechInsights data, China-based suppliers went from 2 to 7 percent of deposition and from under 3 to almost 9 percent of etch and clean[12]. In ALD they gained no share and held less than 1 percent in 2024[12].

CSET's newer analysis of data that TechInsights published in July 2026 puts them at 9.8 percent of deposition in 2025[3]. They held 10.1 percent of etch and clean that year[4]. In dry etch alone their share was 12.4 percent[8]. In ALD they held 2.6 percent in 2025, most of it from Piotech[6].

The Chinese toolmaker AMEC alone held 12.4 percent of dielectric etch in 2025, by CSET's count from TechInsights data[11]. Its revenue was RMB 12.385 billion that year, up about 36.62 percent, and it spent RMB 3.744 billion on research and development, 30.2 percent of sales[36].

The United States tightened its export controls at least five times between 2022 and 2025.

- **October 2022.** The 7 October interim final rule required a license for certain chipmaking equipment, and for some activities by US persons that support the making of certain chips in China[37]. It also extended US license rules to more foreign-made items bound for 28 Chinese entities on the Entity List, the US list of foreign firms that exporters need a license to supply[37].
- **October 2023.** The October rule, effective 17 November, rewrote the equipment entries[38]. It added sideways (isotropic) and vertical (anisotropic) dry etch (ECCN 3B001.c.1), spatial ALD (3B001.d.9), low-fluorine tungsten ALD and CVD (3B001.d.10) and carbon hard mask PECVD (3B001.d.5)[5].

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
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11 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
12 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu
31 ASM reports Q4 and full-year 2025 results. ASM International, 3 March 2026, p. 2. asm.com
32 FY2026 (April 2025-March 2026) Financial Announcement, transcript. Tokyo Electron, 30 April 2026, p. 27. tel.com
33 Q1 FY2027 (April-June 2026) Financial Announcement, transcript. Tokyo Electron, 30 July 2026, p. 6. tel.com
34 LAM RESEARCH CORP, Form 10-K annual report for the period ended 2026-06-28... U.S. Securities and Exchange Commission, 7 August 2026. sec.gov
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37 Implementation of Additional Export Controls: Certain Advanced Computing and... U.S. Government Publishing Office, 13 October 2022, p. 1. govinfo.gov
38 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 1. govinfo.gov

- **December 2024.** BIS controlled 24 more types of equipment and three types of software, and added 140 entities to the Entity List[14]. The new equipment entries include TSV etch, and deposition into features with aspect ratios above 200:1[39]. BIS also wrote a foreign direct product rule that reaches some tools built outside the United States[14]. The rule largely exempts foreign-made tools exported by firms located in Japan, the Netherlands and the 31 other countries it lists, whose governments can control these tools themselves[40]. Japan added advanced chipmaking equipment to its control list on 23 July 2023[41].
- **September 2025.** BIS removed the Chinese units of Samsung and SK hynix, and Intel Semiconductor (Dalian), from the Validated End-User program, effective 31 December 2025[42]. From that date, US tools bound for them need a license[42]. BIS estimated that the change would add 1,000 license applications a year[42].
- **September 2025.** The Affiliates Rule extended Entity List restrictions to any company at least 50 percent owned by a listed entity[43]. BIS then suspended it until 9 November 2026[44].

The rules apply to the toolmakers as well as to their customers. In fiscal 2026 Applied Materials took a $253 million charge to settle an export controls compliance matter with BIS[35].

Of the tools in the chart below, China-based suppliers have their largest share in dry stripping, which clears leftover photoresist off the wafer and was a $314.6 million market in 2025, by CSET's count from TechInsights data[45]. Kokusai held 35.7 percent of it, Mattson Technology 22.6 percent and Naura 8.3 percent[45]. Mattson is a US-based firm that Beijing E-Town bought in 2016[12].

12 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

14 Commerce Strengthens Export Controls to Restrict China's Capability to Produce... U.S. Bureau of Industry and Security, 2 December 2024. bis.gov

35 Applied Materials Announces Third Quarter 2026 Results. GlobeNewswire (release by Applied Materials), 13 August 2026. globenewswire.com

39 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 16. govinfo.gov

40 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 28. govinfo.gov

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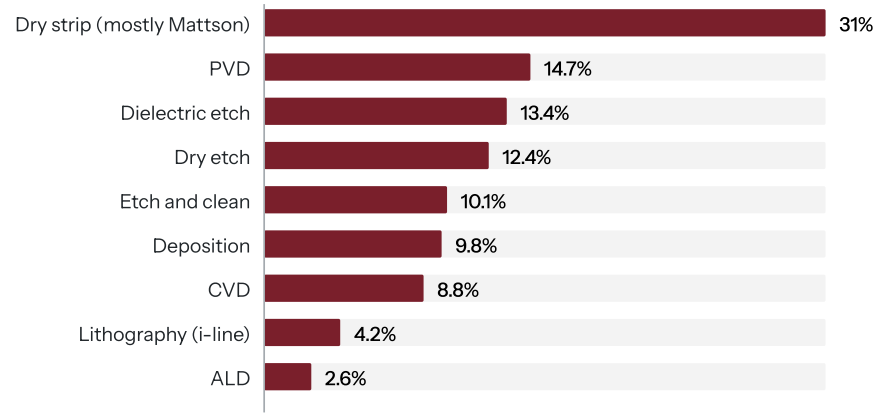
42 Revocation of Validated End-User Authorizations in the People's Republic of... U.S. Government Publishing Office, 2 September 2025. govinfo.gov

43 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. U.S. Government Publishing Office, 30 September 2025, p. 1. govinfo.gov

44 One Year Suspension of Expansion of End-User Controls for Affiliates of Certain... U.S. Government Publishing Office, 12 November 2025. govinfo.gov

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China-based suppliers' share of global tool segments, 2025 %



Source: CSET analysis of TechInsights data (published July 2026), ETO Chip Explorer

The chokepoint

Deposition and etch tools are a chokepoint, but a weaker one than lithography. Most deposition and etch steps have three or four credible vendors. Atomic layer deposition has at least five[6]. The December 2024 rule targets tools for advanced atomic layer deposition and for coating holes more than 200 times deeper than they are wide[7].

FOUR THINGS TO REMEMBER

- Deposition lays down thin films and etch cuts them into patterns. Making one chip takes 1,000 to 1,500 steps, by a 2025 industry estimate.
- Four American, Japanese and Dutch firms make most of these tools, and none dominates.
- Chinese makers held 12.4 percent of dry etch and 9.8 percent of deposition in 2025, on CSET's count.
- Chinese makers held 2.6 percent of atomic layer deposition in 2025, most of it from Piotech.

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CHAPTER 07

STAGE 7 OF 15

Metrology and Inspection

KLA, an American firm, held 56.8 percent of the market for tools that measure and inspect wafers in 2025, by CSET's count.

IN PLAIN TERMS

Metrology (measurement) and inspection are a chip factory's quality control. Between the steps that build a chip, another machine looks the wafer over. It checks whether the lines are the right width, whether this layer landed squarely on the last one, and whether a speck of dirt has killed a circuit. The main chip in an AI accelerator, the one that does the calculating, is one of the largest cut from a wafer. A bigger chip is a bigger target for a speck. In a textbook example with the same density of specks, 49.2 percent of chips 1 centimeter on a side come out flawless, but only 11.3 percent of chips 2 centimeters on a side[1]. In 2025 one American firm, KLA, held 56.8 percent of the market for these tools, by CSET's count from TechInsights data[2]. So a chip factory, or fab, cut off from KLA would find fewer of the faults that make its chips fail.

IN SHORT

By the count of CSET, a Georgetown University research center working from TechInsights data, defect inspection was the largest kind of measuring and inspection tool in 2025, a $6.7 billion market[2]. KLA held 85.9 percent of it[2]. Chinese makers held 0.7 percent of the whole market that year by the same count[2]. Yet China took 26 percent of KLA's revenue in the June 2026 quarter[3].

Since October 2022 the US has required a license for any tool subject to US export rules bound for Chinese fabs that make advanced logic or memory chips[4]. A December 2024 rule added specific tools to the export control list, among them inspection tools that can find defects of 21 nm or smaller[5].

1 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara, 11 October 2020, p. 8. web.ece.ucsb.edu

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

3 Letter to Shareholders Q4 Fiscal 2026. KLA Corporation, 28 July 2026, p. 5. d1io3yog0oux5.cloudfront.net

4 Implementation of Additional Export Controls: Certain Advanced Computing and... U.S. Government Publishing Office, 13 October 2022, p. 5. govinfo.gov

5 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 18. govinfo.gov

CHOKEPOINT CARD	HIGH CONCENTRATION	MODERATELY HARD TO SUBSTITUTE
CONCENTRATION High	SUBSTITUTABILITY Moderate Every kind of measuring and inspecting tool had a second maker in 2025[2].	PRICE OR MARKET SIZE $15.7B process control tool market in 2025[2]. KLA puts the 2025 wafer equipment market at about $120B[6]

WHO LEADS	US **KLA** 56.8% of process control tools in 2025 by CSET's count[2] US **Applied Materials** 11.2% of process control tools in 2025[2] JP **Lasertec** 6.4% of process control tools in 2025[2] JP **Hitachi High-Tech** 6.2% of process control tools in 2025[2] NL **ASML** 5.9% of process control tools in 2025[2] IL **Nova** 4.3% of process control tools in 2025[2] US **Onto Innovation** 2.6% of process control tools in 2025[2]
WHERE IT IS MADE	US **United States** KLA (Milpitas), Onto Innovation (Wilmington, MA), Applied Materials NL **Netherlands** ASML YieldStar and HMI e-beam, Veldhoven and San Jose JP **Japan** Hitachi High-Tech (CD-SEM), Lasertec (mask inspection), Rigaku (X-ray) IL **Israel** Nova (Rehovot) and Camtek (Migdal Haemek) CN **China** Skyverse (Shenzhen), RSIC and other domestic makers, with 0.7% of the market in 2025 by CSET's count
WHY SUBSTITUTION IS POSSIBLE	Eight firms besides KLA each held more than 1 percent of the market in 2025[2]. But KLA alone held 85.9 percent of defect inspection, the largest kind, that year[2].

WHERE CHINA STANDS	WHERE THE US STANDS
Chinese firms held 0.7 percent of the world market for these tools in 2025, by CSET's count[2]. Skyverse, a Shenzhen maker of inspection and metrology tools, reported 2,053,298,171.74 yuan of revenue in 2025, up 48.75 percent on 2024[7].	In 2025 the American firm KLA was the world's largest maker of metrology and inspection tools[2]. Citing Gartner, KLA puts its 2024 share at 56.5 percent, or 6.5 times its nearest rival[8].

Process control tools, the machines that measure and inspect wafers and masks, were a $15.7 billion market in 2025, by CSET's count from TechInsights data[2]. KLA puts the whole wafer equipment market, including advanced packaging, at about $120 billion that year[6].

CSET's count for 2025, built on TechInsights data, puts KLA at 56.8 percent and Applied Materials, the next largest, at 11.2 percent[2]. At its March 2026 investor day KLA cited Gartner's figures for 2024, which put its share at 56.5 percent, 6.5 times its nearest competitor[8]. In April 2026 KLA said its 2025 share was about 7 times the nearest competitor's[9].

How it works

An inspection tool sweeps a beam of light across the wafer, photographs it, and compares the image of each chip with the image of its neighbors. The

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
6 Letter to Shareholders Q4 Fiscal 2026. KLA Corporation, 28 July 2026, p. 7. d1io3yog0oux5.cloudfront.net
7 深圳中科飞测科技股份有限公司2025年年度报告摘要. CNINFO, 24 April 2026, p. 12. static.cninfo.com.cn
8 Compounding Sustainable Outperformance. KLA Corporation, 12 March 2026, p. 35. d1io3yog0oux5.cloudfront.net
9 Letter to Shareholders Q3 Fiscal 2026. KLA Corporation, 29 April 2026, p. 3. d1io3yog0oux5.cloudfront.net

chips on a wafer are meant to be identical, so the tool flags anything that shows in one and not the others as a possible defect. It records where each one is. For faults too small for light to show, a second tool scans a beam of electrons across a small patch and builds a picture from the electrons the beam knocks loose from the surface.

A metrology tool measures known shapes. It shines light on a test pattern printed beside the circuits and works out from the reflected light how wide the lines are and how thick the film is. It also reads marks printed in this layer and the last to measure how far the two are out of line, which chipmakers call overlay.

The fab pays for all this to protect yield, the share of chips on a wafer that work. A wafer loses value two ways: whole regions ruined by mishandling, misalignment or over-etching, and single circuits killed by stray material or an airborne particle[10].

A bigger chip is a bigger target for the second kind, so the cost of each good chip grows faster than its area[1]. Chipmakers call each chip on a wafer a die. In the standard formula, die yield is (1 + D0A/a) to the power minus a. Here D0 is defects per square centimeter, A is die area, and a is a clustering constant of 3 to 4 that says how much defects bunch together. At a defect density of 0.8 per square centimeter, with a set to 3, a 1 square centimeter die yields 49.2 percent and a 2 by 2 centimeter die 11.3 percent[1]. Because the bigger die yields less and fits fewer times on a wafer, it costs 20.1 times as much as the small one[1].

In another textbook example, a fab makes 5,000 wafers a week, each holding 200 dies, and earns $15 from each good die[11]. Each percentage point of yield is then worth $7.8 million a year[11]. Engineers raise yield by finding the process step that is losing dies and fixing it, work called yield learning.

The formula's percentages are the share of dies with no killer defect. A chip built from many identical blocks can carry spares and switch off a faulty block[12]. Nvidia's full H100 design has 144 of its main computing blocks, called streaming multiprocessors, and the SXM5 version it ships has 132 turned on[13]. So the share of dies that can be sold is higher than the formula gives. But the formula shows how fast flawless dies get scarce as chips grow.

1 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara, 11 October 2020, p. 8. web.ece.ucsb.edu

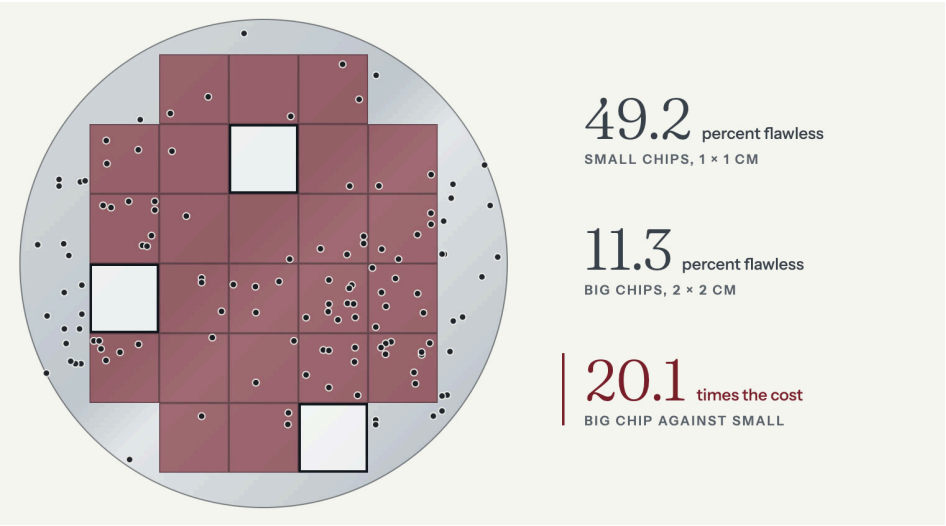
10 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara, 11 October 2020, p. 10. web.ece.ucsb.edu

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12 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara, 11 October 2020, p. 23. web.ece.ucsb.edu

13 NVIDIA Hopper Architecture In-Depth. Nvidia, 22 March 2022. developer.nvidia.com

Why big chips lose more to defects



1. **Specks land.** Specks of dust and tiny flaws land at random on every wafer. A textbook example assumes 0.8 per square centimeter.
2. **Small chips.** Of chips 1 centimeter on a side, 49.2 percent miss every speck and come out flawless.
3. **Big chips.** Chips 2 centimeters on a side are bigger targets for the same specks. Only 11.3 percent come out flawless.
4. **The cost.** The big chip yields less and fits fewer times on a wafer, so it costs 20.1 times as much as the small one.

Simulated wafer laid out like the textbook example: 120 chips 1 centimeter on a side, or 26 chips 2 centimeters on a side. Specks land at random in loose clusters, 0.8 per square centimeter. The share of flawless chips follows the formula above, with a clustering constant of 3, as closely as whole chips allow.

Variants and trade-offs

The tools trade sensitivity against speed. All the market sizes and shares below are CSET's, for 2025, from TechInsights data[2].

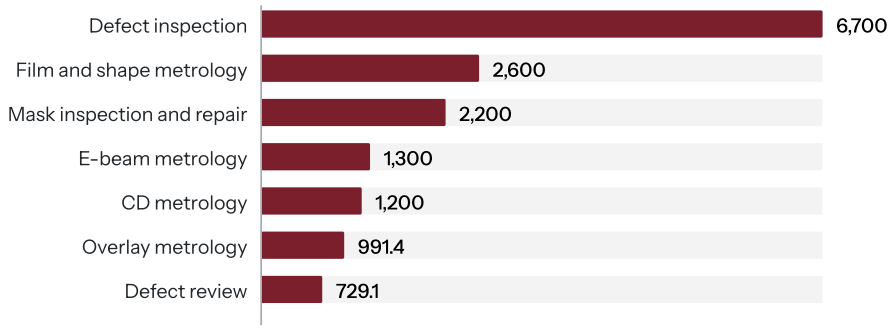
- **Defect inspection** scans a whole wafer with light, or small areas of it with electrons, and compares each die against its neighbors. It is by far the largest family, at $6.7 billion, and KLA held 85.9 percent of it. ASML's HMI eScan 1100 runs 25 electron beams at once, up to 15 times the throughput of single-beam inspection tools, and finds patterning defects down to 7 nm[14].
- **Film and shape metrology** measures how thick each film is and how flat the wafer is, without cutting it open. In this $2.6 billion family KLA held 45.6 percent, Nova 24.6 and Onto Innovation 10.9. Onto paid about $720 million in August 2026 for 27 percent of the Japanese X-ray tool maker Rigaku[15].
- **Mask inspection and repair** has its own tools, because a defect on a mask, the stencil the circuit pattern is printed from, repeats on every die. This $2.2 billion family is one of two where KLA has a close rival. KLA held 45.9 percent, Lasertec 42 and Zeiss 10.1. Lasertec also makes actinic inspection tools, which check EUV masks with the same 13.5 nm light that prints

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
14 The HMI eScan 1100. ASML, accessed 9 September 2026. asml.com
15 Onto Innovation Completes Strategic Investment in Rigaku Holdings Corporation. Onto Innovation, 10 August 2026. investors.ontoinnovation.com

them[16]. Lasertec calls its ACTIS A150 the world's first EUV patterned mask inspection system[16]. KLA's own actinic mask inspection tool was still at an early demo stage at its March 2026 investor day[17].

- **Overlay metrology** measures how well this layer lines up with the one below. Its $991.4 million market was split almost evenly between KLA at 51.5 percent and ASML at 47.9.
- **Critical dimension metrology**, also called line-width measurement, measures how wide a printed feature came out. It is the one family Japan leads. Hitachi held 71.9 percent of this $1.2 billion market and Applied Materials 27.5.
- **Electron-beam metrology** resolves detail too fine for light. Applied Materials led this $1.3 billion family at 45.1 percent, with ASML at 35.6 and KLA at 18.8.

Process control market by tool family, 2025 $M

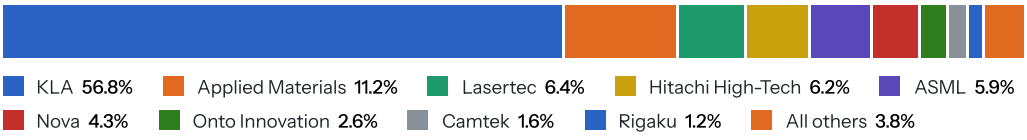


Source: ETO Supply Chain Explorer, CSET

Who makes it

KLA says its share grew 3.6 percentage points from 2021 to 2025[9]. It says it gained share again in 2025 across mask, optical wafer and e-beam inspection[9]. Its revenue for fiscal 2026, the year to June 2026, was $13.579 billion, up from $12.156 billion in fiscal 2025[18].

Process control tool revenue share, 2025 %



Source: ETO Supply Chain Explorer, CSET

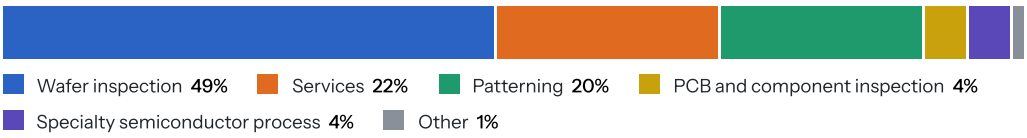
The rest are specialists, and none holds more than 11.2 percent, by CSET's count from TechInsights data for 2025[2]. Eight of them each held more than 1 percent of the market that year[2].

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
9 Letter to Shareholders Q3 Fiscal 2026. KLA Corporation, 29 April 2026, p. 3. d1io3yog0oux5.cloudfront.net
16 ACTIS A150. Lasertec, 12 September 2019. lasertec.co.jp
17 Compounding Sustainable Outperformance. KLA Corporation, 12 March 2026, p. 91. d1io3yog0oux5.cloudfront.net
18 Revenue from Contract with Customer, Excluding Assessed Tax... U.S. Securities and Exchange Commission, accessed 9 September 2026. data.sec.gov

- **Applied Materials** is the largest of them. It sells e-beam defect review tools, CD-SEMs (electron microscopes that measure line width) and e-beam metrology tools next to the deposition and etch tools whose work they check.
- **Lasertec** holds 6.4 percent, almost all of it mask inspection[2]. **Hitachi High-Tech** holds 6.2 percent, most of it from CD-SEMs[2]. **ASML** holds 5.9 percent through YieldStar overlay and HMI e-beam metrology[2].
- **Nova** holds 4.3 percent, through film metrology[2]. It took $880.6 million of revenue in 2025[19].
- **Onto Innovation** holds 2.6 percent[2]. It took $1,005.3 million of revenue in 2025[20].
- **Camtek** holds 1.6 percent[2]. It took $496.1 million of revenue in 2025[21].

KLA's patterning line, which includes several metrology products and mask inspection, grew 61 percent year on year in the June 2026 quarter[3]. In that quarter Taiwan took 31 percent of KLA's revenue and China 26 percent[3].

KLA revenue by product line, June 2026 quarter %



Source: KLA letter to shareholders, Q4 FY2026

KLA says process control's share of spending on chipmaking equipment is still rising[22]. In memory chips, KLA credits the rise to higher performance specifications, less redundancy (fewer spare circuits) and more logic content in devices built for high-performance computing[23].

KLA's systems analyze data with AI and also filter out noise[23]. In April 2026 KLA said it had added AI to every product line over the previous five years[23].

Packaging, AI and China

KLA says the AI build-out is making advanced packaging, and the process control it needs, more important[24]. KLA became the largest supplier of process control tools for advanced wafer-level packaging in 2025, adding 14 percentage points of share[9]. It expects about $1.1 billion of packaging process control revenue in 2026, up more than 70 percent[24]. In February 2026 Onto said it had signed a volume purchase agreement estimated at over $240

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
3 Letter to Shareholders Q4 Fiscal 2026. KLA Corporation, 28 July 2026, p. 5. d1io3yog0oux5.cloudfront.net
9 Letter to Shareholders Q3 Fiscal 2026. KLA Corporation, 29 April 2026, p. 3. d1io3yog0oux5.cloudfront.net
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20 Onto Innovation Reports 2025 Fourth Quarter and Full Year Results. Onto Innovation, 19 February 2026. investors.ontoinnovation.com
21 CAMTEK LTD, Form 6-K report of foreign private issuer for the period ended... U.S. Securities and Exchange Commission, 18 February 2026. sec.gov
22 Letter to Shareholders Q3 Fiscal 2026. KLA Corporation, 29 April 2026, p. 12. d1io3yog0oux5.cloudfront.net
23 Letter to Shareholders Q3 Fiscal 2026. KLA Corporation, 29 April 2026, p. 4. d1io3yog0oux5.cloudfront.net
24 Letter to Shareholders Q4 Fiscal 2026. KLA Corporation, 28 July 2026, p. 2. d1io3yog0oux5.cloudfront.net

million with a high-bandwidth memory (HBM) maker for 2D inspection and 3D bump metrology through 2027[20].

Chinese makers are still small in process control, but one of them is growing fast. CSET's count, built on TechInsights data, puts all Chinese firms at 0.7 percent of the market in 2025[2]. Skyverse (Shenzhen Zhongke Feice), a maker of wafer inspection and metrology tools, reported 2,053,298,171.74 yuan of revenue in 2025[7]. That was up 48.75 percent on 2024[7]. In its 2025 annual report Skyverse says Chinese makers still hold a low share of China's market for these tools and that KLA holds the largest share of it[25].

By CSET's 2025 count from TechInsights data, Chinese firms held 0.6 percent of defect inspection tools[26]. They held 0.4 percent of line-width measurement tools in 2025[27]. In film and shape metrology they held 2 percent in 2025, all of it from Raintree Scientific Instruments, or RSIC[28]. They held no share of mask inspection and repair tools in 2025[29]. Nor did they hold any share that year of the tools that inspect wafers during advanced packaging[30].

In October 2022 the US began requiring a license for any item subject to US export rules, process control tools included, bound for certain Chinese fabs[4]. The rule covered fabs that made logic chips at 16/14 nm or below, NAND memory with 128 or more layers, or DRAM at 18 nm half-pitch or below[31].

A December 2024 rule added specific process control tools to the export control list. It controls tools for defect inspection and measurement of patterned 300 mm wafers if they can find defects of 21 nm or smaller[32]. They must do this using light with a wavelength below 400 nm, an electron beam with a resolution of 1.65 nm or better, a cold field emission source, or two or more electron-beam sources[32]. The same rule controls overlay metrology accurate to 0.5 nm or better if the tool is built into the track (the machine that coats and develops the light-sensitive resist) or can switch wavelengths quickly[32]. It also controls stand-alone wafer shape metrology that feeds corrections to immersion or EUV scanners[32]. A separate new entry, 3B994, covers optical film and line-width metrology for 300 mm wafers with software for measuring non-planar transistors[33].

BIS, the US Commerce Department bureau that wrote the rule, placed these tools in new entries for items it says also have legitimate uses in older

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
4 Implementation of Additional Export Controls: Certain Advanced Computing and... U.S. Government Publishing Office, 13 October 2022, p. 5. govinfo.gov
7 深圳中科飞测科技股份有限公司2025年年度报告摘要. CNINFO, 24 April 2026, p. 12. static.cninfo.com.cn
20 Onto Innovation Reports 2025 Fourth Quarter and Full Year Results. Onto Innovation, 19 February 2026. investors.ontoinnovation.com
25 深圳中科飞测科技股份有限公司2025年年度报告摘要. CNINFO, 24 April 2026, p. 11. static.cninfo.com.cn
26 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
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29 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
30 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
31 Implementation of Additional Export Controls: Certain Advanced Computing and... U.S. Government Publishing Office, 13 October 2022, p. 16. govinfo.gov
32 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 37. govinfo.gov
33 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 38. govinfo.gov

chipmaking[34]. So they carry narrower license requirements than tools built only for advanced chips[34]. A new foreign direct product rule extends US jurisdiction to specified foreign-made chipmaking tools bound for Macau or for a Country Group D:5 destination such as China[35].

The chokepoint

One firm, KLA, held more than half the process control market in 2025. Every kind of process control tool also had a second maker that year, by CSET's count from TechInsights data[2]. Several firms can build a good film metrology tool, but KLA led four of the eight kinds of process control tool in 2025, by CSET's count[2]. KLA says its share of the whole market grew from 2021 to 2025[9]. A fab that can buy scanners and etchers but not sensitive inspection tools finds fewer of the defects that lower its yield, so its yield rises more slowly.

FOUR THINGS TO REMEMBER

- Metrology and inspection tools check each wafer between steps and help a fab find out why its chips fail.
- AI chips are among the largest made, so finding defects matters most for them.
- CSET's count puts KLA of the United States at 56.8 percent of this tool market in 2025, and Applied Materials, the next largest, at 11.2 percent.
- Chinese makers held 0.7 percent of this tool market in 2025 on CSET's count, though one Chinese maker, Skyverse, grew its revenue 48.75 percent that year.

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CHAPTER
08
STAGE 8 OF 15

Transistors and the Front End

Only TSMC, Intel and Samsung make 2 nm-class transistors in volume. TSMC plans its A14 process for 2028, with more than 20 percent higher logic density than N2.

IN PLAIN TERMS

The front end of a chip factory is the part that builds the transistors, the billions of tiny switches in every chip. Each switch works like a valve. Current runs along a narrow strip of silicon called the channel, and a small voltage on a gate above the strip opens or shuts the flow. For decades chipmakers made the strips shorter, so they switched faster and more of them fit on a chip. But once the strip gets short enough, the valve stops sealing. Current leaks through even when the gate is off, and the chip burns power doing nothing. The newest kind of transistor, called gate-all-around, cuts this leak by wrapping the gate around all four sides of the strip. Building them takes a long sequence of steps that have to run in exact order. Only three companies, TSMC, Intel and Samsung, make these switches in volume. In 2024 the United States put a worldwide license requirement on the know-how for building them, with an exception for countries that impose matching controls.

IN SHORT

Intel's 18A process shipped gate-all-around transistors together with backside power, a second power network built underneath them[1]. TSMC says its A14 process will give up to 15 percent more speed or up to 30 percent less power than N2[2]. Rapidus, a state-backed newcomer in Japan, said in July 2025 that it would begin 2 nm mass production in 2027[3].

China's SMIC makes 7 nm-class chips without extreme-ultraviolet (EUV) lithography, the most advanced way to print chip patterns[4]. China has announced $43 billion to develop its own EUV[5]. In September 2025 the Center for Strategic and International Studies judged that EUV remains firmly out of reach[5].

1 Intel Foundry Details Process Milestones and Future Innovation at VLSI Symposium. Intel, 16 June 2026. intc.com
2 TSMC Unveils Next-Generation A14 Process at North America Technology Symposium. TSMC, 24 April 2025. pr.tsmc.com
3 Rapidus Achieves Significant Milestone at its State-of-the-Art Foundry with... Rapidus, 18 July 2025. rapidus.inc
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CHOKEPOINT CARD EXTREME CONCENTRATION | HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	Hard TSMC, Intel and Samsung all make chips with gate-all-around transistors, the newest kind[6].	$17,000 per 300 mm wafer on TSMC's 4NP process, Epoch AI's December 2025 estimate[7]

WHO LEADS	TW **TSMC** N2 in volume production since 4Q25, TSMC's first nanosheet node[8] US **Intel** 18A in high-volume production in the United States, as of 2026[9] KR **Samsung Foundry** Started initial production of its 3 nm gate-all-around process in June 2022[10] CN **SMIC** N+3, a 7 nm-class process made without EUV, SemiAnalysis found in June 2026[4] JP **Rapidus** Said in July 2025 it would begin 2 nm mass production in 2027[3]
WHERE IT IS MADE	TW **Taiwan** TSMC N2 at Fab 20 Hsinchu and Fab 22 Kaohsiung US **United States** Intel 18A at Fab 52, Arizona KR **South Korea** Samsung Hwaseong and Pyeongtaek JP **Japan** Rapidus Chitose, Hokkaido CN **China** SMIC Shanghai and Beijing 300 mm lines
WHY SUBSTITUTION IS SLOW	Rapidus, a newcomer, is building on IBM's 2 nm technology under a December 2022 agreement[11]. Since September 2024 the United States has required a license to share the know-how for building these transistors, except with countries that impose matching controls[12].

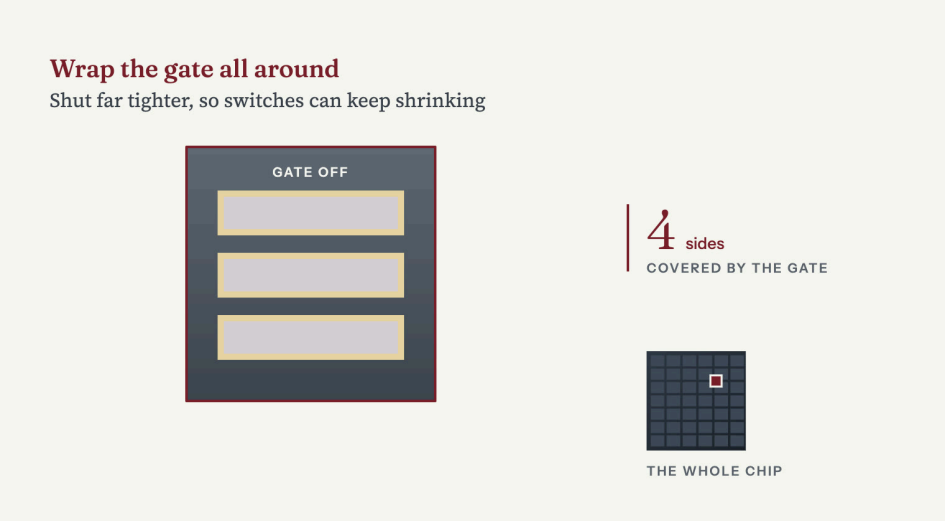
WHERE CHINA STANDS	WHERE THE US STANDS
SMIC ships its N+3 process, which SemiAnalysis measured in June 2026 at 113.4 million transistors per square millimeter[4]. It builds its tightest layers in several patterning steps because it cannot buy an extreme-ultraviolet scanner[4].	In August 2025 Intel called itself the only chipmaker that both develops and manufactures leading-edge logic chips in the United States[13]. It says its 18A process brought gate-all-around transistors and backside power to market together in 2025[1].

Nvidia's Blackwell GPU holds 208 billion switches on two dies, each die the largest a lithography machine can print in one shot[14]. Every switch has to turn on fully, let almost no current through when off, and keep doing it for years. Making the switches is called the front end of line. Wiring them together is the back end of line.

1 Intel Foundry Details Process Milestones and Future Innovation at VLSI Symposium. Intel, 16 June 2026. intc.com
3 Rapidus Achieves Significant Milestone at its State-of-the-Art Foundry with... Rapidus, 18 July 2025. rapidus.inc
4 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com
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7 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai
8 2nm Technology. TSMC, accessed 9 September 2026. tsmc.com
9 Intel 18A: See Our Biggest Process Innovation. Intel, accessed 18 October 2026. intel.com
10 Samsung Begins Chip Production Using 3nm Process Technology With GAA... Samsung Global Newsroom, 30 June 2022. news.samsung.com
11 IBM and Rapidus Form Strategic Partnership to Build Advanced Semiconductor... Rapidus, 13 December 2022. rapidus.inc
12 Commerce Control List Additions and Revisions; Implementation of Controls on... Federal Register, 6 September 2024. federalregister.gov
13 Intel and Trump Administration Reach Historic Agreement to Accelerate American... Intel, 22 August 2025. intc.com
14 NVIDIA Blackwell Architecture. Nvidia, 18 December 2025. nvidia.com

How it works

Why the gate wraps around the channel



1. **Inside a chip.** An AI chip does all its work with transistors, tiny switches built into the silicon. One Nvidia Blackwell GPU holds 208 billion of them. Making the switches is the front end of the factory, and wiring them together comes after.
2. **On.** Each one is a switch. Seen end on, current flows through a strip of silicon, the channel, and the gate sits on top.
3. **Off, leaking.** Switching off, the gate squeezes the flow shut, but only near itself. In a tiny transistor, current still leaks through the part it cannot reach.
4. **Fin.** Stand the channel on edge as a fin, and the gate covers three sides. Much less gets through.
5. **All around.** Split it into thin sheets and wrap the gate around every side, and it shuts far tighter. That lets the switches keep shrinking, so each new chip can pack in more of them.

Simplified. The channel is seen end on, cut across the gate, with current drawn as points of light. Not to scale.

The front end builds the transistors in a fixed order, starting with the channel. In a FinFET, the older design, the fab etches the silicon into thin fins that stand up from the wafer. The gate then goes over the top and both sides of each fin, three sides instead of one.

In a gate-all-around transistor, the fab instead grows alternating layers of silicon and silicon-germanium and cuts the stack into narrow bars[15]. It lays a placeholder, called a dummy gate, across each bar where the real gate will go[16]. On either side of the placeholder it cuts deep trenches where the source and drain, the two ends of the channel, will go[17]. Through those trenches it etches the silicon-germanium layers back a little and fills the gaps with an insulator, the inner spacer, which keeps the gate apart from the source and drain[15]. Then it grows the source and drain as crystal inside the trenches, a step called epitaxy[17].

15 Entering the nanosheet transistor era. Imec, 11 February 2022. imec-int.com
16 Development of SiGe Indentation Process Control for Gate-All-Around FET... ArXiv, 12 January 2022. arxiv.org
17 Applied Materials Unveils Next-Gen Chipmaking Products to Supercharge AI... Applied Materials, 7 October 2025. ir.appliedmaterials.com

Only then does the fab remove the placeholder and etch away the silicon-germanium, leaving thin silicon sheets that hang between the source and drain like shelves[15]. The inner spacer shields the source and drain from that etch[16]. The fab coats every sheet with an insulating film a few atoms thick, so the gate can control the channel without touching it. Gate metal then fills the gaps above, below and beside each sheet, covering all four sides[18]. Last come the contacts, small metal plugs down to the source, the drain and the gate, ready for the wiring layers above.

Each of those steps is a separate deposition, etch or implant. An implant step, called ion implantation, fires atoms of boron or phosphorus into chosen patches of silicon so they conduct the way the design needs. The steps have to run in exact order, because later layers cover earlier ones. Once gate metal surrounds a sheet, the insulator can no longer go between them.

Current leaks in two main ways, straight through the gate insulator and along the channel from end to end while the gate is off. The second grows as the channel gets shorter, and wrapping the gate around the channel reduces it. Three fixes deal with the two leaks:

- **Better insulator.** A thicker hafnium-based film, paired with a metal gate, lets fewer electrons slip straight through yet controls the channel as tightly as before. Intel replaced the silicon dioxide insulator with it at 45 nm in 2007[19]. It said the new film cut leakage through the insulator more than tenfold[19]. With the metal gate, it could either raise the current a switch carries when on by more than 20 percent or cut leakage along the channel more than fivefold[19].
- **More gate.** Fins and nanosheets wrap gate metal around more sides of the channel.
- **Thinner channel.** The fab thins the silicon until no current can flow where the gate cannot reach it.

Variants and trade-offs

Planar and FinFET

A planar transistor lays the gate on one side of a flat channel. As the channel shortens, the gate loses control of it. A fin stands the channel on edge so the gate covers three sides. But each fin adds a fixed amount of current, so a designer who needs more current has to add a whole fin.

Gate-all-around nanosheets

A nanosheet transistor lays the channel flat again, stacks two to four thin sheets and wraps gate metal around each one. The sheets can be drawn at different widths, so current can be set in finer steps. TSMC sells this as NanoFlex, which lets designers mix short logic cells for small area with tall cells for performance[20].

15 Entering the nanosheet transistor era. Imec, 11 February 2022. imec-int.com
16 Development of SiGe Indentation Process Control for Gate-All-Around FET... ArXiv, 12 January 2022. arxiv.org
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19 Intel's Transistor Technology Breakthrough Represents Biggest Change to... Intel, 27 January 2007. intel.com
20 TSMC Celebrates 30th North America Technology Symposium with Innovations... TSMC, 24 April 2024. pr.tsmc.com

TSMC's N2, its first nanosheet node, entered volume production in the fourth quarter of 2025[8]. TSMC rates the follow-on N2P at 18 percent more speed at the same power and 36 percent less power at the same speed than its earlier N3E[21]. It gives N2P 1.2 times the logic density of N3E and 1.15 times its chip density, two separate measures[21].

Intel's version, called RibbonFET, is built into its 18A process, which Intel says is in high-volume production in the United States[9]. An improved version, 18A-P, adds another 9 percent performance at the same power[1]. Samsung got there first, starting initial production of its 3 nm gate-all-around process in June 2022[10]. But SemiAnalysis reported in February 2023 that, because of yield problems, high-volume chips on Samsung's 3 nm gate-all-around technology were expected only in 2024[22].

Those numbers are the foundries' own claims until someone measures the silicon. Each foundry picks its own baseline, and none publishes a density figure that can be compared with a rival's.

TSMC's published chip density gains (a separate measure from the logic density it also quotes), each over the baseline TSMC names, with A16 at its stated maximum %



Source: TSMC advanced technologies, HPC platform

Backside power delivery

In a conventional chip, power and signal share one stack of copper wiring above the transistors, where wide power rails crowd out the signal wires. With backside power, the fab grinds the wafer thin and builds a second network for power underneath it.

Intel's version, PowerVia, is part of 18A, and Intel calls it an industry first[9]. It cut routed area by 11 percent and dynamic voltage droop tenfold[1]. Droop is the sag in supply voltage when a block of logic switches on at once. TSMC's Super Power Rail connects the backside network directly to each transistor's source and drain, the two ends of its channel[23]. Against N2P, TSMC rates A16, which uses Super Power Rail, at 8 to 10 percent more speed, 15 to 20 percent less power and up to 1.10 times the chip density[21].

TSMC plans to start A14 production in 2028, with more than 20 percent higher logic density than N2[2]. It said in April 2026 that two more processes, A13 and A12, are scheduled to enter production in 2029[24]. A13 is a direct shrink of A14

1 Intel Foundry Details Process Milestones and Future Innovation at VLSI Symposium. Intel, 16 June 2026. intc.com
2 TSMC Unveils Next-Generation A14 Process at North America Technology Symposium. TSMC, 24 April 2025. pr.tsmc.com
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21 HPC Platform – Advanced Technologies. TSMC, accessed 9 September 2026. tsmc.com
22 The Future of the Transistor. SemiAnalysis, 21 February 2023. newsletter.semianalysis.com
23 Clash of the Foundries: Gate All Around + Backside Power at 2nm. SemiAnalysis, 1 October 2024. newsletter.semianalysis.com
24 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com

with 6 percent area savings, and A12 adds Super Power Rail to the A14 platform[24].

TSMC's logic roadmap beyond N2, from its own announcements

- 2028
  N2U, 3-4% faster or 8-10% lower power than N2P
- 2028
  A14, the base node for A13 and A12
- 2029
  A13, a direct shrink of A14 for 6% area savings
- 2029
  A12, the A14 platform with Super Power Rail backside power

Source: TSMC, 2026 North America Technology Symposium

CFET and what comes after

A basic logic circuit needs two transistors that switch on opposite signals, and in production chips the two sit side by side. Complementary field-effect transistors, or CFETs, stack the two on top of each other[25]. The research institute imec sees them as an attractive design for logic nodes beyond 1 nm[25].

The back end of line

Wiring has become as hard to make as the transistor. Copper seeps into the insulator around it, so every wire needs a barrier, a liner and a cap[26]. As the most critical wiring levels drop below a 20 nm pitch, those extra layers take a large share of the wire's width[26]. Pitch is the distance from the center of one wire to the center of the next. Two metals are moving into those narrow levels.

- **Ruthenium** shows promise for use without a barrier, which would keep the wire's resistance low[26]. imec has demonstrated ruthenium lines at a 16 nm pitch[27].
- **Molybdenum** conducts worse than copper in bulk, with a resistivity of 5.3 microhm-cm against copper's 1.68[28]. But in a thin film it has up to 30 percent lower resistivity than tungsten and needs no barrier either, so chipmakers are putting it into some contacts and local wiring layers[28].

Who makes it

TSMC, Intel and Samsung build gate-all-around nodes with extreme-ultraviolet (EUV) lithography. China's SMIC builds 7 nm-class nodes without it[4].

4 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com
24 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com
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27 16nm Ru lines using semi-damascene integration approach. Imec, 3 June 2025. imec-int.com
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SemiAnalysis reported in September 2026 that Intel's Panther Lake, Samsung's Exynos 2600 and Apple's A20 Pro, made on TSMC's N2, all use gate-all-around transistors[6]. In August 2025 Intel called itself the only semiconductor company that both develops and manufactures leading-edge logic chips in the United States[13].

In July 2025 Rapidus, a state-backed newcomer in Japan, said it had started making 2 nm prototypes at its plant in Chitose, Hokkaido, and would begin mass production in 2027[3]. Its process builds on IBM's 2 nm technology, which the two firms agreed in December 2022 to develop further for production at Rapidus's plant in Japan[11]. By June 2026 Rapidus had raised 424.95 billion yen in equity, the latest 150 billion yen from a state agency[29]. It has also received government research subsidies since fiscal 2022[29].

A node's name, such as 7 nm, does not measure any feature on the chip. Applied Materials, a toolmaker, said in 2022 that chipmakers raise density in two ways, by printing smaller features and by laying out each logic cell more tightly[18]. SMIC's N+3, its third-generation 7 nm process, sets its narrowest wires 32.5 nm apart, tighter than the 36 nm in Intel's Panther Lake chips on 18A[4]. But 18A allows 32 nm, and SemiAnalysis called the comparison an incomplete, cherry-picked metric[4]. N+3 packs 113.4 million transistors per square millimeter, against 107.7 for TSMC's N6[4].

AI accelerators have usually trailed the leading edge. A die as large as theirs is more likely to hold a defect[30]. So the share of dies that work, called the yield, gets high enough only once a process has run long enough to cut its defect rate.

Blackwell uses a custom TSMC 4NP process[14]. Epoch AI estimated in December 2025 that a 300 mm wafer on that process costs $17,000[7]. AMD's MI455X, launched on 23 July 2026, lists TSMC 2 nm and 3 nm silicon[31]. It spreads its compute across eight separate dies[32].

SRAM is the fast memory on the logic die. SemiAnalysis reported in 2024 that the cell that holds one bit had barely shrunk since the 5 nm node[23]. TSMC said in 2025 that its 2 nm SRAM uses a 0.021 square-micrometer cell and gets 1.1 times the density of its previous node by tuning design and process together[33]. See Memory and HBM.

3 Rapidus Achieves Significant Milestone at its State-of-the-Art Foundry with... Rapidus, 18 July 2025. rapidus.inc
4 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com
6 Intel Panther Lake Teardown. SemiAnalysis, 26 September 2026. newsletter.semianalysis.com
7 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai
11 IBM and Rapidus Form Strategic Partnership to Build Advanced Semiconductor... Rapidus, 13 December 2022. rapidus.inc
13 Intel and Trump Administration Reach Historic Agreement to Accelerate American... Intel, 22 August 2025. intc.com
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23 Clash of the Foundries: Gate All Around + Backside Power at 2nm. SemiAnalysis, 1 October 2024. newsletter.semianalysis.com
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30 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara, 11 October 2020, p. 8. web.ece.ucsb.edu
31 AMD Instinct™ MI455X GPUs. AMD, accessed 9 September 2026. amd.com
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The chokepoint

TSMC, Intel and Samsung do not sell their leading-edge process flows, which set the steps, their order and the tools for each. In September 2024 the United States put a worldwide license requirement on the know-how for building gate-all-around transistors, with an exception for countries that impose matching controls[12].

China launched its third National Integrated Circuit Fund in May 2024 with 344 billion yuan, about $47 billion[34]. But China cannot buy EUV scanners. Its 7 nm chips have been made instead on deep-ultraviolet machines, a generation behind EUV, stockpiled from ASML, the Dutch scanner maker, before export restrictions took effect[5]. SMIC gets fine features out of them by building its tightest layers in several patterning steps[4].

In July 2026 Reuters reported that a Shanghai state firm had begun making home-grown immersion deep-ultraviolet scanners[35]. Citing The Information, it said the firm planned about five that year and roughly 20 in 2027[35]. Reuters's own source said the firm's machine still needed testing and was far from matching ASML's models[35].

Domestic EUV is further off. China has announced $43 billion to develop it[5]. In September 2025 the Center for Strategic and International Studies (CSIS) judged that EUV remains firmly out of reach despite massive state investment[5].

See Lithography and Who Controls the Chain.

FOUR THINGS TO REMEMBER

- The newest transistors wrap the gate around all four sides of the channel to cut the current that leaks when the switch is off.
- Only TSMC, Intel and Samsung make them in volume.
- TSMC, Intel and Samsung do not sell their leading-edge process flows, so a newcomer has to develop its own or build on another firm's, as Rapidus is doing with IBM's.
- China's SMIC makes 7 nm-class chips without EUV by building its tightest layers in several patterning steps.

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CHAPTER
09
STAGE 9 OF 15

Foundries and Fabs

TSMC's six largest plants are all in Taiwan. In 2025 they could make more than 13 million wafers a year, out of more than 17 million for the whole company.

IN PLAIN TERMS

A foundry is a company that makes chips for other companies in its chip factories, or fabs. Those companies send in their designs, and the foundry runs them all through the same building on the same tools. The building costs tens of billions of dollars, and the air inside is cleaner than in an operating room. Even so, each wafer, the silicon disc the chips are built on, moves from machine to machine inside sealed boxes and never touches that air. No chip company sells enough of one product to fill such a factory on its own, so chip companies share a handful of them. TSMC runs the biggest, and all six of its largest plants are in Taiwan, so a political crisis over Taiwan would affect the whole chip supply.

IN SHORT

In 2022 Taiwan held 69 percent of the world's capacity for advanced logic chips, those below 10 nanometers, and the United States almost none[1]. In July 2026 TSMC raised its planned investment in Arizona to $265 billion across 12 facilities[2].

The industry association SEMI projected in 2026 that China would keep the world's largest installed fab capacity in 2026 and 2027[3]. China's SMIC cannot buy an EUV scanner, the machine that prints the finest layers, yet it makes 7 nm-class chips by printing those layers in several passes on older deep-ultraviolet scanners[4].

1 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org
2 TSMC Announces Additional $100 Billion Investment In Arizona. City of Phoenix, 16 July 2026. phoenix.gov
3 World Fab Forecast. SEMI, accessed 9 September 2026. semi.org
4 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

CHOKEPOINT CARD	EXTREME CONCENTRATION	HARD TO SUBSTITUTE
CONCENTRATION Extreme	SUBSTITUTABILITY Hard TSMC has not built one of its six largest fabs outside Taiwan[5].	PRICE OR MARKET SIZE $35-43 billion over ten years to own a new 2 nm-class fab completed in 2026, BCG's 2023 estimate[6]

WHO LEADS	TW **TSMC** 79% of its wafer revenue from 7 nm and below, 3Q26[7] US **Intel Foundry** $5.8bn segment revenue in 2Q26, only $293m of it from outside customers[8] KR **Samsung Foundry** Over $37bn planned in Texas (CHIPS award, Dec 2024)[9] CN **SMIC** $9.327bn revenue in 2025[10] JP **Rapidus** Raised 267.6 billion yen in February 2026 to reach 2 nm mass production by 2027[11]
WHERE IT IS MADE	TW **Taiwan** TSMC (six gigafabs, N2 and N3 lines) and UMC KR **South Korea** Samsung Foundry, Hwaseong and Pyeongtaek CN **China** SMIC and Hua Hong US **United States** Intel Foundry, GlobalFoundries, TSMC Arizona, Samsung Taylor JP **Japan** TSMC Kumamoto, Rapidus Chitose
WHY SUBSTITUTION IS SLOW	In March 2026 SIA put the cost to build and equip a leading-edge fab at as much as $20 to $25 billion[12]. SemiAnalysis judged in December 2025 that finding the highly skilled engineers a foundry needs remained a major challenge for TSMC in Arizona[13].

WHERE CHINA STANDS	WHERE THE US STANDS
Georgetown University's CSET reported in June 2024 that Huawei's Ascend 910B AI chip is believed to be made on SMIC's 7 nm-class N+2 process[14]. But SemiAnalysis judged in April 2025 that most 910B and 910C chips were made at TSMC[15].	The US government agreed in August 2025 to buy a passive 9.9 percent stake in Intel[16]. Announcing the deal, Intel called itself the only chipmaker that both develops and manufactures leading-edge logic chips in the United States[16]. Intel Foundry lost $2.089 billion in the second quarter of 2026[17].

TSMC calls a plant a GIGAFAB once it can run more than 100,000 wafers a month, each a 300 mm disc, twelve inches across[18]. It operates six of them, and every one is in Taiwan[5].

How it works

Inside, a chip is built up in layers, and a similar loop of steps repeats for each one. Wafers travel in batches, called lots, inside a sealed pod called a FOUP,

5 Fab Capacity. TSMC, accessed 9 September 2026. tsmc.com
6 Navigating the Costly Economics of Chip Making. Boston Consulting Group, 28 September 2023. bcg.com
7 TSMC Reports Third Quarter EPS of NT$29.43. U.S. Securities and Exchange Commission, 15 October 2026. sec.gov
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17 Intel Reports Second-Quarter 2026 Financial Results. Intel, 23 July 2026. intc.com
18 GIGAFAB® Facilities. TSMC, accessed 9 September 2026. tsmc.com

short for front-opening unified pod. Entegris, a US maker of these pods, offers a standard version that holds 25 wafers[19]. An overhead track carries the pod from tool to tool.

At each tool a robot lifts one wafer out and the tool does its one job, such as depositing a film, printing a pattern, etching it, cleaning it or measuring it. Then the robot puts the wafer back. The pod moves on to the next tool, often across the building. When a layer is finished, the loop starts again with the next one.

Turning silicon into a finished, packaged chip takes 1,000 to 1,500 steps, depending on how complex the chip is, the Semiconductor Industry Association said in 2025[20]. In 2021 the association put the time to make a finished wafer at about 12 weeks on average, and up to 14 to 20 weeks for advanced processes[21].

The pod keeps the wafer clean for all of that time because the wafer never meets the room's air. Intel said in 2018 that the air in its fab cleanrooms is filtered to 1,000 times fewer airborne particles than a sterile hospital operating room[22].

Every design in the building runs on the same tools because each tool is set by a recipe, the list of gases, temperatures and times for one step. A foundry offers each customer the same list of recipes on shared machines. The tools run day and night, because an idle tool earns nothing. The finished wafer is tested (see Test and assembly), then cut and packaged.

TSMC, founded in 1987, pioneered the pure-play model, in which a foundry makes only its customers' products[23]. In 2023 the Boston Consulting Group (BCG) estimated that a new fab for chips at 2 nm and below, completed in 2026, would cost $35 to $43 billion to own over ten years[6]. That is 33 to 66 percent above 2023 costs[6]. In March 2026 the Semiconductor Industry Association (SIA) told the US Senate that building and equipping a leading-edge fab takes as much as $20 to $25 billion of capital[12].

Together, the plants TSMC and its subsidiaries run had capacity for more than 17 million wafers a year in 2025, counted as if each were the standard 12-inch size[5]. The six largest plants, called gigafabs, had a combined capacity of more than 13 million[18]. The other plants are five smaller fabs in Taiwan, two in China, two in the United States and one in Japan[5].

5 Fab Capacity. TSMC, accessed 9 September 2026. tsmc.com

6 Navigating the Costly Economics of Chip Making. Boston Consulting Group, 28 September 2023. bcg.com

12 Testimony of the Semiconductor Industry Association (SIA) before the Senate... U.S. Senate Committee on Environment and Public Works, 4 March 2026, p. 3. epw.senate.gov

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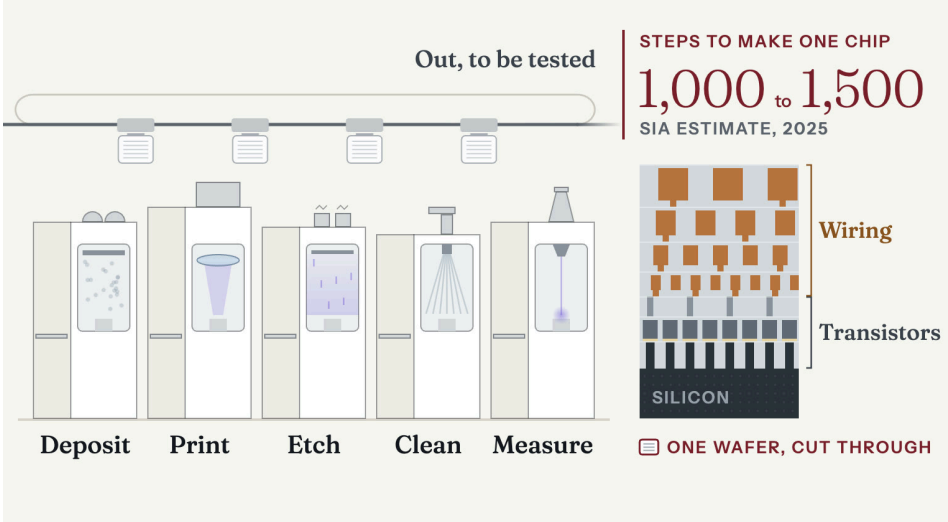
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23 TSMC, company profile page. TSMC, accessed 9 September 2026. tsmc.com

How a fab builds a chip, layer by layer



1. **The fab.** Sealed pods carry wafers from tool to tool on a track near the ceiling.
2. **Deposit.** The first tool lays a thin film on the wafer.
3. **Print.** The next prints a pattern on it with light.
4. **Etch.** An etch tool cuts the pattern into the film.
5. **Clean, measure.** The wafer is cleaned, then measured to check the layer came out right.
6. **Next layer.** Then the loop starts again for the next layer. The transistors come first, then the wiring above them.
7. **Finished.** Making one chip takes 1,000 to 1,500 steps, the Semiconductor Industry Association estimated in 2025.

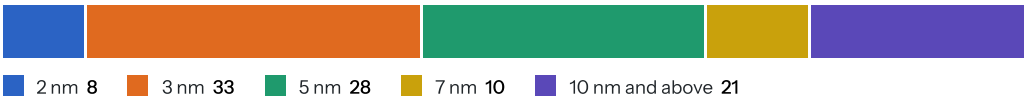
Simplified. A real fab has many tools of each kind, spread across the building, and a chip has far more layers than the seven shown. The layers are drawn far thicker than they are.

Variants and trade-offs

Pure-play foundries

TSMC and UMC, both based in Taiwan, make chips only for other companies. Chips at 7 nm and below, meaning the newest processes, were 79 percent of TSMC's wafer revenue in the third quarter of 2026[7]. That share split into 33 percent at 3 nm, 28 at 5 nm, 10 at 7 nm and 8 at 2 nm[7]. TSMC's revenue for the quarter was $46.72 billion, at a 66.4 percent gross margin[7].

TSMC wafer revenue by node, 3Q26 % OF WAFER REVENUE



Source: TSMC 3Q26 results, filed with the SEC

Integrated device makers turned foundry

Intel and Samsung design and sell their own chips and rent out capacity too. Renting out capacity is harder for them, because customers dislike handing

7 TSMC Reports Third Quarter EPS of NT$29.43. U.S. Securities and Exchange Commission, 15 October 2026. sec.gov

designs to a competitor and the company's own products take the best capacity. Intel Foundry reported $5.8 billion of segment revenue in the second quarter of 2026 and an operating loss of $2.089 billion[17]. Only $293 million of that revenue came from outside customers[8]. Intel says substantially all of its foundry business still makes Intel's own products[8].

National champions

China's SMIC and Hua Hong and Japan's Rapidus have government backing. SMIC's revenue rose 16.2 percent to $9.327 billion in 2025, at a 21 percent gross margin[10]. Its lines ran 93.5 percent full, on monthly capacity above a million wafers counted in the smaller 8-inch size[10]. SMIC cannot buy an EUV scanner, yet it makes 7 nm-class chips on its N+2 and N+3 processes by printing the tightest layers in several passes on older deep-ultraviolet scanners[4].

Georgetown University's Center for Security and Emerging Technology (CSET) reported in June 2024 that Huawei's Ascend 910B AI chip is believed to be made on N+2[14]. It also reported that SMIC's limited 7 nm capacity forced Huawei to choose between its most advanced phone chips and its AI chips[14]. But the research firm SemiAnalysis judged in April 2025 that most 910B and 910C chips were made on TSMC's 7 nm process, from wafers Huawei bought through another company[15].

In February 2026 Rapidus raised 267.6 billion yen from the Japanese government and 32 companies to reach 2 nm mass production by 2027[11].

Mature-node capacity

Most wafers come off older lines. These make power devices, analog chips and microcontrollers, and they compete on price. In January 2025 the industry association SEMI forecast that foundry capacity at all nodes would rise from 11.3 million wafers a month in 2024 to a record 12.6 million in 2025, counted in the 8-inch size[24]. Its June 2024 forecast had credited much of that growth to Intel's new foundry business and China's expansion[25]. But in July 2025 Intel scaled back its own expansion plans, saying it would no longer go ahead with planned projects in Germany and Poland and would slow construction in Ohio[26].

Capacity and cost

In March 2025 TSMC announced plans to add $100 billion to the $65 billion it was already investing in Phoenix, for three more fabs, two packaging plants and an R&D center[27]. In July 2026 it announced another $100 billion for four

4 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

8 INTEL CORP, Form 10-Q quarterly report for the period ended 2026-06-27 (10-Q). U.S. Securities and Exchange Commission, 24 July 2026. sec.gov

10 SMIC Announces 2025 Annual Results. SMIC, 26 March 2026. smics.com

11 Rapidus Secures 267.6 Billion Yen in Funding from Japan Government and Private... Rapidus, 27 February 2026. rapidus.inc

14 Pushing the Limits: Huawei's AI Chip Tests U.S. Export Controls. Center for Security and Emerging Technology, 17 June 2024. cset.georgetown.edu

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17 Intel Reports Second-Quarter 2026 Financial Results. Intel, 23 July 2026. intc.com

24 Eighteen New Semiconductor Fabs to Start Construction in 2025, SEMI Reports. SEMI, 7 January 2025. semi.org

25 Global Semiconductor Fab Capacity Projected to Expand 6% in 2024 and 7% in... SEMI, 18 June 2024. semi.org

26 Intel Reports Second-Quarter 2025 Financial Results. Intel, 24 July 2025. intc.com

27 TSMC Intends to Expand Its Investment in the United States to US$165 Billion to... TSMC, 4 March 2025. pr.tsmc.com

more fabs at 2 nm or below, taking its planned investment in Arizona to $265 billion across 12 facilities[2].

Some customers pay TSMC in advance to hold capacity at its fabs[28]. These deposits stood at NT$189,858.2 million at the end of 2025, down from NT$291,101.8 million a year earlier[28]. Epoch AI estimated that in 2025 the four largest AI chip designers used around 90 percent of the world's CoWoS packaging capacity and HBM memory supply, against only 12 percent of advanced logic production[29]. See Advanced packaging.

Who makes it

In August 2025 the US government agreed to buy 433.3 million newly issued Intel shares at $20.47 each, or $8.9 billion in all, for a 9.9 percent stake[16]. It would pay with $5.7 billion of CHIPS Act grants that Intel had not yet received and $3.2 billion from the Secure Enclave program[16]. Announcing the deal, Intel's chief executive, Lip-Bu Tan, called Intel the only semiconductor company that both develops and manufactures leading-edge logic chips in the United States[16]. The deal closed on 27 August 2025 and made the government one of Intel's largest stockholders, Intel said in its annual report for 2025[30].

The stake is passive, with no board seat[16]. The government agreed to vote with Intel's board on matters that need shareholder approval, with limited exceptions[16]. The deal also gave the government a five-year warrant, a right to buy a further 5 percent of Intel at $20 a share[16]. The government can use that right only if Intel's ownership of its foundry business falls below 51 percent[16].

In 2026 Intel began high-volume manufacturing of some of its Panther Lake processors using ASML's High-NA extreme ultraviolet scanners[17]. It also put its 18A-P process into risk production, the trial runs a process goes through before it is signed off[17].

In 2022 Samsung broke ground in Taylor, Texas, on a plant with an initial minimum investment of $17 billion, split into $6 billion for buildings and $11 billion for equipment[31]. In December 2024 the Commerce Department awarded Samsung up to $4.745 billion of CHIPS Act funding, to support more than $37 billion of planned investment in Central Texas[9]. That plan covers two new leading-edge logic fabs and a research fab in Taylor, and an expansion of Samsung's existing plant in Austin[9].

2 TSMC Announces Additional $100 Billion Investment In Arizona. City of Phoenix, 16 July 2026. phoenix.gov

9 Biden-Harris Administration Announces CHIPS Incentives Award with Samsung... NIST, 20 December 2024. nist.gov

16 Intel and Trump Administration Reach Historic Agreement to Accelerate American... Intel, 22 August 2025. intc.com

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29 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai

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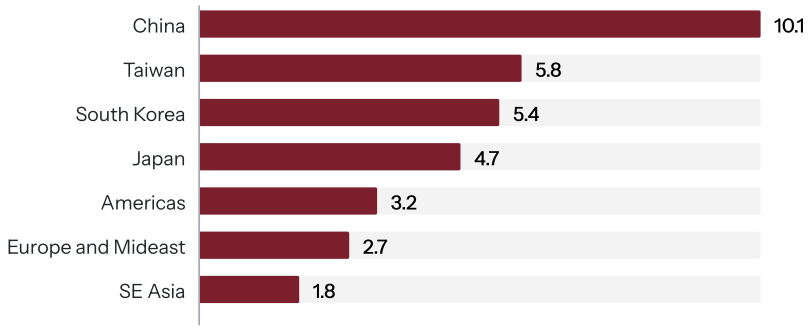
31 Taylor, US Fab. Samsung Electronics, 11 September 2025. semiconductor.samsung.com

The chokepoint

In 2022 South Korea and Taiwan held almost all of the world's logic capacity below 10 nm[32]. The United States had almost no capacity below 10 nm that year[32]. In May 2024 SIA and BCG projected that the US share of that capacity would reach 28 percent by 2032[1]. They also projected that new fab investment worldwide would reach around $2.3 trillion from 2024 to 2032[33].

Across all nodes, old and new, SEMI forecast in June 2024 that China would have installed capacity of 10.1 million wafers a month in 2025, the most of any region[25]. It forecast 5.8 million for Taiwan, out of a world total of 33.7 million, all counted in the 8-inch size[25]. In the third-quarter 2026 update of its World Fab Forecast, SEMI projected that China would keep the largest installed capacity in 2026 and 2027[3]. Little of China's capacity is advanced[34].

SEMI's June 2024 forecast of installed wafer fab capacity by region in 2025, all nodes MILLION 8-INCH-EQUIVALENT WAFERS PER MONTH



Source: SEMI World Fab Forecast, June 2024

Chip production in Taiwan faces three risks:

- **Seismic.** Taiwan sits on an active plate boundary, and after an earthquake in January 2025 TSMC had to scrap some wafers in process[35].
- **Power.** A gigafab is a large industrial load on a small island grid.
- **Political.** Taiwan held 69 percent of the world's logic capacity below 10 nm in 2022[1]. In December 2025 SemiAnalysis put Taiwan's share of the world's advanced-node capacity at over 90 percent, though it did not say which node sizes count as advanced[13]. So a blockade or conflict would cut most of that output off from the world market at once. See Who Controls the Chain.

1 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

3 World Fab Forecast. SEMI, accessed 9 September 2026. semi.org

13 TSMC Overseas Fabs – A Success? SemiAnalysis, 1 December 2025. newsletter.semianalysis.com

25 Global Semiconductor Fab Capacity Projected to Expand 6% in 2024 and 7% in... SEMI, 18 June 2024. semi.org

32 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 13. semiconductors.org

33 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 4. semiconductors.org

34 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 30. semiconductors.org

35 TSMC January 2025 Revenue Report and the Statement on the Impact of Earthquake. TSMC, 10 February 2025. pr.tsmc.com

Building capacity outside Taiwan is slow, and TSMC has yet to build a gigafab anywhere else[5]. SemiAnalysis wrote in December 2025 that a foundry needs highly skilled engineers to get the most working chips from each wafer[13]. It judged that finding enough workers remained a major challenge for TSMC in Arizona[13].

FOUR THINGS TO REMEMBER

- A foundry makes chips for other companies, because no chip company sells enough of one product to fill a fab alone.
- All six of TSMC's largest plants are in Taiwan.
- SemiAnalysis judged in December 2025 that finding enough skilled workers remained a major challenge for TSMC in Arizona.
- SEMI projected in 2026 that China would keep the world's largest installed fab capacity in 2026 and 2027, but little of it is advanced.

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CHAPTER
10
STAGE 10 OF 15

Memory and HBM

SK hynix, Samsung and Micron make nearly all high bandwidth memory (HBM), the stacked memory beside AI chips. SK hynix took 63.2 percent of HBM revenue in 2025.

IN PLAIN TERMS

High bandwidth memory, or HBM, is where an AI chip keeps the numbers it is working on, and it is built to hand them over fast. A memory maker etches tiny holes into each memory wafer, fills them with copper and grinds the wafer until the copper shows through its back. It tests each of the wafer's chips and stacks up to sixteen good ones into a cube, with every layer wired to the ones below. The cube then sits beside the processor, the chip that does the calculating. Stacking is the hard part because each chip is ground thinner than a human hair and the finished cube has to stay within a set height. One chip that fails after stacking can scrap the whole cube. Only SK hynix, Samsung and Micron make HBM in volume, and a single entry in the US export rules covers HBM from all three.

IN SHORT

US-based Micron went from 5.8 percent of HBM revenue in 2024 to 23.1 percent in the first quarter of 2026, according to IDC figures in SK hynix's US prospectus[1]. Over the same period, Samsung's share fell from 37.8 percent to 20.5 percent[1].

Since December 2024, a US license has been required, unless an exception applies, to export to China any HBM moving more than 2 GB per second per square millimeter of its area[2]. The Institute for Progress reported in October 2025 that China's CXMT planned to produce HBM3 in 2026[3]. Even so, working from TrendForce's estimate of world HBM shipments, the institute projected that producers outside China would make 70 times as much HBM as China in 2026[3]. In June 2026 SemiAnalysis judged that CXMT was still struggling to stabilize supply of eight-high HBM3[4].

1 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

2 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... Federal Register, 5 December 2024. federalregister.gov

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CHOKEPOINT CARD	EXTREME CONCENTRATION	HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	Hard Three firms took nearly all HBM revenue in 2025[1].	About $35B HBM market in 2025, Micron's December 2025 estimate[5]

WHO LEADS	KR **SK hynix** 63.2% of HBM revenue in 2025[1] KR **Samsung** 19.3% of HBM revenue in 2025[1] US **Micron** 17.4% of HBM revenue in 2025[1]
WHERE IT IS MADE	KR **South Korea** SK hynix in Cheongju and Icheon, Samsung in Pyeongtaek and Hwaseong US **United States** Micron headquarters and R&D, and an SK hynix packaging plant under construction in Indiana TW **Taiwan** Micron DRAM and HBM production JP **Japan** Micron Hiroshima, 1-gamma DRAM SG **Singapore** Micron HBM advanced packaging plant, adding capacity from 2027
WHY SUBSTITUTION IS SLOW	SK hynix says only a few firms have the expertise and resources to design, make and package memory chips[1]. It says HBM also needs advanced processes such as TSV packaging and must pass customers' rigorous testing and approval[1].

WHERE CHINA STANDS	WHERE THE US STANDS
The Institute for Progress estimated in October 2025 that China would make virtually no HBM that year[3]. Huawei said in September 2025 that its Ascend 950 chips would use two HBM designs of its own, without saying who would manufacture the memory[6].	Micron is the only American HBM maker[1]. As of October 2026, exporters headquartered in the United States or a Country Group A:5 ally can ship HBM below 3.3 GB per second per square millimeter under License Exception HBM[7].

An AI accelerator moves numbers from memory to its arithmetic units and back. The cost of arithmetic has fallen faster than the cost of moving numbers. The US export control office wrote in December 2024 that a processor cannot reach its full capabilities unless its memory capacity and bandwidth rise with its speed[2].

How it works

Each layer in the cube is a DRAM chip, the ordinary kind of computer memory. Each layer is ground thin and bonded to the one below through tiny metal bumps called microbumps. The whole stack sits on a base die, a chip that manages it. The copper-filled holes are called through-silicon vias, or TSVs. They let a signal reach any layer without a wire running around the outside.

1 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

2 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... Federal Register, 5 December 2024. federalregister.gov

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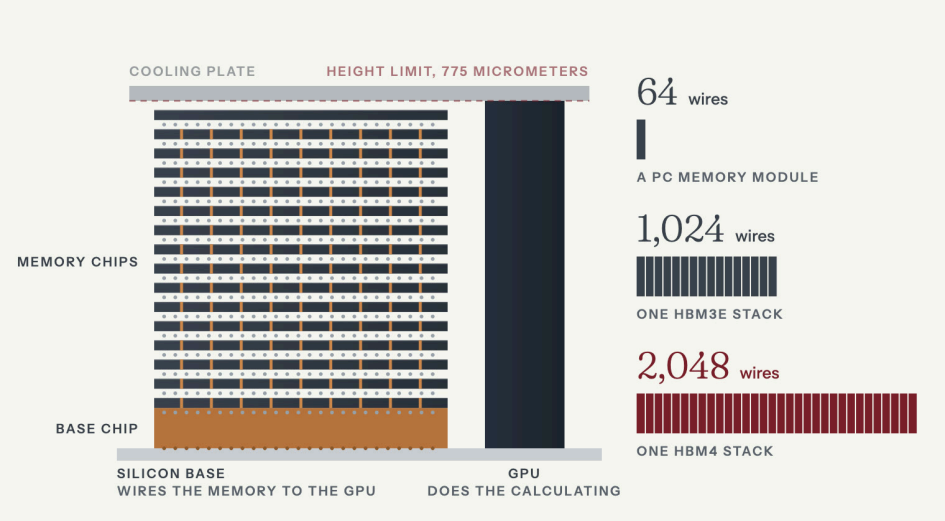
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7 15 CFR 740.25 -- License Exception High Bandwidth Memory (HBM). Electronic Code of Federal Regulations, 5 December 2024. ecfr.gov

An HBM3E cube has a 1,024-bit interface, meaning it exchanges data with the processor over 1,024 wires at once[8]. That is sixteen times wider than a standard DDR5 memory module, the memory stick in a PC[8]. HBM4 doubles that to 2,048 wires[9]. Ordinary memory cannot be made that wide, because the pins would not fit on a circuit board. The stack can, since its wires run down through the TSVs and across a silicon interposer to the processor beside it, with no board in the way[8].

Inside a memory stack



1. **Base chip.** A memory stack sits beside the GPU on a shared silicon base. It starts with a base chip, which controls the memory chips above it.
2. **Thin.** Each memory chip is ground to about 30 micrometers, thinner than a human hair.
3. **Stack.** Twelve chips are joined by dots of solder, and copper-filled holes run straight down through all but the top one.
4. **Height.** The stack has to fit under the cooling plate. Twelve chips reach 0.72 mm, the old limit. HBM4 raised the limit to 775 micrometers and fits up to sixteen chips.
5. **Width.** An HBM3E stack talks to the processor over 1,024 wires at once, sixteen times as many as a PC memory module. HBM4 doubles that to 2,048.

Simplified cross-section. Heights follow the published limits and are drawn to one vertical scale. Widths are not to scale.

A chip needs the width most when it answers a prompt. Training reuses each stored number across a huge multiplication, so a chip can fetch little and compute a lot. But to generate one word of an answer, the chip reads the whole model from memory, plus the whole record of the conversation so far. Then it does a little arithmetic and waits for the next read. That record grows with the length of the conversation and the number of users served at once. So memory size and speed usually set how many people one chip can serve.

8 High-bandwidth memory (HBM). Micron Technology, accessed 9 September 2026. micron.com

9 JEDEC® and Industry Leaders Collaborate to Release JESD270-4 HBM4 Standard... JEDEC Solid State Technology Association, 16 April 2025. jedec.org

Variants and trade-offs

GENERATION	INTERFACE	BANDWIDTH PER STACK	CAPACITY PER STACK
HBM3E	1,024-bit, 16 channels (32 pseudo-channels)	over 1.2 TB/s	24 GB (8 chips), 36 GB (12 chips)
HBM4	2,048-bit, 32 channels	over 2.0 TB/s, up to 3.3 TB/s	up to 64 GB (16 chips, 32 Gb dies)

HBM3E

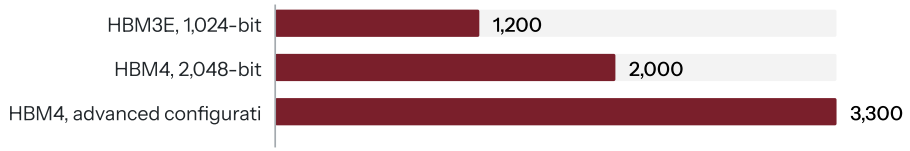
Micron's HBM3E comes in two sizes, 24 GB with eight chips stacked and 36 GB with twelve[10].

HBM4

The standards body JEDEC published JESD270-4 in April 2025, setting the 2,048-bit interface and 2 TB/s per stack at 8 Gb/s per pin[9]. SK hynix says it was first to complete HBM4 development, in September 2025, at over 10 Gb/s per pin[11]. Samsung says it began mass production and shipped the first commercial HBM4 in February 2026, at 11.7 Gb/s per pin, with a 4 nm logic base die[12]. Micron's parts run above 11 Gb/s for more than 2.8 TB/s per stack[13]. Micron says it began volume production of its 36 GB twelve-high HBM4 in fiscal 2026 and expects to ramp volume of HBM4E, the next generation, in calendar 2027[14].

Through HBM3E, SK hynix made its own base dies, the chips at the bottom of each stack[15]. For HBM4 it chose TSMC's advanced logic process, so the base die can hold more functions[15]. As a result, TSMC, which does not make HBM, makes the base die for SK hynix's HBM4.

Bandwidth per HBM stack GB/S



Source: Siemens EDA, HBM3E and HBM4 IC design guide, April 2026

How a stack is built

- **Etch and thin.** Etch vertical holes, the through-silicon vias, into a memory wafer, fill them with copper, then grind the wafer from behind until the

9 JEDEC® and Industry Leaders Collaborate to Release JESD270-4 HBM4 Standard... JEDEC Solid State Technology Association, 16 April 2025. jedec.org

10 Micron, HBM3E product page. Micron Technology, accessed 9 September 2026. micron.com

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14 MICRON TECHNOLOGY INC, Form 10-K annual report for the period ended 2026-09-03... U.S. Securities and Exchange Commission, 9 October 2026. sec.gov

15 SK hynix Partners with TSMC to Strengthen HBM Technological Leadership. SK hynix, 19 April 2024. news.skhynix.com

copper comes through.

- **Stack and join.** Thermocompression bonding joins the chips through metal bumps under heat and pressure, and it comes in several forms[17]. SK hynix uses Advanced MR-MUF, which stacks the chips, injects a liquid protective material between them and hardens it[11]. SK hynix says the process controls warping and dissipates heat well[11]. Samsung said in February 2024 that its twelve-high HBM3E uses TC-NCF, thermocompression with a thin non-conductive film between the chips[18]. Samsung expected TC-NCF to help limit warping as chips get thinner[18].
- **Stay under the height limit.** HBM4 raised the height allowed for a finished cube from 720 micrometers to 775, still under a millimeter[17].

Hybrid bonding is the next step, but it is not ready. It would join the chips copper to copper and insulator to insulator, with no solder and no filler[17]. In chip packaging generally, microbumps sit about 20 micrometers apart, and hybrid bonds can sit less than 1 micrometer apart[17]. A tech note on SK hynix's newsroom says full-scale adoption in HBM is most likely at HBM4E or HBM5, where stacks are projected to pass twenty layers[17]. Heat builds up in the stack, and removing it becomes a major challenge at 16 to 20 layers[17].

Who makes it

Three firms took more than 90 percent of the revenue from ordinary computer memory, or DRAM, in the first quarter of 2026, according to IDC figures in SK hynix's US prospectus[1]. HBM comes from the same three, SK hynix, Samsung and Micron[1]. No other company has both leading-edge DRAM and a stacking line of its own. SK hynix says only a few firms have the expertise and resources to design, make and package memory chips[1].

SK hynix says it was the first company to mass-produce several generations of HBM, including HBM3 and HBM3E[1].

SK hynix broke ground in August 2026 on an advanced packaging plant in West Lafayette, Indiana, an investment of over $4 billion[19]. Micron is modernizing and expanding its DRAM and HBM production capacity in Taiwan[14]. It has also broken ground on an HBM advanced packaging plant in Singapore to expand its packaging capacity from early 2027[14].

Of 2025 HBM revenue, SK hynix took 63.2 percent, Samsung 19.3 and Micron 17.4[1]. The split is public because SK hynix registered shares in the United States[1]. Its US prospectus credits the research firm IDC. Market shares move fast. In 2024 the same three stood at 56.4, 37.8 and 5.8 percent[1]. In the first quarter of 2026 they stood at 56.4, 20.5 and 23.1 percent, which put SK hynix back at its 2024 share[1].

1 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

11 SK hynix Completes World's First HBM4 Development and Readies Mass Production. SK hynix, 12 September 2025. news.skhynix.com

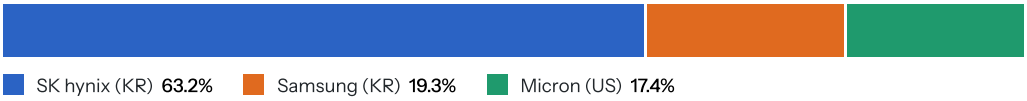
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17 [Tech Note] Hybrid Bonding: Evolving into a Foundational Technology for... SK hynix, 25 August 2026. news.skhynix.com

18 Samsung Develops Industry-First 36GB HBM3E 12H DRAM. Samsung Global Newsroom, 27 February 2024. news.samsung.com

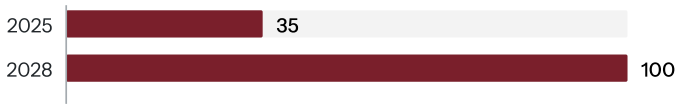
19 SK hynix breaks ground on $4 billion advanced packaging production facility in... Purdue University, 27 August 2026. purdue.edu

Global HBM revenue by supplier, 2025 %



Source: SK hynix Form 424B4 prospectus, 10 July 2026, on IDC data

HBM market, Micron's estimate for 2025 and forecast for 2028 $B



Source: Micron prepared remarks, fiscal first quarter 2026, 17 December 2025

In December 2025 Micron put the HBM market at about $35 billion in 2025 and forecast that it would reach around $100 billion in 2028[5]. Micron says HBM needs more wafers than traditional DRAM, the kind used in PCs and phones, to make the same number of bits[14]. SK hynix says the industry's shift of production capacity to HBM has left traditional DRAM undersupplied[1]. The shortage, it says, helped drive a significant price recovery for that DRAM from the third quarter of 2025[1].

The chokepoint

The leading accelerators made outside China use HBM from SK hynix, Samsung or Micron. Nvidia's GB300 NVL72 rack carries 20 TB of HBM3E at up to 576 TB/s[20]. Each Nvidia Rubin GPU carries up to 288 GB of HBM4 at up to 22 TB/s[21]. AMD's MI355X carries 288 GB of HBM3E at 8 TB/s[22].

On 5 December 2024 the Bureau of Industry and Security (BIS), the US export control office, published a rule at 89 FR 96790. It added an export classification, ECCN 3A090.c, for "High bandwidth memory (HBM) having a 'memory bandwidth density' greater than 2 gigabytes per second per square millimeter". Its technical note defines that density as the memory bandwidth in gigabytes per second divided by the area of the package or stack in square millimeters. The rule also covers stacks made outside the United States, through a foreign direct product rule, which applies US controls to goods made abroad with American technology[2]. BIS stated in the rule that all HBM stacks then in production exceeded that threshold[2].

Under 15 CFR 742.6(b)(10)(ii), BIS reviews license applications to send 3A090.c items to Macau or a Country Group D:5 destination such as China[23]. It reviews

1 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

2 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... Federal Register, 5 December 2024. federalregister.gov

5 Fiscal Q1 2026 Earnings Call Prepared Remarks. Micron Technology, 17 December 2025. s25.q4cdn.com

14 MICRON TECHNOLOGY INC, Form 10-K annual report for the period ended 2026-09-03... U.S. Securities and Exchange Commission, 9 October 2026. sec.gov

20 NVIDIA GB300 NVL72. Nvidia, 23 July 2026. nvidia.com

21 Inside NVIDIA Rubin GPU Architecture: Powering the Era of Agentic AI. Nvidia, 21 July 2026. developer.nvidia.com

22 AMD Instinct™ MI355X GPUs. AMD, accessed 9 September 2026. amd.com

23 15 CFR 742.6, Regional stability. Electronic Code of Federal Regulations, 23 July 2026. ecfr.gov

them with a presumption of approval when neither the buyer nor its ultimate parent is headquartered in one of those places, and with a presumption of denial otherwise[23].

Two provisions narrow the control. License Exception HBM, at 15 CFR 740.25, applies to exporters headquartered in the United States or a Country Group A:5 ally[7]. It lets them ship HBM below 3.3 GB per second per square millimeter without a license[7]. The HBM must be bought directly by the designer of the chip it will be packaged into and must go straight to the packaging site[7]. The rule also excludes co-packaged chips that combine HBM and logic and whose main function is processing, which other chip controls may cover instead[2].

The think tank CSIS said in December 2024 that the rules still allowed HBM2, a generation introduced in 2016, to go to China, with limits on who could buy it and what for[24]. It also said Nvidia's H20, whose HBM3 is packaged with the processor, was then still exportable under the performance limits for AI chips[24].

In December 2024 CSIS called China's strongest DRAM maker, CXMT, one of only two Chinese firms with a credible path to producing advanced HBM[24]. The Institute for Progress reported in October 2025 that CXMT was planning to produce HBM3 in 2026[3]. It also estimated that China would make virtually no HBM in 2025[3]. In June 2026 SemiAnalysis judged that CXMT was still struggling to stabilize supply of eight-high HBM3, with even greater challenges at twelve-high[4].

Huawei said in September 2025 that its Ascend 950 chips would use two HBM designs of its own, HiBL 1.0 and HiZQ 2.0[6]. It said the 950DT, due in the fourth quarter of 2026, would carry 144 GB of HiZQ 2.0 at 4 TB/s[6]. The announcement did not say who would manufacture the memory[6]. SK hynix says HBM needs advanced production processes, such as TSV packaging, and must pass the rigorous testing and approval processes its customers set[1].

FOUR THINGS TO REMEMBER

- HBM is memory stacked into a cube beside the processor and wired to pass data fast.
- An AI chip reads its whole model from memory for each word, so memory speed limits it.
- SK hynix, Samsung and Micron make nearly all HBM.
- Since December 2024 US controls have covered the HBM stacks then in production, with two exceptions. The Institute for Progress estimated

1 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

2 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... Federal Register, 5 December 2024. federalregister.gov

3 Should the US Sell Blackwell Chips to China? Institute for Progress, 25 October 2025. ifp.org

4 China's CXMT Is Set to Challenge DRAM Incumbents. SemiAnalysis, 23 June 2026. newsletter.semianalysis.com

6 Groundbreaking SuperPoD Interconnect: Leading a New Paradigm for AI... Huawei, 18 September 2025. huawei.com

7 15 CFR 740.25 -- License Exception High Bandwidth Memory (HBM). Electronic Code of Federal Regulations, 5 December 2024. ecfr.gov

23 15 CFR 742.6, Regional stability. Electronic Code of Federal Regulations, 23 July 2026. ecfr.gov

24 Understanding the Biden Administration's Updated Export Controls. Center for Strategic and International Studies, 11 December 2024. csis.org

that China made virtually no HBM in 2025.

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5. [A] Fiscal Q1 2026 Earnings Call Prepared Remarks. Micron Technology · 17 December 2025
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21. [A] Inside NVIDIA Rubin GPU Architecture: Powering the Era of Agentic AI. Nvidia · 21 July 2026
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23. A 15 CFR 742.6, Regional stability. Electronic Code of Federal Regulations · 23 July 2026
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CHAPTER
11
STAGE 11 OF 15

Advanced Packaging

Epoch AI estimates that in 2025 Nvidia, Google, AMD and Amazon used around 90 percent of the capacity for CoWoS, TSMC's packaging for AI chips.

IN PLAIN TERMS

Packaging joins finished chips into one component and mounts them on a small circuit board, called a substrate, wired far more coarsely than the chips themselves. A lithography machine, which prints a chip's circuits, can cover only a rectangle about the size of a postage stamp in one shot. Many leading accelerators, Nvidia's Blackwell among them, need more logic than fits in that rectangle. So they are built from several chips, called dies, set side by side on an interposer, a layer of much finer wiring that links them and sits on the substrate. Stacks of memory chips sit on the same interposer, a few millimeters away. The interposer is either one piece of silicon or layers of fine copper wiring that often carry small silicon bridges where the links are densest. Silicon expands less than the resin beneath it when heated, so the package can warp and its joints crack. Preventing that in high volume is hard. Nvidia, Google, AMD and Amazon all use TSMC's version, called CoWoS, made in Taiwan.

IN SHORT

Epoch AI estimates that in 2025 Nvidia used 60.3 percent of all capacity for CoWoS, TSMC's interposer packaging[1]. On the July 2026 earnings call, TSMC's chief executive, C.C. Wei, said its packaging capacity was so tight that it limited customers' growth[2]. Asked about Intel's EMIB-T packaging, he said he hoped it would take some of the load off TSMC[2].

In a TrendForce ranking by 2024 revenue, which CSET cites, four of the top ten outsourced firms that package and test chips are headquartered in China[3]. But in 2025 Chinese firms had 5.3 percent of the market for advanced packaging fabrication tools, against 65.8 percent for US firms, by CSET's analysis of TechInsights data[4].

1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai
2 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 10. investor.tsmc.com
3 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu
4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech

CHOKEPOINT CARD | EXTREME CONCENTRATION | MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	Moderate Rated on CoWoS capacity, which TSMC said in July 2026 was so tight that it limited its customers' growth[2].	About $1,100 for the CoWoS-L packaging of one Nvidia B200, against about $900 for its two logic dies, by Epoch AI's December 2025 estimate[5]

WHO LEADS

- TW **TSMC** CoWoS packaging used by Nvidia, Google, AMD and Amazon in 2025[1]
- US **Intel** Foveros 3D packaging in volume at Rio Rancho, New Mexico, since January 2024[6]
- TW **ASE Technology** NT$389.2B assembly, test and materials revenue, 2025[7]
- US **Amkor** $6.71B net sales, 2025[8]
- CN **JCET** RMB 38.87B revenue, 2025[9]
- SG **ASMPT** Thermocompression bonders, with $532.1M of advanced packaging revenue in 2025[10]

WHERE IT IS MADE

- TW **Taiwan** TSMC advanced packaging fabs and a cluster of outsourced assembly and test firms
- US **United States** Intel Foveros packaging in volume at Rio Rancho, New Mexico
- CN **China** Four of the top ten outsourced assembly and test firms by 2024 revenue are Chinese, in a TrendForce ranking cited by CSET
- NL **Netherlands** Besi hybrid bonders
- JP **Japan** Disco grinders and dicers

WHY SUBSTITUTION IS POSSIBLE

CoWoS packaging has been outsourced to assembly firms such as ASE and Amkor before[11]. Intel's EMIB is another 2.5D option, and Amazon's Trainium and Google's TPU were adopting it to different degrees as of March 2026[11]. But a newcomer would have to match TSMC, which said in April 2026 that it was producing CoWoS at 5.5 times the area a scanner prints in one shot[12].

WHERE CHINA STANDS

Chinese firms had 5.3 percent of the market for fabrication tools built for advanced packaging in 2025, by CSET's analysis of TechInsights data[4]. Between 2019 and 2024 they gained 5 percentage points of share in test tools but lost share in bonding and other conventional assembly and packaging tools[3].

WHERE THE US STANDS

Intel opened Fab 9 in Rio Rancho, New Mexico, in January 2024 to make its Foveros 3D packages in volume[6]. Amkor broke ground in October 2025 on a $7B packaging and test campus in Peoria, Arizona, and expects production to start in early 2028[13].

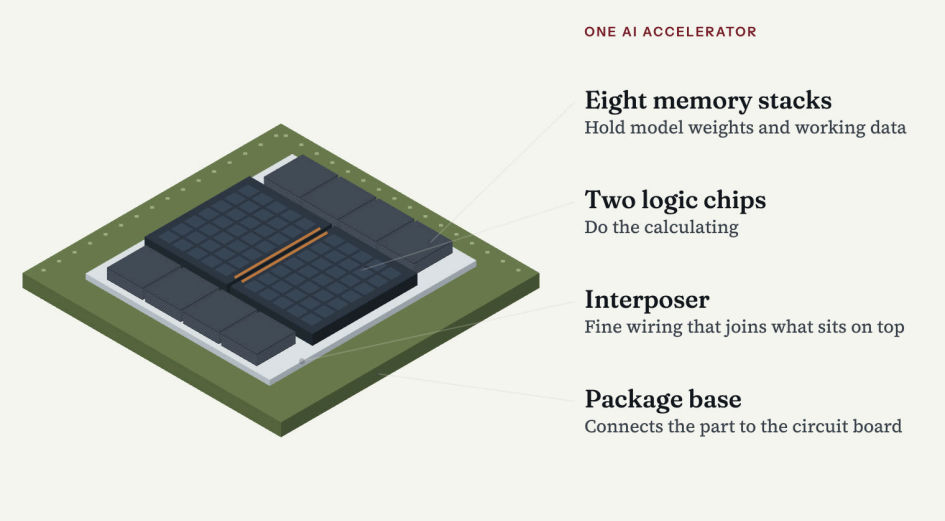
1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai
2 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 10. investor.tsmc.com
3 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu
4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech
5 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai
6 Intel Opens Fab 9 in New Mexico. Intel, 24 January 2024. intel.com
7 ASE Technology Holding Co., Ltd. Reports Its Unaudited Consolidated Financial... PR Newswire, 5 February 2026. prnewswire.com
8 AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-02-09... U.S. Securities and Exchange Commission, 9 February 2026. sec.gov
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12 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com
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A lithography scanner prints a rectangle 26 mm by 33 mm, or 858 mm2, in one exposure[14]. That area is called the reticle field. Every leading AI accelerator is bigger than that. Packaging is the stage that lets an accelerator be larger than one reticle field, and the packaging method limits how much larger it can be.

How it works

A second reason for the interposer is high bandwidth memory (HBM). An HBM4 stack connects through 2,048 data signals, packed too tightly for a standard resin substrate, so the memory sits beside the logic on the interposer[15]. Even Nvidia's H100, whose single logic die is 814 mm2 and fits inside one reticle field, carries five HBM stacks[16]. Epoch AI counts TSMC's CoWoS packaging among the three inputs the H100 needs, along with its logic die and its HBM[1].

The parts of an AI accelerator



1. **Package base.** An AI chip is built from several parts. First comes the package base, a layered board that connects everything to the circuit board below.
2. **Interposer.** On it sits an interposer, a thin layer of very fine wiring that joins everything placed on top.
3. **Logic chips.** Two logic chips go in the middle. They do the calculating, and each is as large as a chipmaking machine can print.
4. **Memory.** Eight stacks of memory go beside them, a few millimeters away. They hold the AI model's weights and the record of each conversation so far.
5. **One package.** Together they make one AI accelerator. Most are put together by TSMC, in Taiwan.

Simplified view of a two-die, eight-stack accelerator. Footprints follow the sizes given in this chapter and chapter 10. Heights are exaggerated so each layer shows.

CoWoS stands for chip on wafer on substrate, the order in which the parts are joined. For CoWoS-S, TSMC makes a silicon wafer that carries wiring and

1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai

14 TWINSCAN NXE:3400C. ASML, accessed 9 September 2026. asml.com

15 New JEDEC® SPHBM4 Standard Enables HBM4-Class Bandwidth on Organic Substrates. JEDEC Solid State Technology Association, 13 July 2026. jedec.org

16 NVIDIA Hopper Architecture In-Depth. Nvidia, 22 March 2022. developer.nvidia.com

vertical copper vias but no transistors. Vias are metal-filled holes that join one layer of wiring to the next.

TSMC bonds the logic dies and memory stacks face down onto that wafer with microbumps. These are dots of solder, and each dot is one electrical joint. In Intel's version of this packaging the microbumps sit 36 microns apart, less than half a hair's width[17]. Then TSMC cuts the interposer out of the wafer and attaches it to the resin substrate, which later goes onto a circuit board.

One cracked joint can scrap a package holding eight tested HBM stacks and two large, expensive logic dies. Epoch AI estimated in December 2025 that CoWoS-L packaging added about $1,100 to the cost of building one Nvidia B200, more than the $900 of its two logic dies[5].

Variants and trade-offs

FAMILY	CARRIER	SIZE LIMIT	USED FOR
CoWoS-S	Full silicon interposer	Up to 3.3x reticle	H100-class parts
CoWoS-R	Resin interposer with fine redistribution wiring	Above 3.3x reticle	Cost-sensitive, fewer HBM stacks
CoWoS-L	Redistribution wiring with local silicon bridges	5.5x reticle in production, 9.5x in development	Blackwell, Rubin

2.5D: silicon, RDL and bridges

In 2.5D packaging the chips sit side by side and connect through an interposer or silicon bridges beneath them. TSMC has run CoWoS in volume since 2012[18]. CoWoS-S has the finest wiring, but its interposer is one unbroken piece of silicon whose yield (the share of interposers that work) falls as the area grows. CoWoS-L replaces that piece of silicon with redistribution layers (RDL) of fine copper wiring and embeds small silicon bridges where the wiring is densest[19].

Intel's EMIB embeds similar silicon bridges in the resin substrate[17]. SemiAnalysis reported in March 2026 that Amazon's Trainium and Google's TPU were adopting it to different degrees[11].

Blackwell was the first high-volume design on CoWoS-L, and packaging problems at TSMC and in Nvidia's design limited its supply[20]. The bridges that carry the 10 TB/s link between the two logic dies must be placed very precisely[20]. The dies, bridges, interposer and substrate also expand at different rates, and that mismatch caused warpage[20].

3D: SoIC and Foveros

Intel's Foveros stacks dies face to face and joins them with solder microbumps[17]. Its newer Foveros Direct drops the solder and bonds the copper pads directly, a method called hybrid bonding[17]. Hybrid bonding joins

5 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai
11 The Great AI Silicon Shortage. SemiAnalysis, 12 March 2026. newsletter.semianalysis.com
17 Foveros Direct 3D Technology Brief. Intel, 6 February 2026. intel.com
18 TSMC, CoWoS technology page. TSMC, accessed 9 September 2026. 3dfabric.tsmc.com
19 TSMC 2025 Annual Report, chapter 5. TSMC, 16 April 2026, p. 3. investor.tsmc.com
20 Nvidia's Blackwell Reworked - Shipment Delays & GB200A Reworked Platforms. SemiAnalysis, 4 August 2024. semianalysis.com

the insulating layer around the pads as well as the copper, with no bumps in between[21]. That lets the connections sit far closer together than microbumps allow[17]. TSMC's SoIC also uses hybrid bonding, and AMD's 3D V-Cache processors use it to bond an extra cache die to the processor die[22].

TSMC put SoIC into volume production in 2022, stacking SRAM memory chips on logic wafers[23]. Its 3 nm SoIC entered volume production in 2025[19]. An A14-on-A14 version, stacking dies made on TSMC's 1.4 nm process, is due in 2029[12]. It will carry 1.8 times the die-to-die connection density of the 2 nm version[12].

In a stack the bottom die is the hardest to cool and the top die is the hardest to power. Because of the heat, 3D designs often put cache memory underneath and compute circuits on top.

SoIC bonding also brings optics into the package. TSMC's COUPE uses SoIC to join a silicon photonics chip, which sends data as light, to its electrical control chip[19]. TSMC said in April 2026 that COUPE on substrate would begin production in 2026[12]. It also said that putting COUPE inside the package gives twice the power efficiency of a pluggable version on the circuit board[12].

Two ways off the silicon interposer

Panel-level packaging replaces the round wafer with a rectangular panel, which wastes far less area. See Test and assembly. JEDEC, the semiconductor standards body, published a standard called SPHBM4 on 13 July 2026. It puts HBM4 memory on a base chip carrying 512 data signals instead of 2,048, each running four times as fast to keep the same bandwidth[15]. With fewer signals, the connections can be spaced widely enough to mount the stack on a standard resin substrate instead of the silicon interposer[15].

Capacity

TSMC does not publish its CoWoS capacity[1]. Industry estimates from SemiWiki and TrendForce, compiled by Epoch AI, put it at about 65,000 to 75,000 wafers a month at the end of 2025, up from 35,000 to 40,000 a year earlier[1].

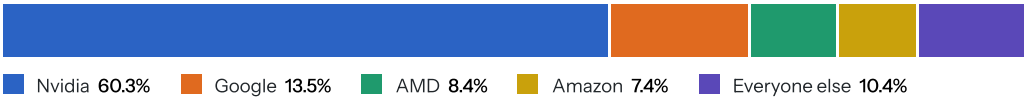
At its 2024 symposium TSMC said CoWoS capacity would grow at a 60 percent compound annual rate over the next few years[24]. But when asked about that rate on the July 2024 earnings call, TSMC's chief executive, C.C. Wei, did not repeat it[24]. He said he could not yet balance supply and demand[24]. He also said TSMC was more than doubling CoWoS capacity in 2024 and would more than double it again in 2025[24].

1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai
12 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com
15 New JEDEC® SPHBM4 Standard Enables HBM4-Class Bandwidth on Organic Substrates. JEDEC Solid State Technology Association, 13 July 2026. jedec.org
17 Foveros Direct 3D Technology Brief. Intel, 6 February 2026. intel.com
19 TSMC 2025 Annual Report, chapter 5. TSMC, 16 April 2026, p. 3. investor.tsmc.com
21 [Tech Note] Hybrid Bonding: Evolving into a Foundational Technology for... SK hynix, 25 August 2026. news.skhynix.com
22 Hybrid Bonding Process Flow - Advanced Packaging Part 5. SemiAnalysis, 9 February 2024. newsletter.semianalysis.com
23 TSMC 2022 Annual Report, page 97. TSMC, accessed 18 October 2026. investor.tsmc.com
24 Q2 2024 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 18 July 2024, p. 6. investor.tsmc.com

In October 2025 Wei would say only that TSMC was working to increase capacity in 2026 and would probably give the real number in 2026[25]. He said front-end capacity (making the chips) and back-end capacity (packaging them) were both, in his words, very tight[25].

Epoch AI compared each chip designer's use of components with world supply[1]. It puts Nvidia at 60.3 percent of all CoWoS capacity in 2025, with a 90 percent confidence interval of 56.5 to 64.3[1]. Google took 13.5 percent, AMD 8.4 and Amazon 7.4[1]. Together those four used around 90 percent of global CoWoS capacity and HBM supply, with HBM measured by value, but only 12 percent of advanced logic die production[1].

Estimated share of CoWoS capacity used, 2025 %



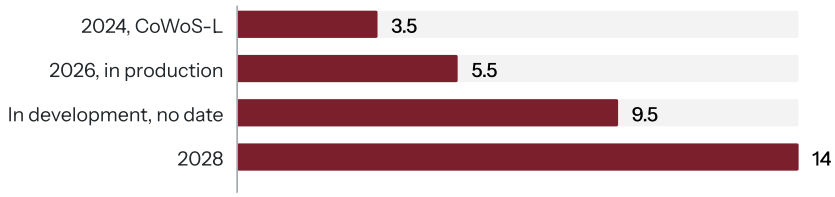
Source: Epoch AI, advanced packaging and HBM were the bottlenecks on AI chip production in 2025

In March 2026 the research firm SemiAnalysis judged that front-end capacity had become the dominant bottleneck and that CoWoS constraints were easing[11]. It said TSMC was planning CoWoS capacity around how many wafers N3, its 3 nm process, could produce, because added packaging capacity is of no use without front-end wafers to fill it[11]. It also noted that CoWoS packaging can be, and has been, outsourced to assembly firms such as ASE and Amkor[11].

TSMC gave a different account four months later. On the July 2026 earnings call, Wei said TSMC's packaging capacity was so tight that it limited customers' growth[2]. Asked about Intel's EMIB-T, he said he welcomed the extra flexibility and hoped it would take some of the load off TSMC[2]. Later in the call he said back-end capacity was still in shortage[26].

Who makes it

CoWoS interposer size on TSMC's roadmap RETICLES



Source: TSMC 2025 annual report and 2026 technology symposium

1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai
2 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 10. investor.tsmc.com
11 The Great AI Silicon Shortage. SemiAnalysis, 12 March 2026. newsletter.semianalysis.com
25 Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 16 October 2025, p. 7. investor.tsmc.com
26 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 17. investor.tsmc.com

Three kinds of firm package chips. TSMC, which makes chips for others, and Intel, which makes its own, do it in their own back-end fabs. Outsourced assembly and test firms do the rest. Intel opened Fab 9 in Rio Rancho, New Mexico, in January 2024 to make its Foveros 3D packages in volume[6]. It called the plant the only US factory producing the most advanced packaging at scale[6].

ASE Technology Holding, the leading provider of assembly and test services, had revenue of NT$645.4 billion in 2025, about 40 percent of it from electronics manufacturing services[7]. Its assembly, test and materials business brought in NT$389.2 billion, up from NT$325.9 billion in 2024[7]. Amkor, the largest US-headquartered assembly and test firm, had net sales of $6.71 billion in 2025[8].

The packaging tools are a narrower chokepoint than the assembly firms.

- **Thermocompression bonding.** Pressing dies together under heat and force. ASMPT's advanced packaging revenue reached $532.1 million in 2025, with thermocompression bonder revenue up about 146 percent[10]. ASMPT targets 35 to 40 percent of a market it puts at $1.6 billion in 2028[10].
- **Hybrid bonding.** Copper pads and the insulator around them bond directly, with no solder. Applied Materials co-developed its Kinex hybrid bonding system with Besi, which Applied calls the industry leader in hybrid bonding[27].
- **Grinding and dicing.** Disco held 87 percent of the $1.9 billion world market for dicing tools in 2025, by CSET's count from TechInsights data[28]. The category covers the grinders that thin wafers and the saws that cut them into chips[28]. See Test and assembly.

TSMC's Arizona commitment reached $265 billion in July 2026 and covers two advanced packaging plants alongside 10 fabs[29]. Amkor started building a $7 billion campus in Peoria in October 2025 and expects production from early 2028[13]. The Commerce Department awarded Amkor up to $407 million in CHIPS Act funding for the site in December 2024[30]. In October 2025 Wei said a large outsourced packaging partner, already breaking ground in Arizona, was on an earlier schedule than TSMC's own two packaging fabs there[31].

6 Intel Opens Fab 9 in New Mexico. Intel, 24 January 2024. intel.com

7 ASE Technology Holding Co., Ltd. Reports Its Unaudited Consolidated Financial... PR Newswire, 5 February 2026. prnewswire.com

8 AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-02-09... U.S. Securities and Exchange Commission, 9 February 2026. sec.gov

10 ASMPT Announces 2025 Annual Results AI-Driven Structural Growth Underpins Group... ASMPT, 4 March 2026. asmpt.com

13 Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test... Amkor Technology, 6 October 2025. ir.amkor.com

27 Applied Materials, Kinex integrated die-to-wafer hybrid bonding system product... Applied Materials, accessed 9 September 2026. appliedmaterials.com

28 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

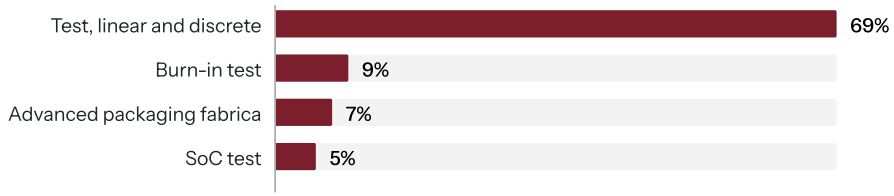
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30 AMKOR TECHNOLOGY, INC., Form 10-K annual report for the period ended 2025-12-31... U.S. Securities and Exchange Commission, 20 February 2026. sec.gov

31 Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 16 October 2025, p. 13. investor.tsmc.com

The chokepoint

China-based suppliers' share of selected tool segments, 2024 %



Source: CSET, Inside Beijing's Chipmaking Offensive

Four of the top ten outsourced assembly and test firms by 2024 revenue are headquartered in China, by a TrendForce ranking that CSET cites[3]. JCET, a leading Chinese assembly and test firm, had record revenue of RMB 38.87 billion in 2025, up 8.1 percent[9]. JCET says its XDFOI platform has entered high-volume manufacturing with redistribution-layer interposer, silicon interposer and silicon bridge options[32].

Between 2019 and 2024, Chinese firms gained 5 percentage points of share in test tools and 4 points in fabrication tools built for advanced packaging, by CSET's analysis of TechInsights data[3]. But they lost share in bonding and other conventional assembly and packaging tools, the category where CSET counts hybrid and thermocompression bonders[3]. Within the advanced packaging fabrication tools, Chinese firms gained 10 points in etch and clean tools and 3 points in deposition tools[3]. They lost 2 points each in lithography and polishing (CMP) tools[3]. In 2024 Chinese firms had 7 percent of the market for advanced packaging fabrication tools[3]. In 2025 they had 5.3 percent of the $6.2 billion world market for these tools, against 65.8 percent for US firms and 15.3 percent for Japanese firms, by CSET's analysis of TechInsights data[4].

TSMC said in April 2026 that it was producing CoWoS at 5.5 times the size of a reticle field and planning larger versions[12]. Each generation on TSMC's roadmap needs a larger interposer.

- **2028.** A 14-reticle interposer, about the area of a compact disc, carrying about ten large compute dies and 20 HBM stacks[12].
- **2029.** More than 14 reticles, plus a 40-reticle System-on-Wafer[12].

FOUR THINGS TO REMEMBER

- Many leading accelerators, Nvidia's Blackwell among them, need more logic than one chip can hold, so packaging joins several chips and their memory.
- Nvidia, Google, AMD and Amazon all rely on TSMC, in Taiwan, to package their accelerators.

3 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu
4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech
9 JCET Releases 2025 Annual Report, Posts Record-High Full-Year Revenue and... JCET Group, 9 April 2026. jcetglobal.com
12 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com
32 JCET Accelerates Strategic Shift Toward High-End Advanced Packaging, 2025... JCET Group, 21 April 2026. jcetglobal.com

- Packaging was one of the main limits on AI chip output in 2025. In July 2026 TSMC said its packaging capacity was still so tight that it limited customers' growth.
- No US site yet packages Nvidia-class accelerators with HBM on CoWoS, though Intel makes its own Foveros 3D packages in volume in New Mexico.

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CHAPTER 12
STAGE 12 OF 15

Substrates and PCBs

Ajinomoto, a Japanese food company, told investors in June 2026 that it holds over 95 percent of the market for the insulating film inside high-end chip packages.

IN PLAIN TERMS

A package substrate is the adapter between a chip and the circuit board under it. The chip's connections are finer than a human hair, but the board is wired in millimeters. The substrate is built in layers, like plywood, around a stiff core of glass cloth and resin. Thin sheets of epoxy film are pressed onto both faces of the core. A laser burns holes through each sheet, and copper plated into the holes carries signals from one layer to the next. Every added layer is another set of holes that can fail. A handful of firms build these substrates. Japan's Ajinomoto dominates the film, and Japan's Nitto Boseki dominates the low-expansion glass cloth in the core.

IN SHORT

Ajinomoto's third site for making its build-up film, in Gifu Prefecture, is scheduled to start operating in 2032 to meet demand after 2030[1]. Nitto Boseki says market demand seems to have risen about two to three times since it set its plan for fiscal 2024 to 2027[2]. In January 2026 Nikkei Asia reported that Apple and Qualcomm were among the customers worried about Japanese glass cloth supply[3]. In August 2026 Ibiden said it had prospects of a third and fourth glass cloth supplier and expected its material procurement risks to decrease[4].

1 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 20. ajinomoto.co.jp
2 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 4. nittobo.co.jp
3 Apple and Qualcomm fret over strained supplies of Japan's glass cloth. Nikkei Asia, 14 January 2026. asia.nikkei.com
4 Ibiden, questions and answers from the financial presentation for the first... Ibiden, 5 August 2026. ibiden.com

CHOKEPOINT CARD EXTREME CONCENTRATION | HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	Hard Rated on the insulating film inside the substrate, because chipmakers and device makers would have to approve any rival film before using it[5].	$7.5 billion market for FC-BGA substrates in 2025, up 18 percent, in a Prismark estimate cited by AT&S[6]

WHO LEADS

- JP **Ajinomoto** Told investors in June 2026 that its build-up film has held over 95% of its market since launch[7]
- JP **Nitto Boseki** Claimed an overwhelming advantage in low thermal expansion glass in its 2025 integrated report[2]
- JP **Ibiden** Said in May 2026 that it typically takes close to 100% of each new generation of interposer-type substrates at launch[8]
- TW **Unimicron** Build-up substrates were 52% of sales in Q2 2026[9]
- KR **Samsung Electro-Mechanics** Package solution sales KRW 771.6bn in Q2 2026, up 37%[10]

WHERE IT IS MADE

- JP **Japan** Build-up film, low-expansion glass cloth, Ibiden and Shinko substrate plants in Gifu and Nagano
- TW **Taiwan** Unimicron and Nan Ya PCB, substrate makers that sell to outside customers
- KR **South Korea** Samsung Electro-Mechanics and a planned glass-core joint venture with Sumitomo Chemical's Dongwoo Fine-Chem
- CN **China** Most of TTM's circuit board competitors are based in China and Taiwan
- AT **Austria** AT&S

WHY SUBSTITUTION IS SLOW

Ajinomoto makes the thin insulating film between the substrate's wiring layers[11]. Chipmakers and device makers approve substrate materials such as laminate, so a rival film would need their approval too[5].

WHERE CHINA STANDS

TTM, an American circuit board maker, said in its annual report for 2025 that its circuit board competitors are mostly based in China and Taiwan[12]. Nitto Boseki, a Japanese glass fiber maker, said in its 2025 integrated report that Chinese and Taiwanese companies are aggressively catching up with it in the special glass sector[2].

WHERE THE US STANDS

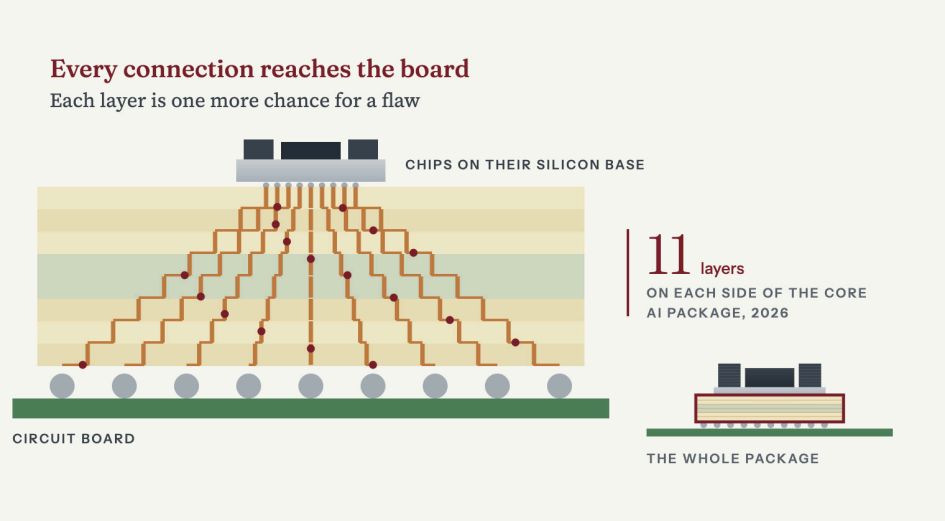
The White House's 2021 supply chain review listed no American firm among the key substrate makers[13]. In December 2024 the Commerce Department finalized an award of up to $75 million to Absolics, an affiliate of Korea's SKC, for a glass substrate plant in Covington, Georgia[14].

The silicon interposer wires chips together inside a package. Beyond the interposer, the wiring runs through a build-up substrate made of epoxy and glass, then a printed circuit board of thirty layers, then a server rack.

2 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 4. nittobo.co.jp
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How it works

How a substrate fans out



1. **The package.** An AI chip is more than the chip itself. It sits in a package: the chips on a silicon base, and under them the substrate, which connects everything to the circuit board.
2. **Tight to wide.** The connections on top are packed tight. The circuit board below is wired in millimeters. The substrate has to join the two.
3. **Layer by layer.** The substrate does it in layers, built outward from a stiff core. Each layer carries the copper a little further out.
4. **Connected.** Stacked together, the layers fan the tight grid out to the board's wide one. Ajinomoto counts 11 layers of its film on each side of the core in a 2026 AI package substrate, against three on each side for a PC in 1999. Every layer is another chance for a flaw, which is what makes these substrates hard to build.

Simplified. Three layers of film are drawn on each side of the core, and every thickness is exaggerated.

The substrate maker drills the core and plates it with copper before the build-up starts. In each cycle the maker then presses on a film of epoxy, burns holes through it with a laser to connect the layers, and adds copper wiring on top[15]. Each film is about 10 micrometers thick, and a human hair is about 80 micrometers across[16].

Ajinomoto counts 11 layers of its film on each side of the core in a 2026 AI package substrate about 100 mm square[17]. A high-performance computing substrate about 70 mm square had nine on each side around 2023[17]. PC substrates had three layers of the film on each side when PCs first adopted it in 1999[17]. Silicon and epoxy expand at different rates when heated, so the larger the package, the more it warps. Nitto Boseki says its low-expansion T-glass cloth is highly rated as an effective fix for that warping, and that its use in substrates has increased significantly[18].

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Variants and trade-offs

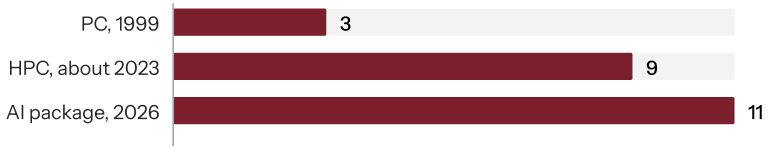
BT substrates

Substrates built on bismaleimide-triazine laminate, or BT, are the other main family. Mitsubishi Gas Chemical, which makes BT materials, says they go mostly into thin packages for smartphone processors, radio modules and memory[19]. The company says BT also goes into flip-chip substrates for computer CPUs and GPUs[19]. In these the chip is mounted face down, and BT laminate forms the core between layers of build-up film[20]. In an October 2024 investor presentation, the company estimated that its BT resins held about 40 percent of the substrate materials market[21].

ABF build-up substrates

Ajinomoto Build-up Film, or ABF, is the insulating film between the layers of package substrates for logic chips, used in PCs, network equipment, servers and AI equipment[11]. Ajinomoto first developed it in 1999 and says major chipmakers have used it as the de facto standard ever since[22]. On Ajinomoto's own site, its electronic materials division said practically all high-performance computers and servers rely on the film[16]. In June 2026 Ajinomoto told investors that the film has held a share of over 95 percent since its launch[7].

Layers of build-up film on each side of a package substrate's core LAYERS



Source: Ajinomoto, Growth Strategy for the Electronic Materials Business, June 2026

Glass core substrates

A glass core expands less with heat than a resin core and stays flatter, so chipmakers and substrate makers are both trying it[23].

- **Absolics**, a subsidiary of Korea's SKC, has a final Commerce Department award of up to $75 million for a 120,000-square-foot glass substrate plant in Covington, Georgia[24]. The department finalized the award in December

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2024[14]. As of October 2026, the award's project page at NIST, a Commerce Department agency, expected first deliveries to customers in 2025 and production capacity to come online in 2027[24].

- **Intel** announced glass substrates in September 2023 for the latter part of this decade[25].
- **Samsung Electro-Mechanics** signed a joint venture agreement with Sumitomo Chemical's subsidiary Dongwoo Fine-Chem in July 2026 to make glass core substrates[26]. The partners plan to have a supply system in place by the second half of fiscal 2027[26].

The materials underneath

Ajinomoto's Healthcare and Others segment, which includes electronic materials such as the film, lifted business profit 45.1 percent to JPY 66.2 billion in the year to March 2026[27]. Ajinomoto attributes the rise mainly to higher electronic materials revenue and higher profit in Bio-Pharma Services & Ingredients[27]. It has set an investment budget of JPY 25 billion between 2023 and 2030 to expand film production[28]. It says there are no concerns about its current supply chain overall[7]. In May 2026 it announced the acquisition of land in Gifu Prefecture for a third production site, scheduled to start operating in 2032 to meet demand after 2030[1].

The glass cloth in the core also has a dominant supplier. Nitto Boseki's T-glass is stiff and expands little when heated, and the company calls it essential for making high-density package substrates[29]. Laminate makers such as Mitsubishi Gas Chemical coat the glass cloth with resin and press it with copper foil into the core laminate that substrate makers build on[30].

Nitto Boseki claims an overwhelming advantage in quality and capacity for low-expansion glass[2]. It planned about JPY 80 billion of capital spending for fiscal 2024 to 2027[31]. It says market demand seems to have risen about two to three times since that plan was set[2]. In August 2025 it decided to expand glass fiber cloth production at its Fukushima Enterprise Center, with production scheduled to begin in the fourth quarter of its fiscal 2026[32]. In May 2026 it forecast capital investment of JPY 45,000 million for the year to March 2027, up from JPY 21,702 million in the year to March 2026[33].

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32 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 12. nittobo.co.jp
33 Consolidated Financial Results for the Fiscal Year Ended March 31, 2026. Nitto Boseki (Nittobo), 12 May 2026, p. 5. ssl4.eir-parts.net

In January 2026 Nikkei Asia reported that Apple and Qualcomm were among the customers worried about Japanese glass cloth supply[3]. In August 2026 Ibiden, a Japanese substrate maker, said it had used only two glass cloth suppliers, in Japan and Taiwan, but now had prospects of a third and fourth for mass production[4]. Because a Japanese supplier also plans to add capacity, Ibiden said it expected its material procurement risks to decrease[4].

Nitto Boseki says Chinese and Taiwanese companies are aggressively catching up with it in the special glass sector[2]. In July 2026 Nikkei Asia reported that Chinese chip materials makers were competing with Japanese rivals for a market that the industry association SEMI put at $73.2 billion in 2025[34].

Who makes it

Most of the firms that build substrates for others are Asian. The White House's 2021 hundred-day supply chain review listed build-up substrate among the materials that commenters called vulnerable[35]. It named key substrate makers including Ibiden, Nanya, Shinko, Samsung, Unimicron, Shennan Circuits, Zhuhai Yueya and AKM, none of them American[13].

A May 2024 report by the Semiconductor Industry Association and Boston Consulting Group counted the major package substrate vendors by headquarters[36]. It found two in Japan, two in Taiwan, one in mainland China and none in the United States[36]. TTM Technologies, an American circuit board maker, says it also offers package substrates[12].

- **Ibiden** will invest about JPY 500 billion in high-performance package substrates from fiscal 2026 to fiscal 2028, starting with about JPY 220 billion, mainly at its Gama plant in Gifu Prefecture[37]. In August 2026 it said the Gama plant was scheduled to start operating in stages from the end of fiscal 2027 and to add to its results from fiscal 2028[4]. In May 2026 management said Ibiden typically takes close to 100 percent of a new generation of interposer-type substrates at launch[8]. It added that this share may decline by 20 to 30 percent after about three to six months[8]. On silicon-bridge substrates, it said wiring the underside is hard enough that Ibiden expects to keep its technical lead at least through fiscal 2030[8].
- **Unimicron** reported NT$42.9 billion of sales in the second quarter of 2026, 52 percent of it build-up substrates and 61 percent of it AI data center work[9].

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34 China chip material makers battle Japan rivals for $73bn market. Nikkei Asia, 1 July 2026. asia.nikkei.com

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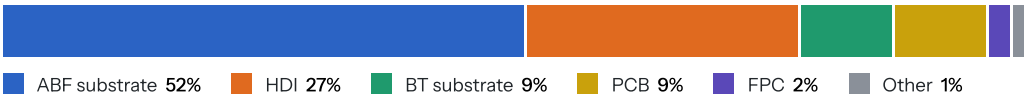
37 Ibiden, Notice Regarding Capital Investment Plan for High-Performance IC... Ibiden, 3 February 2026. ibiden.com

- **Samsung Electro-Mechanics** grew package solution sales 37 percent year on year to KRW 771.6 billion in the second quarter of 2026, on substrates for AI accelerators, server processors and cars[10].
- **AT&S**, an Austrian firm, calls itself a globally leading maker of high-end package substrates[38]. It makes them in China and Malaysia, and since its 2025/26 financial year also in Austria[39]. Citing the research firm Prismark, its annual report for 2025/26 valued the market for flip-chip ball grid array (FC-BGA) substrates at $7.5 billion in 2025, up 18 percent from 2024[6].

In February 2025 a company owned mainly by funds of JIC Capital, a subsidiary of Japan Investment Corporation, launched a tender offer to take Shinko, a Japanese substrate maker, private[40]. The offer covered the shares Fujitsu did not own, and Fujitsu agreed to sell its 50.02 percent stake back to Shinko[41]. The tender offer succeeded in March 2025[42].

In September 2026 Shinko agreed to buy a plant in Mobara, Chiba Prefecture, from Japan Display, a display maker[43]. Shinko said it would make the site a new core plant for its flip-chip package substrates and would also expand capacity at its plants in Nagano[43]. Production at the Mobara plant is scheduled to start in fiscal 2028[43].

Unimicron sales by technology, second quarter 2026 %



ABF substrate 52% HDI 27% BT substrate 9% PCB 9% FPC 2% Other 1%

Source: Unimicron 2026 Q2 earnings conference

The boards under the GPU

TTM Technologies says it regularly builds printed circuit boards of more than 30 layers and can go past 70[12]. Those layers carry power to the GPUs and route fast signals between them, so how much signal the laminate absorbs is a main design constraint. Nitto Boseki also makes NE-glass, a low-dielectric glass cloth that absorbs less signal, used in switches and routers in data centers[44].

Data center computing was 24 percent of TTM's fiscal 2025 revenue of about $2.9 billion, against 14 percent two years earlier[12]. TTM says its circuit board competitors are mostly based in China and Taiwan[12]. In its annual report for 2025, TTM said it was building a plant in Syracuse, New York, to make ultra-high-density circuit boards for national security needs[12]. It expected volume

6 AT&S Annual Report 2025/26. AT&S, 10 June 2026, p. 33. ats.net

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production there in the second half of 2026[12]. Thailand aims to be the biggest circuit board hub outside China as board makers diversify production[45].

The chokepoint

The few firms that build substrates are spending to add capacity, but it takes years. In August 2026 Ibiden said that, with supply short across the industry, the average price of package substrates had risen more than it expected[4]. The materials inside the substrate are harder to replace. Mitsubishi Gas Chemical, a laminate maker, says chipmakers and electronic device makers further down the chain approve the materials it sells to substrate makers[5]. Ajinomoto dominates the film, and Nitto Boseki dominates the low-expansion glass cloth in the core. Both are adding capacity on plans that run for years.

Packages are also getting bigger. TSMC told investors in July 2026 that its CoWoS packaging roadmap goes beyond 14 times the area a lithography machine can print in one shot[46]. A bigger package warps more and needs more layers, so it has more holes that can break the circuit. That lowers yield, the share of substrates that come out working, and each substrate is already expensive before a chip is mounted on it.

FOUR THINGS TO REMEMBER

- A package substrate connects a chip's fine contacts to a circuit board wired in millimeters.
- One Japanese supplier leads the market for the substrate's insulating film, and another leads the market for its glass cloth.
- Replacing the film would mean re-approval by every substrate maker and chip customer.
- The White House's 2021 supply chain review named no American firm among the key substrate makers. Absolics is building a glass substrate plant in Georgia, and a Commerce Department agency expects its production capacity to come online in 2027.

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CHAPTER
13
STAGE 13 OF 15

Test and Assembly

Testing finds bad chips before they go into an expensive package. Japan's Advantest took about 65 percent of 2025 spending on chip testers, by its own estimate.

IN PLAIN TERMS

Before a chip can be sold, a machine called a tester has to check that it works. While the chips are still on the wafer, thousands of needles press onto small metal pads on each one. The tester sends signals through the needles and reads what comes back, and any chip that fails goes no further. A modern AI accelerator is a dozen expensive pieces bonded into one package, and if one piece is bad, all of them are thrown away. So the industry now tests chips earlier, at higher temperatures and for longer, to make weak ones fail during testing before they reach a finished product. Two firms, Advantest in Japan and Teradyne in the United States, sell most of the testers.

IN SHORT

In a TrendForce ranking cited by CSET, four of the ten largest chip assembly and test contractors by 2024 revenue are headquartered in China[1]. Georgetown's CSET, analyzing TechInsights data, finds that between 2019 and 2024 Chinese toolmakers gained 5 percentage points of share in test tools[1]. Across assembly, test and packaging tools, they still lag significantly behind foreign suppliers[1]. Advantest says that switching to another tester maker means rebuilding a customer's whole test setup, at significant extra cost[2].

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CHOKEPOINT CARD	HIGH CONCENTRATION	MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
High	Moderate Rated on test tools, where Advantest held 62.7 percent of the market in 2025 and Teradyne 21.8 percent, by CSET's analysis of TechInsights data[3].	About $9.0 billion market for chip testers in 2025, by Advantest's estimate[4]

WHO LEADS	JP **Advantest** About 65% of the tester market in 2025, by its own estimate[4] US **Teradyne** $2.52bn of semiconductor test revenue in 2025, service included[5] TW **ASE** Ran 8,348 testers in Q2 2026, up from 6,797 a year earlier[6] US **Amkor** $6.71bn of 2025 revenue[7] JP **Disco** 87% of the $1.9bn market for dicing saws and grinders in 2025, on CSET's analysis of TechInsights data[8]
WHERE IT IS MADE	TW **Taiwan** Taiwanese firms such as ASE and Powertech lead outsourced assembly and test JP **Japan** Advantest testers, Accretech and Tokyo Electron probers, Disco and Accretech dicing and grinding US **United States** Test equipment from Teradyne, FormFactor, Cohu and Aehr, and Amkor's planned Arizona packaging plant CN **China** Four of the top ten assembly and test firms by 2024 revenue, in a TrendForce ranking cited by CSET MY **Malaysia** Malaysia said in July 2026 that it ships 13% of the world's packaged chips
WHY SUBSTITUTION IS POSSIBLE	Customers such as ASE already buy testers from both firms[9]. The barrier is the cost of switching, which Advantest says means rebuilding a customer's whole test setup[2].

WHERE CHINA STANDS	WHERE THE US STANDS
Georgetown's CSET, analyzing TechInsights data, finds that between 2019 and 2024 Chinese toolmakers gained 5 percentage points of share in test tools and 4 points in fabrication tools built for advanced packaging[1]. But they lost share in bonding and other conventional assembly and packaging tools[1].	American firms held 23.5 percent of the market for test tools in 2025 and Japanese firms 63.0 percent, by CSET's analysis of TechInsights data[3]. Amkor raised its planned investment in an Arizona packaging and test campus to $7 billion in October 2025[10]. In June 2026 it announced a long-term partnership with TSMC, which will buy packaging and test services from it in Arizona[11].

Turning silicon into a finished, packaged chip takes 1,000 to 1,500 steps, depending on how complex the chip is, the Semiconductor Industry

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Association estimated in 2025[12]. Advantest's investor guide of July 2026 says test is the only step that puts electrical signals through the chip[13]. The earlier stages of chipmaking make dies, the individual chips on a wafer. The back end of the industry, assembly and test, checks which of them work and puts the good ones into packages.

How it works

How a chip is tested



1. **The wafer.** Earlier stages make the chips. Before any of them can be sold, a machine tests each one, starting while the chips are still on the wafer.
2. **Needles.** Thousands of needles on a probe card press onto one chip. They touch small metal pads on its surface.
3. **Questions and answers.** Through the needles, the tester sends signals into the chip and reads what comes back. It checks every answer.
4. **Chip by chip.** The wafer moves under the needles, chip by chip. The tester marks any chip that answers wrong, and that chip goes no further.
5. **Into the package.** Only chips that pass go on to be packaged. A modern accelerator bonds together a dozen expensive pieces, and one bad piece throws away all of them, so each piece is tested before it goes in.

Simplified. Nine needles stand for thousands. In a real prober the card stays still and the wafer is raised to meet the needles. The failed chips are placed for illustration and do not reflect a real yield. The package follows chapter 11: two logic chips and eight memory stacks on an interposer and a package base.

Each round of test catches failures the round before it cannot see.

- **Wafer sort.** Thousands of needles land on each die while it is still on the wafer. A tester sends in test patterns and checks the answers, and dies that fail go no further.
- **Burn-in.** The part runs hot and at raised voltage for minutes to days, on the wafer or in the package, so that early-life failures happen in the factory, before the part ships[14].

12 Testimony of David Isaacs, Semiconductor Industry Association, hearing on... U.S. Senate Committee on Finance, 14 May 2025, p. 2. finance.senate.gov

13 Investors Guide July 29, 2026. Advantest, 30 July 2026, p. 10. advantest.com

14 AEHR TEST SYSTEMS, Form 10-K annual report for the period ended 2026-05-29. U.S. Securities and Exchange Commission, 27 July 2026. sec.gov

- **Final test.** The tester checks the packaged part again, because packaging adds failures of its own.
- **System-level test.** The finished part sits in a socket and runs something close to a real workload, because some failures appear only when everything runs at once.

As the parts in a package grow more valuable, finding a bad die before packaging matters more[15]. Aehr's wafer-level burn-in systems look for early-life and hidden failures before dies are cut from the wafer[15]. Catching those failures early lowers the risk of packaging high-bandwidth memory and an expensive substrate with a processor that may later fail[15]. Aehr says its lead AI customer is moving all burn-in for its AI accelerators from the system level to the wafer[16]. The FOX-XP systems that customer ordered from Aehr in August 2026 each burn in nine 300 mm wafers at a time[15].

Variants and trade-offs

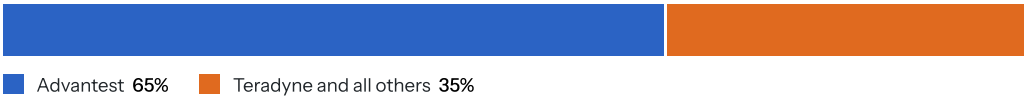
Wafer sort and known-good die

The hard part of wafer sort is the probe card, the circuit board that holds the needles. FormFactor says the high-frequency capability of its probe cards lets it compete favorably in testing high-bandwidth memory, a stack of eight, twelve or sixteen memory dies[17]. For a large logic die every needle must sit in the same plane, survive repeated touchdowns without drifting, and do it hot. Each card is built for one chip design, so every new design needs a new card and a new qualification.

Final test, burn-in and system-level test

The bigger the package, the more one bad die can cost. TSMC has told investors that its CoWoS packaging roadmap runs beyond 14 times the area a lithography machine can print in one shot[18]. A die that fails after assembly wastes the memory and substrate joined to it[15]. ASE, a Taiwanese assembly and test contractor, ran 6,797 testers in the second quarter of 2025 and 8,348 a year later[6].

Automated test equipment: share of the ~$9.0bn tester market, 2025 %



Advantest 65% Teradyne and all others 35%

Source: Advantest FY2025 results briefing, April 2026

6 ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for... U.S. Securities and Exchange Commission, 30 July 2026. sec.gov

15 Aehr Receives $22 Million Follow-On Order for AI Processor Wafer-Level Burn-In... Aehr Test Systems, 12 August 2026. aehr.com

16 Aehr Test Systems Reports Fiscal 2026 Fourth Quarter and Full Year Financial... Aehr Test Systems, 14 July 2026. aehr.com

17 FORMFACTOR INC, Form 10-K annual report for the period ended 2025-12-27 (10-K). U.S. Securities and Exchange Commission, 20 February 2026. sec.gov

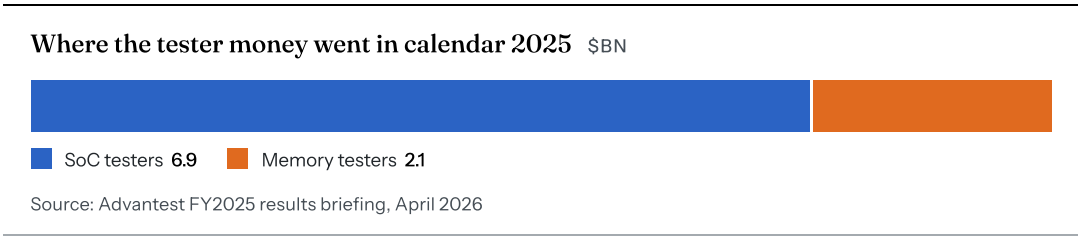
18 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 16. investor.tsmc.com

Who makes it

Advantest puts its own share at about 65 percent of an estimated $9.0 billion tester market in 2025[4]. Its share of system-on-chip (SoC) testers, the machines that test processors and other non-memory chips, rose about ten points in a year to 66 percent[4]. It held about 60 percent of memory test[4].

Teradyne, Advantest's main rival, took $2.52 billion of semiconductor test revenue in 2025, service included[5]. Of that, $1.89 billion came from SoC testers and $505 million from memory testers[5]. Teradyne's total revenue grew 13 percent to $3.19 billion in 2025, with fourth-quarter revenue up 44 percent on AI demand[19].

Georgetown University's CSET, analyzing TechInsights data, gives Advantest 62.7 percent of the $9.1 billion market for test tools in 2025 and Teradyne 21.8 percent[3]. Japanese firms held 63.0 percent of that market and American firms 23.5 percent[3].



Source: Advantest FY2025 results briefing, April 2026

The rest of the test and assembly equipment market splits into niches, each with its own suppliers.

- **Probe cards.** FormFactor took $785.0 million of revenue in fiscal 2025, most of it from probe cards[17]. Technoprobe's revenue grew 15.7 percent to EUR 628.4 million in 2025, about 38 percent of it from AI[20]. It aims to double its manufacturing capacity by the end of 2027[20].
- **Probers**, the machines that move each wafer under the probe card and keep it at a set temperature. Tokyo Seimitsu, which sells under the Accretech name, says its own research gives it the largest global share in probers[21]. Tokyo Electron makes probers too[22]. CSET's analysis of TechInsights data gives Tokyo Electron 19.9 percent and Accretech 17.1 percent of the $2.9 billion market for handlers and probers in 2025[23].
- **Handlers and burn-in**, the machines that feed packaged parts into a tester and hold them at a set temperature, and the systems that run burn-in. Cohu

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech
4 Advantest, presentation notes for the FY2025 (year ended 31 March 2026) results... Advantest, 27 April 2026, p. 13. advantest.com
5 TERADYNE, INC, Form 10-K annual report for the period ended 2025-12-31 (10-K). U.S. Securities and Exchange Commission, 19 February 2026. sec.gov
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19 Teradyne Reports Fourth Quarter and Full Year 2025 Results. Teradyne, 2 February 2026. investors.teradyne.com
20 Technoprobe, press release: the Board of Directors approves the draft statutory... Technoprobe, 18 March 2026, p. 1. technoprobe.com
21 Tokyo Seimitsu Integrated Report 2025. Accretech, 3 December 2025, p. 6. accretech.com
22 Precio™ Series. Tokyo Electron, accessed 9 September 2026. tel.com
23 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech

took $453.0 million of revenue in fiscal 2025[24]. Aehr's revenue was $50.0 million in fiscal 2026, but it booked a record $60.7 million of orders in the fourth quarter alone[16]. It expects revenue of $130 million to $150 million in fiscal 2027[16].

- **Dicing and grinding.** Dicing saws cut wafers into dies, and grinders thin them. CSET's analysis of TechInsights data found that in 2025 Disco held 87 percent and Accretech 8.1 percent of the $1.9 billion market for these machines[8]. In December 2025 Disco announced a saw that cuts panels as large as 400 by 400 mm into individual packages[25].
- **Bonding.** Kulicke and Soffa took $654.1 million of revenue in fiscal 2025[26]. Besi's second-quarter 2026 revenue rose 68.7 percent to EUR 249.9 million on hybrid bonding and data center demand[27].

Thermocompression bonders use heat and force to press memory dies into a stack and to attach chiplets to substrates. ASMPT says it remained the leader in that market in 2025 and aims for a 35 to 40 percent share[28]. In its 2025 results it said it expects the market to reach $1.6 billion by 2028[28]. In July 2026 ASMPT announced new bulk orders for more than 50 of its chip-to-substrate bonders for AI chips, from outsourced assembly and test firms[29]. Hanmi Semiconductor in Korea also sells thermocompression bonders, including dual bonders for high-bandwidth memory[30]. See Memory and HBM.

The back end's geography

Taiwanese firms lead outsourced assembly and test, the OSAT business, and Chinese firms are large challengers. ASE billed $20,573.4 million in 2025[9]. Packaging made up about 48 percent of that, testing about 11 percent and its electronics manufacturing arm about 40 percent[31]. In a TrendForce ranking that CSET cites, four of the top ten OSATs by 2024 revenue are headquartered in China[1].

Malaysia's investment agency said in July 2026 that the country ships 13 percent of the world's packaged chips[32]. It said a consortium of five Malaysian firms aims to help the country capture 7 percent of world advanced packaging shipments by 2035[32]. A 2024 study by the Semiconductor Industry Association

1 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

8 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

9 ASE Technology Holding Co., Ltd., Form 20-F annual report for the period ended... U.S. Securities and Exchange Commission, 1 April 2026. sec.gov

16 Aehr Test Systems Reports Fiscal 2026 Fourth Quarter and Full Year Financial... Aehr Test Systems, 14 July 2026. aehr.com

24 COHU, INC., Form 10-K annual report for the period ended 2025-12-27 (10-K). U.S. Securities and Exchange Commission, 17 February 2026. sec.gov

25 DISCO develops the DFD6080 fully automatic dicing saw for work sizes up to 400... DISCO Corporation, 15 December 2025. disco.co.jp

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27 BE Semiconductor Industries N.V. Announces Q2-26 and H1-26 Results. BE Semiconductor Industries (Besi), 23 July 2026. besi.com

28 ASMPT Announces 2025 Annual Results AI-Driven Structural Growth Underpins Group... ASMPT, 4 March 2026. asmpt.com

29 ASMPT Secures Significant Wins for More than 50 Chip-to-Substrate... ASMPT, 29 July 2026. asmpt.com

30 Hanmi Semiconductor, products page. Hanmi Semiconductor, accessed 9 September 2026. hanmisemi.com

31 ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for... U.S. Securities and Exchange Commission, 5 February 2026. sec.gov

32 Advanced Packaging: How Malaysia is Packaging the Future of AI. Malaysian Investment Development Authority, 1 July 2026. mida.gov.my

and Boston Consulting Group gave Malaysia 7 percent of global assembly, test and packaging capacity in 2022, the largest share in Southeast Asia[33]. The study measured capacity by factory floor space at contractors and chipmakers' own plants[34].

ASE spent $2.7 billion on machinery and $1.4 billion on buildings, facilities and automation in the first half of 2026[35]. It says revenue from leading-edge advanced packaging is running ahead of its $3.5 billion forecast for the year[35]. Powertech, another Taiwanese assembly and test firm, aims to run the first panel-level packaging line for AI chips in 2027[36].

Amkor, an American assembly and test contractor, took $6.71 billion of revenue in 2025[7]. Its revenue in the second quarter of 2026 alone was $1.90 billion, up 26 percent year on year[37]. It has raised the planned investment in its Arizona campus to $7 billion[10].

In October 2024 TSMC and Amkor signed a memorandum of understanding for TSMC to buy packaging and test services from Amkor's planned plant in Peoria, Arizona[38]. TSMC would use those services for its customers, especially those using its Phoenix fabs[38]. The two announced a long-term partnership in June 2026[11].

Testers installed at ASE, a Taiwanese assembly and test contractor TESTERS



Source: ASE Technology Holding, second quarter 2026 earnings release (SEC Form 6-K)

The chokepoint

Advantest says that switching to another tester maker means rebuilding the whole test setup used for development, evaluation and mass production, at significant extra cost to the customer[2]. Still, ASE says it buys testers with similar functions from different makers, including Teradyne and Advantest[9].

2 Investors Guide July 29, 2026. Advantest, 30 July 2026, p. 12. advantest.com

7 AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-02-09... U.S. Securities and Exchange Commission, 9 February 2026. sec.gov

9 ASE Technology Holding Co., Ltd., Form 20-F annual report for the period ended... U.S. Securities and Exchange Commission, 1 April 2026. sec.gov

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36 Powertech eyes world's first panel-level packaging for AI chips in 2027. Nikkei Asia, 27 August 2026. asia.nikkei.com

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CSET, analyzing TechInsights data, finds that between 2019 and 2024 Chinese toolmakers gained 5 percentage points of share in test tools and 4 points in fabrication tools built for advanced packaging[1]. But they lost share in bonding and other conventional assembly and packaging tools[1]. They still lag significantly behind foreign suppliers[1].

ASE's packaging business took $9,829.2 million from outside customers in 2025, and its testing business $2,292.0 million[9]. Amkor took $6.71 billion[7]. Global semiconductor sales that year were $791.7 billion[39].

THREE THINGS TO REMEMBER

- Chips are tested early and often because one bad chip can ruin a package of expensive ones.
- Advantest and Teradyne account for most spending on chip testers.
- In a TrendForce ranking cited by CSET, four of the top ten outsourced assembly and test firms by 2024 revenue are based in China.

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CHAPTER
14
STAGE 14 OF 15

Systems and Networking

AI chips are wired together 72 to a rack and run as one computer. Many firms can assemble the racks, but lasers for the optical links between them were scarce in 2026.

IN PLAIN TERMS

One AI chip is not enough to train a model. So seventy-two of them are wired together in a rack, a cabinet the size of a wardrobe, and run as one computer. Each chip sits on its own board and draws more than a thousand amps, several times the current a whole house is wired to carry. The chips have to swap results with each other all the time. A copper wire can carry those signals only a few meters before they fade, so the longer links send them as light down glass fiber. Many companies can build the rack, but in 2026 the lasers that send the light were scarce.

IN SHORT

In March 2026 Nvidia agreed to invest $2 billion in the laser maker Lumentum, alongside a multibillion-dollar purchase commitment and rights to future capacity for advanced laser components[1]. Indium phosphide is used in high-speed lasers for optical communications, and China produced 70 percent of the world's indium in 2025[2].

On 4 February 2025 China added indium phosphide substrates, the wafers the lasers are grown on, to its export control list[3]. AXT, a US-listed maker of those wafers, does all its manufacturing in China[3]. In its filing for the quarter to June 2026, AXT said it could not estimate when it would resume shipping the wafers to the United States[3].

CHOKEPOINT CARD	HIGH CONCENTRATION	MODERATELY HARD TO SUBSTITUTE
CONCENTRATION **High**	SUBSTITUTABILITY **Moderate** Rated on NVLink and its NVSwitch chips, which Nvidia controls and which join the accelerators in a rack[4]. Lasers for the optical links were also scarce in 2026[5]. Nvidia secured future capacity rights for them from Lumentum in March 2026[1].	PRICE OR MARKET SIZE **$3.1 million** for one GB200 NVL72 rack bought by a typical hyperscaler, or about $3.9 million with networking, storage and other items, by SemiAnalysis's August 2025 estimate[6]

1 NVIDIA Announces Strategic Partnership With Lumentum to Develop... Nvidia, 2 March 2026. nvidianews.nvidia.com
2 Mineral Commodity Summaries 2026: Indium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov
3 AXT INC, Form 10-Q quarterly report for the period ended 2026-06-30. U.S. Securities and Exchange Commission, 13 August 2026. sec.gov
4 NVIDIA CORP, Form 10-K annual report for the period ended 2026-01-25 (10-K). U.S. Securities and Exchange Commission, 25 February 2026. sec.gov
5 AI demand strains supplies of lasers, fiber and other optical tech. Nikkei Asia, 27 May 2026. asia.nikkei.com
6 H100 vs GB200 NVL72 Training Benchmarks - Power, TCO, and Reliability Analysis... SemiAnalysis, 20 August 2025. newsletter.semianalysis.com

WHO LEADS	US **Nvidia** NVLink, InfiniBand and Spectrum-X Ethernet, which it credited for data center growth in the quarter to April 2026[7] US **Broadcom** $16.7B of AI semiconductor revenue in the quarter to August 2026, up 221%[8] TW **Foxconn** World's largest AI server provider, its chairman said in March 2026[9] CN **InnoLight** High-speed optical modules made in Suzhou and, since 2019, Thailand[10] US **Coherent** Lasers and optical networking products, with a $2B Nvidia investment agreed in March 2026[11]
WHERE IT IS MADE	US **United States** Switch and accelerator silicon, system design, indium phosphide lasers TW **Taiwan** Contract design, rack assembly and high-layer-count circuit boards JP **Japan** Low-loss board laminate CN **China** Optical module assembly and test, and most of the world's indium TH **Thailand** Chinese module makers' offshore transceiver plants
WHY SUBSTITUTION IS POSSIBLE	UALink is an open specification for the same job as NVLink[12]. AMD's Helios rack, launched in July 2026, runs it over Ethernet[13]. Since May 2025 Nvidia's NVLink Fusion program has let other firms' custom chips use NVLink, so the dependence on Nvidia stays[14].

WHERE CHINA STANDS	WHERE THE US STANDS
Chinese firms such as InnoLight assemble high-speed optical modules in volume, at sites including Suzhou and, since 2019, Thailand[10]. In February 2025 China put indium under new export restrictions[2]. The same month it began requiring export permits for indium phosphide substrates, the wafers the lasers are grown on[3].	Coherent broke ground in Sherman, Texas, in June 2026 on an expansion of its indium phosphide factory[15]. In August 2026 Nvidia said that Wistron's new plant in Fort Worth, Texas, was producing its GB300 superchips and that Foxconn was building a factory in Houston[16].

Before a chip package can compute, it has to be built into a machine. Building that machine is a separate industry from making the package.

How it works

An accelerator (the AI chip) ships as a module, the chip package on a small board with the circuits that feed it power. Nvidia calls its version SXM, and the open-standards equivalent is the Open Compute Project's accelerator module. Eight modules bolt onto a baseboard that carries their power and the links between them. On Nvidia's baseboard, switch chips pass data between any two modules at full speed.

Rack-scale systems drop the baseboard and put the chips on trays stacked in the rack. An Nvidia GB300 NVL72 holds 18 compute trays plus nine switch

2 Mineral Commodity Summaries 2026: Indium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

3 AXT INC, Form 10-Q quarterly report for the period ended 2026-06-30. U.S. Securities and Exchange Commission, 13 August 2026. sec.gov

7 NVIDIA CORP, Form 10-Q quarterly report for the period ended 2026-04-26 (10-Q). U.S. Securities and Exchange Commission, 20 May 2026. sec.gov

8 Broadcom Inc., Form 8-K current report for the period ended 2026-09-02 (8-K). U.S. Securities and Exchange Commission, 2 September 2026. sec.gov

9 Hon Hai Technology Group (Foxconn) Announces FY2025 & 4Q25 Financial Results. Hon Hai Precision Industry (Foxconn), 16 March 2026. honhai.com

10 InnoLight Technology, company overview page. InnoLight Technology, accessed 9 September 2026. innolight.com

11 NVIDIA and Coherent Announce Strategic Partnership to Develop Optics Technology... Nvidia, 2 March 2026. nvidianews.nvidia.com

12 UALink™ 200G 1.0 Specification Overview. UALink Consortium, 15 May 2025. ualinkconsortium.org

13 AAI 2026: AMD Launches AMD Helios Rackscale Solution for Frontier AI. AMD Newsroom, 23 July 2026. newsroom.amd.com

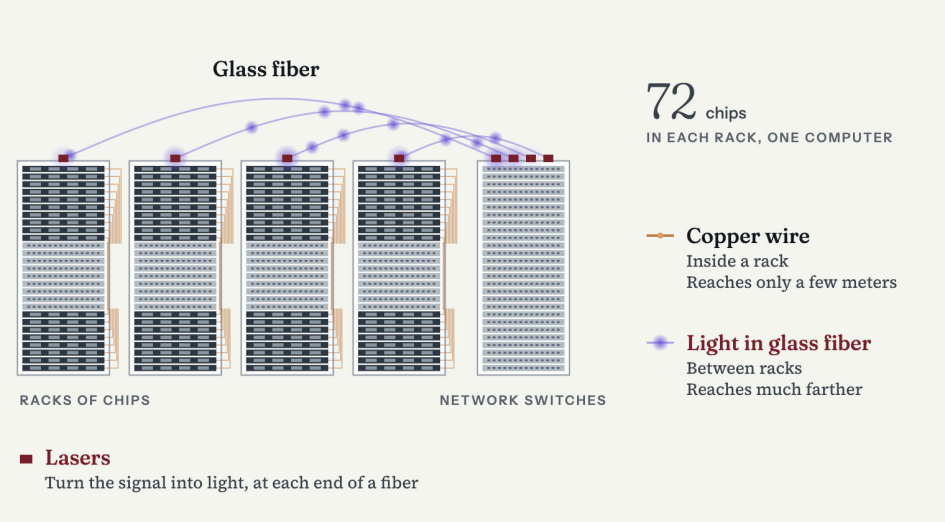
14 NVIDIA Unveils NVLink Fusion for Industry to Build Semi-Custom AI... Nvidia, 18 May 2025. nvidianews.nvidia.com

15 Coherent Announces a CHIPS Letter of Intent for $50 Million to Expand... Coherent, 16 June 2026. coherent.com

16 NVIDIA and Partners Build in America, for America. Nvidia, 5 August 2026. blogs.nvidia.com

trays, 72 GPUs in all, and draws up to 142 kW[17]. The newer Vera Rubin NVL72 uses the same rack footprint[18]. SemiAnalysis estimated in August 2025 that a GB200 NVL72 rack cost a typical hyperscaler $3.1 million, or about $3.9 million with networking, storage and other items[6].

How AI chips are wired together



1. **One rack.** One AI chip is not enough to train a model. So 72 of them are wired together in one rack, a cabinet the size of a wardrobe, and run as one computer.
2. **Copper inside.** Inside the rack, copper wires link each tray of chips to the switch trays, which pass data between any two chips. Copper carries a signal only a few meters.
3. **Light between.** Training a large model takes many racks, most of them too far apart for copper. The links between them carry the signal as flashes of light in glass fiber.
4. **Lasers.** At each end of a fiber, a laser turns the electrical signal into flashes of light, and a detector turns them back. That lets thousands of chips in many racks work on one model.

Simplified. The trays follow the layout of an Nvidia NVL72 rack. Racks and fibers are not drawn to scale.

Inside the rack the trays reach each other over copper[17]. Between neighboring racks, the signal can travel up to 7 meters over active electrical cables, a type of copper cable sold by firms such as Credo[19]. Longer links run on glass fiber, with an optical transceiver at each end. A laser in the sending transceiver turns the electrical signal into pulses of light, and a detector in the receiving one turns the pulses back into current.

The accelerators in an AI cluster exchange data over two networks that are not interchangeable. Scale-up connects anywhere from a few dozen accelerators to thousands closely enough that they can read each other's memory. It runs at terabytes per second. Scale-out joins those groups to one another at hundreds of gigabits per second. A byte is eight bits, so each GPU in a GB300 NVL72 has

6 H100 vs GB200 NVL72 Training Benchmarks - Power, TCO, and Reliability Analysis... SemiAnalysis, 20 August 2025. newsletter.semianalysis.com

17 System Hardware & Components. Nvidia, accessed 9 September 2026. docs.nvidia.com

18 NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for... Nvidia, 13 October 2025. blogs.nvidia.com

19 Credo Technology Group Holding Ltd, Form 10-K annual report for the period... U.S. Securities and Exchange Commission, 15 June 2026. sec.gov

nine times as much bandwidth for scale-up as for scale-out, SemiAnalysis calculated in January 2026[20].

A model too big for one accelerator is split across many. The parts that exchange data constantly stay inside the scale-up group, while whole blocks of layers can sit on different groups across the slower scale-out network[21]. So scale-up bandwidth limits how efficiently a large model trains.

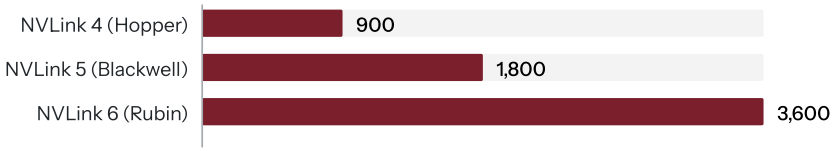
Variants and trade-offs

Scale-up: NVLink, UALink and Google's torus

The designs differ in how many accelerators can share one pool of memory.

- **NVLink and NVSwitch.** Nvidia's per-GPU bandwidth went from 900 GB/s in NVLink 4 to 1,800 in NVLink 5, and the rack's own network moves 130 TB/s on Blackwell[22]. Nvidia's Rubin launch release of January 2026 gives NVLink 6 as 3,600 GB/s per GPU and 260 TB/s for a Vera Rubin NVL72 rack[23]. As of October 2026, Nvidia's NVLink product page still lists preliminary figures of 3,000 GB/s and 216 TB/s[22].
- **UALink.** The open specification runs 200 Gb/s per lane, four lanes to a station for 800 Gb/s, and can address up to 1,024 endpoints in one group[12]. AMD's Helios rack, launched in July 2026, joins 72 GPUs and runs UALink over Ethernet for its scale-up network[13].
- **Google's 3D torus.** Google wires 64 of its Ironwood TPU accelerators into each rack, called a cube[24]. Inside the cube each chip links directly to six others, in a three-dimensional grid that Google calls a 3D torus[24]. Optical circuit switches join cubes into superpods of 9,216 chips that can bypass a failed cube or link[24].

NVLink bandwidth per GPU by generation GB/S



Source: NVIDIA NVLink product page (NVLink 4 and 5); NVIDIA Rubin launch release, January 2026 (NVLink 6)

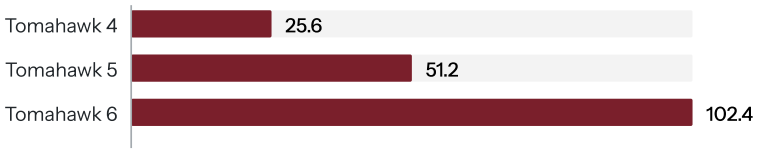
Scale-out: InfiniBand versus Ethernet

InfiniBand, a specialized network long used in supercomputers, was the main scale-out network when AI clusters were small and uniform, but Ethernet switch capacity grew quickly. Broadcom's Tomahawk 4 switched 25.6 Tbps on one chip in December 2019[25]. Tomahawk 5 reached 51.2 Tbps in August 2022,

12 UALink™ 200G 1.0 Specification Overview. UALink Consortium, 15 May 2025. ualinkconsortium.org
13 AAI 2026: AMD Launches AMD Helios Rackscale Solution for Frontier AI. AMD Newsroom, 23 July 2026. newsroom.amd.com
20 Co-Packaged Optics (CPO) Book – Scaling with Light for the Next Wave of... SemiAnalysis, 1 January 2026. newsletter.semianalysis.com
21 Parallelisms Guide — Megatron Bridge. Nvidia, accessed 9 September 2026. docs.nvidia.com
22 NVLink & NVLink Switch for Advanced Multi-GPU Communication. Nvidia, 20 April 2026. nvidia.com
23 NVIDIA Kicks Off the Next Generation of AI With Rubin. Nvidia, 5 January 2026. nvidianews.nvidia.com
24 Inside the Ironwood TPU codesigned AI stack. Google Cloud, 6 November 2025. cloud.google.com
25 Broadcom Ships Tomahawk 4, Industry's Highest Bandwidth Ethernet Switch Chip at... Broadcom, 9 December 2019. investors.broadcom.com

and Broadcom said it had doubled the bandwidth about every two years since 2014[26]. Tomahawk 6 doubled it again to 102.4 Tbps in June 2025[27]. In June 2025 the Ultra Ethernet Consortium published its 1.0 specification, a set of Ethernet-based communication protocols for AI and supercomputing that covers transport, congestion control and direct memory access[28].

Broadcom Tomahawk switching capacity on one chip TBPS



Source: Broadcom product releases for Tomahawk 4, 5 and 6

Nvidia sells switches and network cards for both InfiniBand and Ethernet. Its data center revenue reached $89.0 billion in the quarter to July 2026, up 117 percent from a year earlier[29]. In the quarter before, it credited its data center growth to Blackwell 300 products and to demand for its "InfiniBand, Spectrum-X Ethernet, and NVLink" solutions[7].

Optics: pluggable, then co-packaged

A cluster needs several transceivers, the modules that turn electricity into light and back, for each accelerator. Their power cost is one reason Broadcom and Nvidia are moving the optics out of pluggable modules and into the switch chip's own package, a design called co-packaged optics. Broadcom's Tomahawk 6 Davisson cuts optical interconnect power by 70 percent compared with traditional pluggable modules[30]. It keeps its lasers in field-replaceable modules[30]. Nvidia's Spectrum-X Photonics switches, announced in March 2025, reach 512 ports of 800 gigabits and use four times fewer lasers than traditional designs[31]. Nvidia builds them on its Spectrum-6 switch design[23]. In August 2026 Nvidia said its Spectrum-6 switch systems, with pluggable or co-packaged optics, were arriving at AI data centers around the world[32].

Who makes it

Chinese makers of optical modules have built plants outside China. InnoLight has run a plant in Thailand since 2019, alongside its sites in Suzhou and

7 NVIDIA CORP, Form 10-Q quarterly report for the period ended 2026-04-26 (10-Q). U.S. Securities and Exchange Commission, 20 May 2026. sec.gov

23 NVIDIA Kicks Off the Next Generation of AI With Rubin. Nvidia, 5 January 2026. nvidianews.nvidia.com

26 Broadcom Ships Tomahawk 5, Industry's Highest Bandwidth Switch Chip to... Broadcom, 16 August 2022. investors.broadcom.com

27 Broadcom Ships Tomahawk 6: World's First 102.4 Tbps Switch. Broadcom, 3 June 2025. investors.broadcom.com

28 Ultra Ethernet Consortium (UEC) Launches Specification 1.0 Transforming... Ultra Ethernet Consortium, 11 June 2025. ultraethernet.org

29 NVIDIA CORP, Form 10-Q quarterly report for the period ended 2026-07-26 (10-Q). U.S. Securities and Exchange Commission, 26 August 2026. sec.gov

30 Broadcom Announces Tomahawk 6 – Davisson, the Industry's First 102.4-Tbps... Broadcom, 8 October 2025. investors.broadcom.com

31 NVIDIA Announces Spectrum-X Photonics, Co-Packaged Optics Networking Switches... Nvidia, 18 March 2025. nvidianews.nvidia.com

32 NVIDIA Announces Financial Results for Second Quarter Fiscal 2027. Nvidia, 26 August 2026. nvidianews.nvidia.com

Taiwan[10]. The Thai subsidiary of Eoptolink, another Chinese module maker, lists sites in Chonburi and Rayong[33].

The Federal Communications Commission publishes a Covered List of equipment and services judged to pose an unacceptable risk to US national security or to the security and safety of US persons[34]. Its July 2026 order bars new authorizations for devices that contain logic-bearing hardware components produced by firms on that list[35]. Commenters in the record named optical transceivers as one such component[36]. As of 16 October 2026, the Covered List did not include InnoLight or Coherent[37].

AI data centers were straining supplies of the lasers inside those modules and pushing up their prices, Nikkei Asia reported in May 2026[5]. Nvidia had agreed in March 2026 to invest $2 billion in one laser maker, Lumentum, alongside a multibillion-dollar purchase commitment and future capacity access rights for advanced laser components[1]. Coherent, another laser maker, broke ground in Sherman, Texas, in June 2026 on an expansion of its indium phosphide factory[15].

Indium phosphide is used in the high-speed laser diodes and photodetectors for optical communications[2]. China accounted for 70 percent of world indium production in 2025[2]. In February 2025 China's Ministry of Commerce placed indium under new export restrictions[2].

Broadcom's AI chips include custom accelerators and Ethernet switch chips[38]. In its annual report for fiscal 2025, Broadcom said that distributors and equipment makers account for most of its chip sales, and that it builds custom accelerators to each customer's specifications[38]. Its AI semiconductor revenue reached $16.7 billion in the third quarter of fiscal 2026, up 221 percent from a year earlier[8]. Marvell, another US firm, also builds custom AI chips to each customer's specifications and sells Ethernet switch chips, its annual report for fiscal 2026 says[39]. Marvell's five-year agreement with Amazon Web Services, announced in December 2024, covers custom AI chips alongside the optical and networking parts that go with them[40].

1 NVIDIA Announces Strategic Partnership With Lumentum to Develop... Nvidia, 2 March 2026. nvidianews.nvidia.com

2 Mineral Commodity Summaries 2026: Indium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

5 AI demand strains supplies of lasers, fiber and other optical tech. Nikkei Asia, 27 May 2026. asia.nikkei.com

8 Broadcom Inc., Form 8-K current report for the period ended 2026-09-02 (8-K). U.S. Securities and Exchange Commission, 2 September 2026. sec.gov

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35 Protecting Against National Security Threats to the Communications Supply Chain... U.S. Federal Communications Commission, 23 July 2026, p. 6. docs.fcc.gov

36 Protecting Against National Security Threats to the Communications Supply Chain... U.S. Federal Communications Commission, 23 July 2026, p. 10. docs.fcc.gov

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38 Broadcom Inc., Form 10-K annual report for the period ended 2025-11-02 (10-K). U.S. Securities and Exchange Commission, 18 December 2025. sec.gov

39 Marvell Technology, Inc., Form 10-K annual report for the period ended... U.S. Securities and Exchange Commission, 11 March 2026. sec.gov

40 Marvell Technology, Inc., Form 8-K current report for the period ended... U.S. Securities and Exchange Commission, 2 December 2024. sec.gov

Taiwanese firms do much of the physical assembly. Foxconn booked NT$8.1 trillion of revenue in 2025, up 18 percent, and cloud and networking was its largest product category in the fourth quarter[9]. In the same results release, its chairman called Foxconn the world's largest AI server provider[9].

Some assembly is moving to the United States. In April 2025 Nvidia said it was building supercomputer manufacturing plants in Texas, with Foxconn in Houston and Wistron in Dallas[41]. It expected mass production to ramp up in the following 12 to 15 months[41]. In August 2026 Nvidia said Wistron's new plant in Fort Worth, Texas, near Dallas, was producing its GB300 superchips and Foxconn was building its factory in Houston[16].

The chokepoint

In October 2025 Nvidia said more than 50 makers of systems and components were gearing up for the Vera Rubin NVL72, whose design follows its MGX rack architecture[18]. In its filing for the quarter to July 2026, Nvidia said the complexity of producing its data center systems had caused production delays[29].

Four components that go into the rack take longer to replace.

- **Indium phosphide laser capacity.** In March 2026 Nvidia agreed to invest $2 billion in Coherent, alongside a multibillion-dollar purchase commitment and future access and capacity rights for Coherent's advanced laser and optical networking products[11].
- **Indium phosphide substrates.** China added indium phosphide substrates, the wafers the lasers are grown on, to its export control list on 4 February 2025[3]. AXT, a US-listed maker of these wafers, does all its manufacturing in China[3]. In its filing for the quarter to June 2026, AXT called the export permits the most significant challenge it faces[3]. It said it could not estimate when it would resume shipping the wafers to the United States[3]. The same filing names Sumitomo and JX as AXT's competitors in indium phosphide[3].
- **NVSwitch.** Nvidia's annual report for fiscal 2026 lists NVLink interconnects and switches among its own networking products[4]. UALink is an open standard designed for accelerator communications within one scale-up group[12]. Since May 2025 Nvidia's NVLink Fusion program has let other firms' custom chips use NVLink, including CPUs that Fujitsu and Qualcomm plan to connect to Nvidia GPUs[14].

3 AXT INC, Form 10-Q quarterly report for the period ended 2026-06-30. U.S. Securities and Exchange Commission, 13 August 2026. sec.gov

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- **High-layer-count circuit boards.** TTM Technologies, an American circuit board maker, says its circuit board competitors are mostly based in China and Taiwan[42]. Nitto Boseki, a Japanese glass fiber maker, makes NE-glass, a glass cloth that absorbs less signal, for the boards in data center switches and routers[43]. See Substrates and PCBs.

Optical module assembly can move, as InnoLight's high-volume plant in Thailand, opened in 2019, shows[10].

FOUR THINGS TO REMEMBER

- An AI rack wires 72 chips together to run as one computer.
- Almost every link longer than 7 meters carries its signal as light, because copper fades over longer runs.
- Nvidia's NVLink switch chips and the lasers are the hardest parts to replace. In October 2025 more than 50 makers of systems and components were gearing up to build Nvidia's next rack.
- China produced about 70 percent of the world's indium in 2025 and has required export permits for indium phosphide substrates since February 2025.

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CHAPTER
15
STAGE 15 OF 15

Data Centers and Power

By mid-2026, Texas had requests to connect about 474 GW of new demand, mostly data centers, but only about 205 GW had the engineering studies to plan for it. The large transformers that connect such sites to the grid commonly took three years to arrive in 2024.

IN PLAIN TERMS

A data center feeds electricity to the machines inside it and carries away the heat they give off. One rack of AI chips, a cabinet taller than a door, can draw 142 kilowatts[1]. Every watt of it comes back out as heat, and liquid piped through the rack carries the heat away[1]. A large training site already holds more than a hundred thousand chips[2]. The newest sites are planned in gigawatts[3]. One gigawatt is the whole output of a large power station, and OpenAI's seven Stargate sites in the United States add up to more than nine of them by 2029[3]. Getting that much electricity to the site is harder than putting up the building. Transformers and heavy switches connect a site to the grid, and turbines make the power. All of them come from a few suppliers and take years to arrive.

IN SHORT

Measured by capacity, US factories supplied 29.2 percent of the large power transformers bought in the US in 2023, and imports the rest[4]. GE Vernova's gas equipment backlog and reserved factory slots stood at 116 GW in mid-2026, against planned output of 20 GW a year from the third quarter[5].

ERCOT, the grid operator for most of Texas, counted 9,456 MW of big power users approved to energize in its overview for September 2026[6]. Judging by peak monthly use in September, it believed 4,132 MW of that was operating[6]. On 3 August 2026 Governor Greg Abbott ordered an audit of every data center seeking a connection before any data center project moves forward[7]. In September 2026 ERCOT said it had paused approvals to energize data centers of 75 MW or more until the audit and a review of their impact on local communities are complete[8].

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CHOKEPOINT CARD	HIGH CONCENTRATION	MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
High	Moderate Nvidia named nine data center power system providers for its 800-volt gigawatt designs in October 2025[9].	$580 billion of data center investment expected worldwide in 2025, in the International Energy Agency's November 2025 outlook[10]

WHO LEADS	US	**GE Vernova** Gas equipment, transformers and switchgear, including through Prolec GE[11]
	CH	**Hitachi Energy** Power transformers and switchgear[11]
	DE	**Siemens Energy** Grid technologies, transformers, turbines[11]
	JP	**Mitsubishi Power** Power generation equipment, including gas turbines[11]
	US	**Vertiv** Data center power systems[9]
	FR	**Schneider Electric** Cooling, through its Motivair division[12]
WHERE IT IS MADE	US	**United States** The largest data center operators, and GE Vernova turbines and grid equipment
	CH	**Switzerland** Hitachi Energy transformers and switchgear, and ABB switchgear
	DE	**Germany** Siemens Energy grid technologies and turbines
	JP	**Japan** Mitsubishi Power equipment, including gas turbines
WHY SUBSTITUTION IS POSSIBLE		Buyers of grid equipment can choose among suppliers, and GE Vernova's 2025 annual report names Hitachi Energy, Siemens Energy, Schneider Electric and ABB among its key competitors[11]. But in 2024 the longest large power transformer lead times reached 60 months[13]. An International Energy Agency survey published in 2025 found waits of up to four years worldwide[14].

WHERE CHINA STANDS	WHERE THE US STANDS
China used 25 percent of the world's data center electricity in 2024, second to the United States, by the International Energy Agency's estimate[15]. Data centers made up around 3 percent of the growth in China's electricity demand from 2022 to 2024[16].	US factories supplied 29.2 percent of the large power transformers bought in the United States in 2023, measured by capacity[4]. Constellation said in August 2026 that it expects to restart the Crane nuclear plant in 2027[17]. Microsoft has agreed to buy its power[18].

A wafer passes through a leading-edge fab in months, but in 2024 a large power transformer commonly took three years to arrive[13]. Before the pandemic, the same order took under a year[19].

4 Large Power Transformers from South Korea, Investigation No. 731-TA-1189... U.S. International Trade Commission, August 2024, p. 90. usitc.gov

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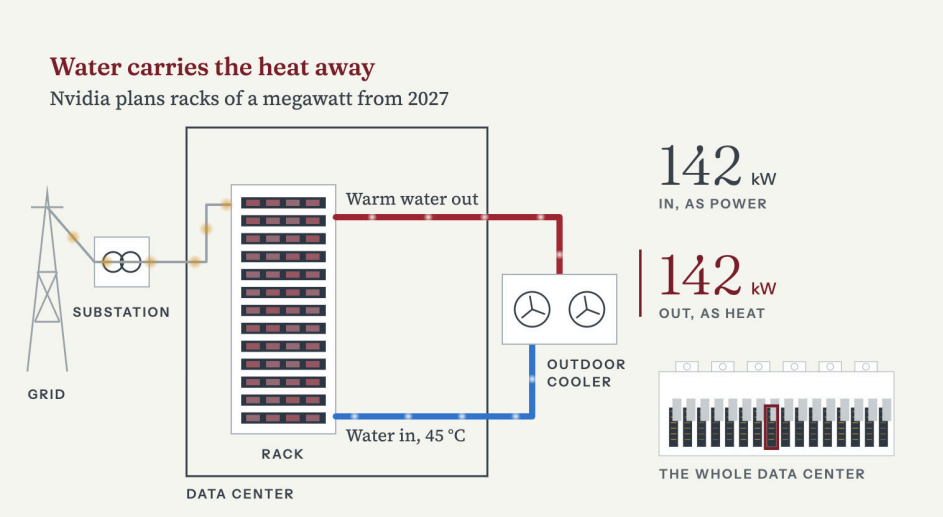
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How it works

Power in, heat out



1. **The site.** Data centers are where AI chips go to work. A large training site holds a hundred thousand of them, in rows of racks.
2. **Power in.** Power comes in from the grid, stepped down by a substation, and runs through the data center to every rack. One rack of AI chips can draw 142 kilowatts.
3. **Heat.** Every watt of it comes back out as heat, more than air can carry away.
4. **Water out.** So water runs through plates on the chips, carries the heat to coolers outdoors and comes back. Nvidia plans for racks of a megawatt each from 2027, so the power coming in and the heat going out keep growing.

Simplified diagram, with illustrative sizes. An Nvidia GB300 NVL72 rack draws at most 142 kW. Nvidia gives 45 °C as the supply-water temperature for its Vera Rubin racks. The megawatt racks from 2027 are Nvidia's plan.

The rack is the basic unit of AI compute. Nvidia's GB300 NVL72 rack draws up to 142 kW through eight 33 kW power shelves, the rack's own power supplies[1]. Colossus 1, an AI data center in Memphis, Tennessee, held an estimated 230,000 Nvidia GPUs by July 2025[2].

Between the grid and the rack, a substation steps the high voltage down with transformers. Switchgear, the building's heavy circuit breakers, can cut any part of the site off in a fault. Battery-backed supplies keep the racks running for the seconds it takes backup generators to start. Distribution boards split the feed among the rows of racks.

At the back of each rack, a busbar carries the current down to the server trays. It is a solid copper bar, because an ordinary cable would overheat at that current. Every piece of equipment from the substation to the busbar is heavy and has its own lead time, the wait between order and delivery.

Heat leaves in water. Cold plates, metal blocks with coolant running through them, sit on the GPU packages and pass the heat to the building's water loop, which releases it outdoors[12]. Nvidia's Rubin racks run coolant at up to 45 °C,

1 System Hardware & Components. Nvidia, accessed 9 September 2026. docs.nvidia.com
2 Colossus 1, AI data centers directory. Epoch AI, accessed 9 September 2026. epoch.ai
12 Hotter Than a Hot Tub: The 45°C Breakthrough to Cool AI's Biggest Machines. Nvidia, 22 June 2026. blogs.nvidia.com

warm enough that in many climates the loop can release its heat without turning on chillers, the refrigeration machines that cool water[12].

Nvidia is also moving the power that feeds its racks to 800 volts of direct current. Raising the voltage cuts the current needed for the same power, and the heat in copper comes from current. Nvidia said in October 2025 that at 800 volts the same wire can carry 157 percent more power than at 415 volts of alternating current[20]. Inside racks such as the GB300 NVL72, power runs at 54 volts, and at that voltage a one-megawatt rack would need up to 200 kg of copper busbar[21].

Variants and trade-offs

Cooling

- **Air.** Heat exchangers in the rack door work when density, the power each rack draws, is moderate, but Nvidia's GB300 NVL72 rack is liquid cooled[1].
- **Direct-to-chip liquid.** Nvidia's Rubin platform is 100 percent liquid cooled, and every cloud provider and data center operator building for it is moving to liquid[12].
- **Immersion.** The servers sit in a bath of fluid, which makes them hard to service.

Nvidia says the cooling supply chain is keeping pace with the move to liquid[12]. Motivair, Schneider Electric's cooling division, has worked alongside Nvidia's product roadmap for nearly a decade[12].

Rack power

Nvidia's 800-volt DC architecture is designed for racks from 100 kW to more than a megawatt, with megawatt racks from 2027[21]. Nvidia is making the change because the 54-volt distribution in existing racks begins to hit physical limits once racks exceed 200 kW[21].

Nvidia names nine data center power system providers for its 800-volt gigawatt designs, among them ABB, Eaton, Hitachi Energy, Mitsubishi Electric, Schneider Electric, Siemens and Vertiv[9].

A megawatt rack changes the busbar, the floor loading, the coolant flow, the leak detection and the energy stored in the rack. Nvidia's Vera Rubin NVL72 rack design holds 20 times more energy storage to keep power steady[9].

Where the electrons come from

- **Grid interconnection.** The slow path in the United States, where the queues to connect are long[3].
- **Behind-the-meter gas**, generation on the site itself. The fast path, which at least three of OpenAI's seven US Stargate sites will use to avoid grid queues[3].

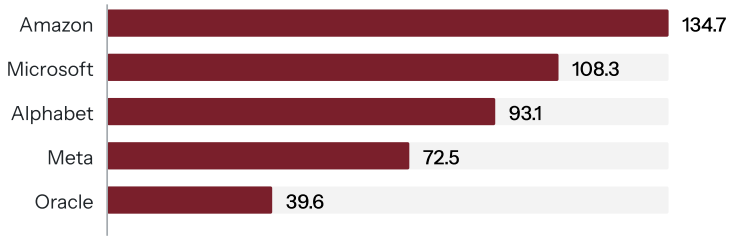
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- **Nuclear.** Existing plants already sell power to data centers[22]. Talen Energy supplies Amazon's data center campus next to the Susquehanna nuclear plant in Pennsylvania under a June 2025 contract expected to reach 1,920 MW no later than 2032[22]. Restarts and new reactors are still to come. Constellation signed a power purchase agreement with Microsoft to restart Three Mile Island Unit 1 as the 835 MW Crane Clean Energy Center[18]. The plant shut in 2019[23]. A $1 billion federal loan that closed in November 2025 partly finances its restart, which is subject to licensing approvals from the US Nuclear Regulatory Commission[23]. No restarted reactor is serving data centers yet.

Who makes it

The largest data center operators pay for the buildout. Capital spending by Alphabet, Amazon, Meta, Microsoft and Oracle came to $448.2 billion in 2025, the sum of quarterly figures compiled by Epoch AI, a research group[24]. If the trend continues, it will reach $770 billion in 2026[24]. The International Energy Agency expected worldwide investment in data centers to reach $580 billion in 2025[10].

Capital spending by five large US data center operators, 2025 $B



Source: Epoch AI, hyperscaler capex trend, February 2026, quarterly figures from company filings summed for 2025

Epoch AI also projects that OpenAI's seven US Stargate sites will exceed 9 GW by 2029[3]. It expects the site in Abilene, Texas, to grow from an estimated 0.3 GW in April to 1.2 GW in the fourth quarter of 2026[3].

Equipment makers limit how fast the buildout can go. GE Vernova builds large gas turbines, and its 2025 annual report names Siemens Energy and Mitsubishi Power among the key competitors of its power generation business[11]. Its electrification business makes power transformers and switchgear, and the report names Hitachi Energy, Siemens Energy, Schneider Electric and ABB among that business's key competitors[11].

3 OpenAI Stargate: where the US sites stand. Epoch AI, 17 April 2026. epoch.ai

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23 Energy Department Closes Loan to Restart Nuclear Power Plant in Pennsylvania. U.S. Department of Energy, 17 November 2025. energy.gov

24 Hyperscaler capex has quadrupled since GPT-4's release. Epoch AI, 26 February 2026. epoch.ai

GE Vernova bought the remaining half of Prolec GE, a transformer maker, in the first half of 2026[5]. GE Vernova's gas equipment backlog and reserved factory slots grew from 100 GW to 116 GW in the second quarter of 2026[5]. In the first half of the year, data center orders for its electrification business, which makes grid equipment, passed $5 billion, more than double its total for 2025[5].

GE Vernova's planned annual gas turbine output GW/YR



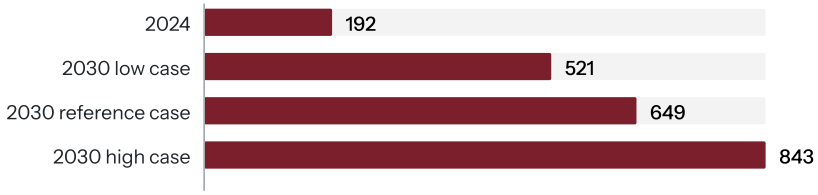
Source: GE Vernova second-quarter 2026 results, July 2026

Capacity and demand

US data centers used 192 TWh in 2024, 4.7 percent of national electricity, according to a June 2026 estimate by Lawrence Berkeley National Laboratory that leaves out cryptocurrency mining[25]. The lab's reference case, its main scenario, projects 464 TWh in 2028 and 649 TWh in 2030, which would be 11.8 percent of forecast US electricity use[25]. Its scenarios for 2030 range from 521 to 843 TWh, or 9.5 percent to 15.3 percent[26].

In its October 2026 Short-Term Energy Outlook, the US Energy Information Administration expects US electricity sales of 4,152.0 TWh in 2026 and 4,216.6 TWh in 2027[27]. It expects electricity demand from the commercial sector, which includes data centers, to grow 2.8 percent in 2027 and demand from industry to grow 2.7 percent[28]. Its September outlook had lowered the 2027 electricity sales forecast for the region that includes Texas by 4.7 percent from August[29]. That outlook noted that Texas had paused connecting new data centers to the grid while it audits proposed projects[30].

US data center electricity use, 2024 and 2030 projection range TWH



Source: Lawrence Berkeley National Laboratory, June 2026

5 GE Vernova Inc., Form 8-K current report for the period ended 2026-07-22 (8-K). U.S. Securities and Exchange Commission, 22 July 2026. sec.gov

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27 Short-Term Energy Outlook, October 2026. U.S. Energy Information Administration, 6 October 2026, p. 46. eia.gov

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By the International Energy Agency's estimate, China accounted for 25 percent of the world's data center electricity use in 2024, second to the United States at 45 percent[15]. Data centers made up around 3 percent of the growth in China's electricity demand from 2022 to 2024, the agency found[16].

Chip efficiency is still improving. Google's Ironwood TPU, an AI chip, delivers twice the performance per watt of Trillium, the TPU before it[31]. A token is the chunk of text a model reads or writes. Energy per token can fall while total energy use still rises, if the number of tokens rises faster.

The chokepoint

On the Texas grid, far more capacity has been requested than is connected and running. ERCOT, the grid operator for most of Texas, was tracking about 474 GW of large loads (big power users) seeking a connection in June 2026, roughly 90 percent of it data centers[32]. In its operational overview for September 2026, ERCOT counted 9,456 MW approved to energize, but judging by peak monthly use in September, it believed only 4,132 MW of it was operating[6].

In a preliminary review in July 2026, ERCOT found about 205 GW of the queue eligible for Batch Zero, its first study of connection requests as a group[33]. On 3 August 2026 Texas Governor Greg Abbott ordered the state's Public Utility Commission and ERCOT to audit every data center in the interconnection process before any data center project moves forward[7]. ERCOT then paused approvals to energize data centers and crypto mining sites of 75 MW or more until the audit and a review of their impact on local communities are complete[8]. In September 2026 ERCOT conditionally classed 66.4 GW of Batch Zero projects as base load and 127.9 GW as studied load, pending final decisions in December[8]. It will report on the audit and the community review to the Public Utility Commission by 10 December 2026[8].

Projects in that queue also wait on equipment.

- **Large power transformers.** In 2024, lead times of 36 months were commonly quoted and the longest reached 60 months, against under a year before the pandemic[13]. By a US Commerce Department estimate, 82 percent of the units put into US service in 2019 were imported[34]. In an industry survey published in 2025, the International Energy Agency found that large power transformers took up to four years to secure worldwide[14]. It also

6 Electric Reliability Council of Texas, ERCOT Monthly Operational Overview... Electric Reliability Council of Texas, 16 October 2026, p. 9. ercot.com

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34 U.S. Department of Energy Large Power Transformer Resilience Report to... U.S. Department of Energy, 10 July 2024, p. 20. energy.gov

found that their average lead times had almost doubled since 2021[14]. Measured by capacity, US factories supplied 29.2 percent of the transformers of 60 MVA and above bought in the US in 2023, and imports supplied the rest[4].

- **Gas turbines.** In July 2026 GE Vernova expected its annual gas turbine output to reach 20 GW in the third quarter of 2026, 24 GW in 2028 and 30 GW in 2030[5].
- **Firm new generation**, the kind that runs whatever the weather. The Nuclear Regulatory Commission has approved a fuel license amendment for the Crane plant, and Constellation expects to restart it in 2027[17].

Memory chips are also in short supply. Micron's quarterly report for the period ended 28 May 2026 says AI-driven growth in demand for memory and storage is outpacing industry supply[35]. The memory shortage and the wait for power equipment both limit how much of the 2026 capital spending turns into working compute.

FOUR THINGS TO REMEMBER

- One AI rack can draw 142 kilowatts, too much heat for air, so racks are cooled with liquid.
- The newest AI sites are planned in gigawatts, and each gigawatt is the output of a large power station.
- Power equipment is one of the slow parts of building a data center. In 2024 lead times of 36 months were commonly quoted for a large transformer.
- The United States used 45 percent of the world's data center electricity in 2024 and China 25 percent, by the International Energy Agency's estimate.

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CHAPTER
16
THE ECONOMICS

The Economics of AI Chips

Epoch AI estimated in December 2025 that Nvidia's B200 AI chip cost about $6,400 to build, mostly for memory and packaging. Its processor silicon cost about $900.

IN PLAIN TERMS

A bill of materials is the list of parts inside a product and what each one costs. Epoch AI estimated in December 2025 that Nvidia's B200, an AI chip from its Blackwell line, cost about $6,400 to build[1]. The two logic dies, the pieces of silicon that do the calculating, cost about $900 of that[1]. The stacked memory beside them accounted for about half, and the packaging that wires the dies and memory together cost more than the dies[1]. Three firms, SK hynix, Samsung and Micron, dominate the world market for that memory[2]. One Taiwanese firm, TSMC, does that packaging[1].

IN SHORT

Reports in March 2024 put the B200's selling price at $30,000 to $40,000 per chip[1]. Epoch AI notes that most Blackwell revenue comes from servers and rack-scale systems, which may carry lower margins[1]. It says Nvidia's Blackwell margin may therefore be lower than the gap between cost and price suggests[1].

The estimate's memory price of $14 to $17 per gigabyte predates the 2026 rise in memory makers' sales and margins[3]. Micron's gross margin, the share of sales left after production costs, rose from 39.8 percent for the year to August 2025 to 86.8 percent in the quarter to September 2026[4]. Nvidia's was 75.0 percent in the quarter to July 2026[5].

In December 2025 Epoch AI estimated that building one of Nvidia's Blackwell B200 chips cost about $6,400[1]. Reports in March 2024 put its selling price at $30,000 to $40,000 per chip[1]. Epoch AI's chip sales model, with data through October 2025, puts Nvidia's compute revenue per GPU in its GB200 rack systems at $33,000 to $42,000[6]. That figure leaves out the networking gear in those racks[6]. Some stages behind that $6,400 are expensive because they are hard to make, some because capacity is short, and some because one firm sets the price.

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How it works

Three numbers set the cost of a die, the finished rectangle of circuitry cut out of a wafer.

- **Wafer price.** TSMC does not publish it. TSMC makes Nvidia's Blackwell chips on a custom-built process called 4NP[7]. Epoch AI estimates $17,000 for a 12-inch (300 mm) wafer on that process[1].
- **Dies per wafer.** A wafer is a circle and dies are rectangles, so the edge of the wafer goes to waste. The bigger the die, the fewer fit. Each of the two logic dies in a B200 is about 800 mm2[1]. Nvidia calls them reticle-limited, meaning they are near the largest size a scanner can print in one shot[7].
- **Yield.** Yield is the share of dies that work. Defects land at random, so a bigger die is more likely to catch one. Epoch AI models Blackwell die yield at 40 to 70 percent, centered on 60[1].

Together, those three numbers put the two dies inside a B200 at about $900[1]. The estimate counts only variable costs and leaves out fixed costs such as research and development[1].

Variants and trade-offs

Memory is the largest line

The largest line on the bill is the 192 GB of high-bandwidth memory beside those dies, a generation called HBM3E[1]. At $14 to $17 per gigabyte it comes to about $2,900, which Epoch AI counts as half of the build cost[1]. Epoch AI says HBM prices are negotiated deal by deal and can vary significantly with volume and contract length[1]. The $14 to $17 range dates from December 2025, before the 2026 rise in memory makers' sales and margins described below[3].

Three firms, SK hynix, Samsung and Micron, dominate the world market for that memory[2]. SK hynix said in July 2026 that demand for its products had exceeded its available supply in recent quarters[8].

Packaging, substrate and test

CoWoS-L, the TSMC process that mounts dies and memory on one carrier and wires them together, adds about $1,100, more than the logic dies it carries[1]. Epoch AI models packaging yield at 65 to 95 percent[1]. A package that fails testing loses its logic dies and memory stacks[1]. Spreading that loss over the good units adds about $1,000 to each B200[1]. Power delivery hardware, the circuit board, and final assembly and testing add $480[1]. See Advanced packaging for why packaging capacity runs out first.

1 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai
2 Understanding the Biden Administration's Updated Export Controls. Center for Strategic and International Studies, 11 December 2024. csis.org
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7 NVIDIA Blackwell Architecture. Nvidia, 18 December 2025. nvidia.com
8 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

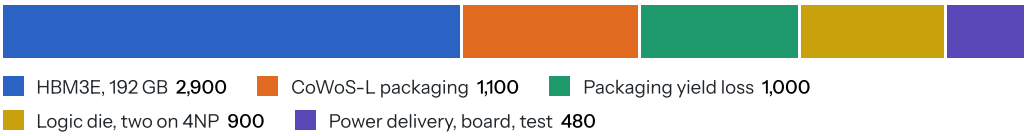
The B200 bill of materials

LINE	ESTIMATE	RANGE
HBM3E, 192 GB	$2,900	$2,800-3,100
CoWoS-L packaging	$1,100	$1,000-1,200
Packaging yield loss	$1,000	$430-1,700
Logic die, two on TSMC 4NP	$900	$720-1,200
Power delivery, board, assembly and test	$480	$370-600
Manufacturing cost	~$6,400	$5,700-7,300
Reported selling price per chip, March 2024	$30,000-40,000	
Nvidia compute revenue per GPU in GB200 racks, data to October 2025	$33,000-42,000	

Epoch AI models each line as a range, deriving the packaging line from TSMC's advanced packaging revenue and Nvidia's share of CoWoS capacity[1]. The estimate covers the accelerator module only and leaves out servers and rack-scale systems[1]. Epoch AI notes that most Blackwell revenue comes from those systems, which may carry lower margins[1]. So, Epoch AI says, Nvidia's actual margin on Blackwell may be lower than the gap between this cost and the reported chip price suggests[1].

The B200 is used here because Epoch AI's line-by-line estimate covers it. In its results for the quarter to July 2026, Nvidia said its Vera Rubin platform was ramping into full production[5]. Each Rubin GPU carries up to 288 GB of HBM4, a newer generation of the stacked memory[9].

Where the ~$6,400 of B200 manufacturing cost goes, Epoch AI estimate, December 2025 $



HBM3E, 192 GB 2,900 · CoWoS-L packaging 1,100 · Packaging yield loss 1,000
Logic die, two on 4NP 900 · Power delivery, board, test 480

Source: Epoch AI B200 cost model, from a $17,000 4NP wafer, 40–70% die yield, $14–17 per GB of memory, and a packaging line derived from TSMC's advanced packaging revenue

From package to rack to cluster

At rack scale, cost is measured per GPU-hour, one GPU running for one hour. For an operator that owns the machine, SemiAnalysis put a Vera Rubin NVL72 at $3.57 per GPU-hour in July 2026, against $1.84 for a GB200 and $2.36 for a GB300[10]. Each generation costs its owner more per hour than the last[10].

1 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai
5 NVIDIA Announces Financial Results for Second Quarter Fiscal 2027. Nvidia, 26 August 2026. nvidianews.nvidia.com
9 Inside NVIDIA Rubin GPU Architecture: Powering the Era of Agentic AI. Nvidia, 21 July 2026. developer.nvidia.com
10 Vera Rubin NVL72 vs GB200 NVL72? Inference TCO & Architecture Analysis. SemiAnalysis, 23 July 2026. inferencex.semianalysis.com

But SemiAnalysis found Rubin cheaper per token than GB200 and GB300 at every interactivity level it compared[10]. A token is a short chunk of text, and interactivity is how many tokens per second each user receives. That finding rests on early results from the cloud provider CoreWeave running the DeepSeek R1 model, and SemiAnalysis said it had not verified them[10].

One level up, SemiAnalysis estimated in June 2024 that a cluster of 100,000 of Nvidia's earlier H100 GPUs cost over $4 billion in servers[11]. It needs more than 150 MW of data center capacity and uses 1.59 TWh a year, which costs $123.9 million at $0.078 per kWh[11]. In the network design Nvidia recommended, 98,304 optical transceivers link its GPUs to the first layer of switches, called leaf switches[11].

Who captures the value

FIRM	STAGE	QUARTER ENDED	GROSS MARGIN
Micron	HBM and DRAM	September 2026	86.8%[4]
Nvidia	Accelerator design	July 2026	75.0%[5]
TSMC	Wafers and CoWoS	September 2026	66.4%[12]
KLA (non-GAAP)	Inspection	June 2026	62.4%[13]
ASML	Lithography	September 2026	57.5%[14]
Lam Research	Etch	June 2026	51.7%[15]
Applied Materials	Deposition	July 2026	50.3%[16]
ASE Technology	Packaging and test segment	June 2026	27.3%[17]
Amkor	OSAT	June 2026	16.8%[18]

Each figure except KLA's comes from the firm's own earnings release. KLA's shareholder letter gives only a non-GAAP gross margin for the quarter, 62.4 percent[13]. KLA says its non-GAAP figures exclude certain costs and expenses that it believes do not reflect its core operating results[19]. For ASE the table shows its packaging and test segment[17]. The company as a whole had a gross margin of 21.0 percent, because about a third of its revenue came from electronics manufacturing services, which earned 8.9 percent[17].

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13 Letter to Shareholders Q4 Fiscal 2026. KLA Corporation, 28 July 2026, p. 3. d1io3yog0oux5.cloudfront.net

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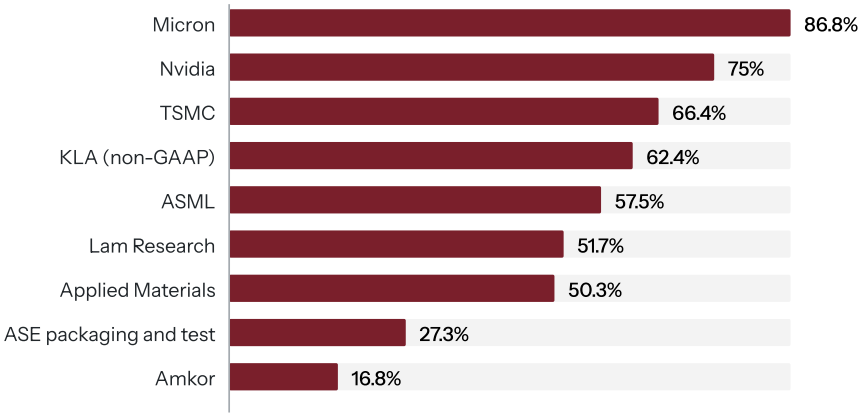
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19 KLA Corporation Reports Fiscal 2026 Fourth Quarter and Full Year Results. KLA Corporation, 28 July 2026. ir.kla.com

SK hynix's earnings release gives no gross margin. Its operating margin was 76 percent in the quarter to June 2026, and its revenue was up 257 percent from a year earlier[3]. The substrate maker Ibiden gives no gross margin either, but its operating profit was 21.8 percent of net sales in the quarter to June 2026[20].

Gross margin, quarters ended June to September 2026 %



Source: Each firm's own quarterly earnings release; quarters end between June and September 2026; ASE's bar is its packaging and test segment; KLA's bar is the non-GAAP figure from its shareholder letter (TSMC shown)

In the table, the firms in design, equipment and leading-edge wafers earn gross margins of 50.3 to 75.0 percent, and those in assembly and test 16.8 to 27.3 percent. In 2026 Micron passed all of them. Its gross margin rose from 39.8 percent for the year to August 2025 to 86.8 percent in the quarter to September 2026[4].

The Semiconductor Industry Association, using monthly data compiled by World Semiconductor Trade Statistics, counted $791.7 billion of global chip sales in 2025[21]. In its June 2026 forecast, World Semiconductor Trade Statistics put 2025 at $795.6 billion and 2026 at $1.51 trillion, with memory growing about 250 percent and logic 37 percent[22].

The industry group SEMI counted $135.1 billion of semiconductor equipment billings in 2025[23]. Capital spending at Alphabet, Amazon, Meta, Microsoft and Oracle came to $448.2 billion in 2025, the sum of Epoch AI's quarterly figures[24]. Epoch AI says that spending would reach $770 billion in 2026 if the trend continues[24]. These totals overlap, so they cannot be added together.

3 SK hynix Announces 2Q26 Financial Results. SK hynix, 29 July 2026. news.skhynix.com

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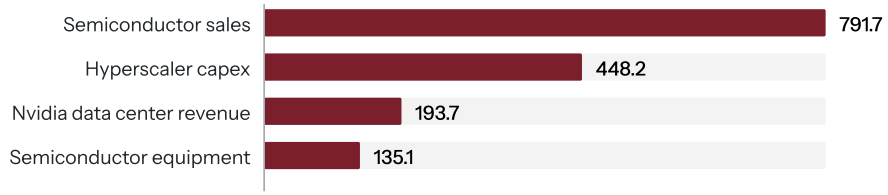
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Selected market totals, 2025 $B



Source: Semiconductor Industry Association on chip sales; Epoch AI quarterly capital spending at Alphabet, Amazon, Meta, Microsoft and Oracle, summed for 2025; Nvidia data center revenue for its fiscal year to January 2026; SEMI on equipment billings

Where the money goes

In Epoch AI's estimate for the B200, packaging cost $1,100 and packaging yield loss another $1,000, against $900 for the logic dies[1].

The table below sorts each stage by whether its cost falls as volume grows. The classes are a judgment drawn from the sources in this chapter and the stage chapters.

STAGE	CLASS	WHY
EUV lithography	Structurally expensive	One supplier, ASML. See Lithography.
Masks and pellicles	Structurally expensive	Every design needs its own mask set, and every EUV layer a flawless mask. See Photomasks and pellicles.
Metrology and inspection	Structurally expensive	KLA held 56.8 percent of the market for these tools in 2025, by CSET's analysis of TechInsights data[25]. See Metrology and inspection.
HBM	Structurally expensive	SK hynix says HBM needs more wafer input than ordinary DRAM[8]. See Memory and HBM.
Advanced packaging, CoWoS-L	Structurally expensive	Modeled packaging yield is 65 to 95 percent, and a failed package loses its dies and memory.
ABF substrates	Structurally expensive	Investment can add substrate lines, but Ajinomoto said in June 2026 that the insulating film it makes for these substrates has held over 95 percent of its market since launch[26]. See Substrates and PCBs.
Power and cooling plant	Structurally expensive	In the United States, grid connections face long queues[27]. GE Vernova's gas equipment backlog and slot reservations reached 116 GW in mid-2026[28]. See Data centers and power.
Leading-edge logic wafers	Compressible with scale	A very large die yields well only once a process matures. See Transistors and the front end.

1 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai

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26 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 3. ajinomoto.co.jp

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Design and EDA	Compressible with scale	A design is paid for once, and its cost is spread across every chip sold. See Chip design, EDA and IP.
Deposition and etch tools	Compressible with scale	Most deposition and etch steps have three or four established tool makers. See Deposition and etch.
Optical transceivers	Compressible with scale	Nvidia agreed in March 2026 to invest $2 billion in the laser maker Lumentum, with rights to future capacity[29]. It agreed a similar $2 billion deal with Coherent[30]. See Systems and networking.
Test and assembly (OSAT)	Already commoditized	Gross margins of 16.8 to 27.3 percent at Amkor and ASE's packaging and test segment, in the table above.
Wafers and bulk gases	Already commoditized	Several firms make approved 300 mm wafers. By Siltronic's estimate, the five largest wafer makers served around 75 percent of the market in the first three quarters of 2025[31]. See Silicon and wafers.
Rack integration (ODM)	Already commoditized	In October 2025 Nvidia said more than 50 makers of systems and components were gearing up for the Vera Rubin NVL72[32]. See Systems and networking.

THREE THINGS TO REMEMBER

- Epoch AI estimated in December 2025 that a B200 cost about $6,400 to build, against a selling price of $30,000 to $40,000 per chip reported in March 2024.
- The stacked memory is the most expensive part, about half of the cost.
- Packaging costs more than the logic chips, and each package that fails wastes the chips and memory inside it.

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31 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 6. siltronic.com
32 NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for... Nvidia, 13 October 2025. blogs.nvidia.com

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17. [A] ASE Technology Holding Co., Ltd. Reports Its Unaudited Consolidated Financial Results for the Second Quarter of 2026. PR Newswire · 30 July 2026
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28. [A] GE Vernova Inc., Form 8-K current report for the period ended 2026-07-22 (8-K). U.S. Securities and Exchange Commission (filing by GE Vernova Inc.) · 22 July 2026
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nvidianews.nvidia.com/news/nvidia-and-coherent-announce-strategic-partnership-to-develop-optics-technology-to-scale-next-generation-data-center-architecture

31. A Investor Presentation: Foundation of Digital Life. Siltronic · April 2026 · p. 6
www.siltronic.com/fileadmin/investorrelations/2026/Q1/20260429_Siltronic_InvestorPresentation__.pdf#page=6

32. A NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for Vera Rubin. Nvidia · 13 October 2025
blogs.nvidia.com/blog/gigawatt-ai-factories-ocp-vera-rubin

CHAPTER 17
THE CONTEST

Who Controls the Chain

Since 2018 the United States has used export controls to limit China's access to advanced chips and chipmaking tools, and China has been building its own.

IN PLAIN TERMS

An export control is a rule that requires the government's permission for a sale. The government names an item and a country or a buyer. From then on its own companies must hold a license before they ship that item to that country or buyer. The American government extends some of these rules to goods made abroad with American technology. Few governments can use such a rule to much effect, because it works only when their own firms sell something the buyer cannot get elsewhere. The machines, software and chemicals used to make an advanced chip come from a short list of firms in the United States, the Netherlands, Japan and South Korea. China can copy the simpler machines within a few years. It cannot yet copy the machine that prints the finest circuits, or the chemicals that machine needs. It has also yet to show it can make high-bandwidth memory, the stacked memory beside an AI chip, in volume.

IN SHORT

Japan and the Netherlands have aligned their own export controls with the American rules on chipmaking equipment[1]. Chinese toolmakers supplied 35 percent of China's own chipmaking equipment market in 2025, up from 10 to 15 percent before 2022[2]. When the United States licensed sales of Nvidia's H200 chip in February 2026, China restricted them[3].

The most advanced chips come mainly from Taiwan. It held 69 percent of the world's capacity for logic chips below 10 nanometers in 2022, by the count of the Semiconductor Industry Association and Boston Consulting Group[4].

The United States is still changing its export controls, and China is still building its response to them.

1 Where the Chips Fall: U.S. Export Controls Under the Biden Administration from... Center for Strategic and International Studies, 12 December 2024. csis.org

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

3 NVIDIA CORP, Form 10-Q quarterly report for the period ended 2026-07-26 (10-Q). U.S. Securities and Exchange Commission, 26 August 2026. sec.gov

4 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

The shape of the contest

In September 2022 National Security Advisor Jake Sullivan said the United States must keep "as large of a lead as possible" in foundational technologies such as advanced logic and memory chips[5]. The October rules that followed were the first to apply his doctrine of a small yard with a high fence, meaning strict controls on a narrow set of technologies[1]. Those rules went beyond the old aim of staying two technology nodes ahead of China and tried to push China's capability back below the level it had already reached[6].

Many license requirements depend on who is buying and why. In October 2018, for example, the Commerce Department put the Chinese chip company Fujian Jinhua Integrated Circuit on its Entity List, so that any US-regulated item needed a license to reach it[7]. The 2022 rules went further and required a license for any advanced AI chip, and any tool for an advanced fab, bound for China[8]. Most applications were presumed denied, but equipment for the China fabs of firms headquartered in the United States or allied and partner countries was reviewed case by case[8].

Chokepoints and who holds them

SEGMENT	LEADER	SHARE	BASIS
EUV lithography	Netherlands	100.0%	revenue, 2025
Immersion DUV lithography	Netherlands	99.3%	revenue, 2025
Metrology and inspection	United States	71.7%	revenue, 2025
Deposition	United States	60.3%	revenue, 2025
Etch and clean	United States	52.6%	revenue, 2025
Photoresist	Japan	78%	revenue, 2023
Silicon wafers	Japan	53%	revenue, 2023
Leading-edge logic, below 10 nm	Taiwan	69%	capacity, 2022
Foundry, all nodes	Taiwan	81.9%	market share, 2025
HBM	South Korea	82.5%	revenue, 2025: SK hynix 63.2, Samsung 19.3, Micron 17.4
Logic chip design	United States	65%	sales, 2022

1 Where the Chips Fall: U.S. Export Controls Under the Biden Administration from... Center for Strategic and International Studies, 12 December 2024. csis.org

5 Remarks by National Security Advisor Jake Sullivan at the Special Competitive... The White House, 16 September 2022. bidenwhitehouse.archives.gov

6 Choking off China's Access to the Future of AI. Center for Strategic and International Studies, 11 October 2022. csis.org

7 Addition of an Entity to the Entity List. Federal Register, 30 October 2018. federalregister.gov

8 Implementation of Additional Export Controls: Certain Advanced Computing and... Federal Register, 13 October 2022. federalregister.gov

Mature logic, 28 nm and above	China	33%	capacity, 2022

No country is self-sufficient, so the United States and China can each use export controls to hurt the other.

Advanced packaging is missing from the table because no public source publishes a split of its capacity by country and the Taiwanese chipmaker TSMC will not give one.

- **Equipment.** TechInsights 2025 revenue by parent-company headquarters, from CSET's ETO Supply Chain Explorer[9].
- **Materials.** Fuji Keizai's survey by parent-company headquarters, as Japan's Ministry of Economy, Trade and Industry published it in December 2025[10].
- **Memory.** Figures from the market research firm IDC, as SK hynix's US listing prospectus relays them[11].
- **Logic capacity.** The Semiconductor Industry Association and Boston Consulting Group's 2022 measurement of logic capacity, below 10 nm for Taiwan and at 28 nm and above for China[4].

The US export-control campaign, 2018 to today

In August 2020 the Commerce Department widened its Foreign Direct Product Rule to require a license for any foreign-made item produced with US technology and bound for Huawei[12]. In 2022 its Bureau of Industry and Security (BIS) blocked China's access to the most advanced AI chips and to the tools to make them, and it tightened those rules in 2023 and 2024[1]. In August 2025 the government began licensing some sales of Nvidia's H20 chip to China, and in February 2026 small amounts of its H200[13].

The Export Controls page has the full timeline, with the legal citation for each rule.

1 Where the Chips Fall: U.S. Export Controls Under the Biden Administration from... Center for Strategic and International Studies, 12 December 2024. csis.org

4 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

9 Documentation: Semiconductor Supply Chain Dataset. Emerging Technology Observatory (ETO), accessed 9 September 2026. eto.tech

10 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

11 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

12 Addition of Huawei Non-U.S. Affiliates to the Entity List, the Removal of... Federal Register, 20 August 2020. federalregister.gov

13 NVIDIA CORP, Form 10-K annual report for the period ended 2026-01-25 (10-K). U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

Eight US actions that shaped chip export controls

- 2019-05-16
  Huawei and 68 affiliates added to the Entity List
- 2022-10-07
  China-wide controls on AI chips and advanced-fab tools
- 2023-10-17
  Performance density added; A800 and H800 captured
- 2024-12-02
  24 tool types, HBM controls, 140 entity listings
- 2025-01-13
  AI Diffusion rule announced, three country tiers
- 2025-05-13
  BIS stops enforcing the AI Diffusion rule
- 2025-08-11
  US officials seek 15% of licensed H20 sales
- 2026-01-15
  H200 and MI325X move to case-by-case review

Source: Federal Register rules, BIS press releases and Nvidia filings, cited for each entry on the Export Controls timeline

Ways around the controls

Renting out time on a chip does not count as exporting it, so a Singapore data center selling time on an H200 already installed there moves nothing across a border[14]. The AI Diffusion rule of January 2025 required a license for advanced AI chips bound anywhere in the world[15]. For the data centers it approved, it said that API access to a model, and rented capacity for inference, "are not prohibited"[15].

The American rules reach that trade through the firms holding the chips. A license under the 15 January 2026 rule for H200-class chips bound for China or Macau requires the applicant to obtain the recipient's know-your-customer procedures[16]. It must also obtain the recipient's list of customers renting computing power in China and seven other places, among them Russia and Iran[16]. The license also depends on a promise that no algorithm trained on the chips is served to blacklisted or military end users[16].

The United States has approved large AI chip sales in the Gulf. In November 2025 Commerce authorized G42 of the United Arab Emirates and HUMAIN of Saudi Arabia to buy the equivalent of up to 35,000 Nvidia GB300 chips each under security and reporting conditions[17]. On 10 July 2026 Commerce moved the Emirates into the export-control group that holds close US allies[18]. It named the Emirati government, G42 and eight US-headquartered AI

14 15 CFR 734.13 -- Export. Electronic Code of Federal Regulations, 3 June 2016. ecfr.gov
15 Framework for Artificial Intelligence Diffusion. Federal Register, 15 January 2025. federalregister.gov
16 Revision to License Review Policy for Advanced Computing Commodities. U.S. Government Publishing Office, 15 January 2026. govinfo.gov
17 Statement on UAE and Saudi Chip Exports. U.S. Department of Commerce, November 2025. commerce.gov
18 Enhanced Favorable Treatment for the United Arab Emirates Under the Export... Federal Register, 14 July 2026. federalregister.gov

companies as recipients that need no license for controlled AI chips and systems[18].

Smugglers route chips past the controls through Southeast Asia. Since 14 July 2025 Malaysia has required a permit to export, transship or transit any high-performance US-origin AI chip[19]. On 25 March 2026 the Justice Department announced charges against three people accused of using Thai companies' names to order servers with export-controlled chips for China[20]. The orders included one for $170 million that was never completed[20].

The House select committee on China cites a median estimate of 140,000 chips smuggled to China in 2024[21]. A bill in Congress, the Chip Security Act, would require location verification on controlled chips within 180 days of enactment[22]. The House Foreign Affairs Committee approved the House version 42-0 on 26 March 2026[23]. It has not become law.

Allies: the Netherlands, Japan, Korea and Europe

US tool controls alone would only shift the sales to Dutch and Japanese suppliers, so the campaign depends on those two governments. Both have aligned their own controls on chipmaking equipment with the American rules[1].

- **The Netherlands.** The Dutch government began licensing advanced deep-ultraviolet tools on 1 September 2023 and widened the controls in September 2024[24]. On 1 April 2025 it extended them to measurement and inspection tools[25].
- **Japan.** Japan's controls on advanced manufacturing equipment took effect on 23 July 2023, apply to all destinations and name no country[26].
- **South Korean fabs in China.** BIS revoked the Validated End-User status that let Samsung's and SK hynix's China subsidiaries and Intel Semiconductor (Dalian) receive US equipment without individual licenses, effective 31 December 2025[27]. BIS said it would license those fabs to keep running, but not to expand capacity or upgrade technology[28]. SK hynix, which bought

1 Where the Chips Fall: U.S. Export Controls Under the Biden Administration from... Center for Strategic and International Studies, 12 December 2024. csis.org

18 Enhanced Favorable Treatment for the United Arab Emirates Under the Export... Federal Register, 14 July 2026. federalregister.gov

19 Export, transshipment and transit of high-performance AI chips of US origin now... Ministry of Investment, Trade and Industry, 14 July 2025. miti.gov.my

20 Chinese National and Two U.S. Citizens Charged with Conspiring to Smuggle... U.S. Department of Justice, 25 March 2026. justice.gov

21 Protecting U.S. Tech: China Committee and Bipartisan, Bicameral Leaders Unite... U.S. House Select Committee on the Chinese Communist Party, 7 July 2025. chinaselectcommittee.house.gov

22 S.1705 – Chip Security Act, 119th Congress (2025-2026), bill text. U.S. Congress, 8 May 2025. congress.gov

23 UNANIMOUS: Bipartisan Huizenga Legislation to Curb AI Chip Smuggling Passes... U.S. House of Representatives, 26 March 2026. huizenga.house.gov

24 The Netherlands expands export control measure for advanced semiconductor... Government of the Netherlands, 6 September 2024. government.nl

25 Klever: export controls on advanced semiconductor manufacturing equipment to be... Government of the Netherlands, 15 January 2025. government.nl

26 CSIS Translation: Updated Japanese Export Controls on High-Performance... Center for Strategic and International Studies, 18 July 2023. csis.org

27 Revocation of Validated End-User Authorizations in the People's Republic of... Federal Register, 2 September 2025. federalregister.gov

28 Department of Commerce Closes Export Controls Loophole for Foreign-Owned... U.S. Bureau of Industry and Security, 29 August 2025. bis.gov

the Dalian fab from Intel, received an annual BIS license in December 2025 to ship approved US equipment to its China fabs during 2026[29].

A BIS rule effective 29 September 2025 extended Entity List restrictions to any company at least 50 percent owned by one or more listed firms[30]. On 30 September the Dutch economy minister invoked the Goods Availability Act against Nexperia, a Nijmegen chipmaker that supplies Europe's car industry[31]. He cited governance failures that risked moving technology out of Europe[31].

Both were suspended in November. On 10 November BIS suspended its rule until 9 November 2026[32]. The minister suspended his order on 19 November[33]. Europe holds a chokepoint in leading-edge lithography, but none in the mature-node chips its car industry uses.

The MATCH Act, a bill introduced on 8 April 2026, would ban sales of immersion deep-ultraviolet scanners to China[34]. It would also name five Chinese companies, CXMT, Hua Hong, Huawei, SMIC and YMTC, in statute[34]. It would give allies 150 days to match those limits before the United States extends the Foreign Direct Product Rule on its own[34].

China's response

In 2024 China's Ministry of Industry and Information Technology reportedly told Chinese telecom operators to strip foreign chips out of their networks by 2027[2].

China has also taken measures against American suppliers. Micron, the American memory maker, failed a review by the Cyberspace Administration of China (CAC) on 21 May 2023 and lost its critical-infrastructure buyers in China[35]. In September 2025 the CAC reportedly told firms including ByteDance and Alibaba to cancel Nvidia orders[36]. When the United States licensed H200 sales in February 2026, China restricted them, as it had restricted licensed H20 sales[3].

China's Ministry of Commerce (MOFCOM) has also restricted exports of minerals.

- **3 December 2024.** MOFCOM banned exports of gallium, germanium, antimony and superhard materials to the United States, one day after the

2 China’s Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

3 NVIDIA CORP, Form 10-Q quarterly report for the period ended 2026-07-26 (10-Q). U.S. Securities and Exchange Commission, 26 August 2026. sec.gov

29 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

30 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. Federal Register, 30 September 2025. federalregister.gov

31 Minister of Economic Affairs invokes Goods Availability Act. Government of the Netherlands, 12 October 2025. government.nl

32 One Year Suspension of Expansion of End-User Controls for Affiliates of Certain... Federal Register, 12 November 2025. federalregister.gov

33 Update on invoking Goods Availability Act. Government of the Netherlands, 19 November 2025. government.nl

34 Risch, Ricketts, Kim Introduce MATCH Act; Level the Global Playing Field for... U.S. Senate Committee on Foreign Relations, 8 April 2026. foreign.senate.gov

35 Micron Aggression: The Right Response to Beijing’s Ban on the U.S. Chipmaker. Center for Strategic and International Studies, 22 June 2023. csis.org

36 Beijing blocks Nvidia sales in vote of confidence for Chinese chipmakers, the... Center for Security and Emerging Technology, 18 September 2025. cset.georgetown.edu

American chip rules[37].

- **4 April 2025.** It required export licenses for seven medium and heavy rare earths[38].
- **9 October 2025.** It announced licensing, from 1 December 2025, for foreign-made goods containing Chinese rare earths worth 0.1 percent or more of their value[39]. MOFCOM suspended that rule on 7 November 2025 until 10 November 2026[40].
- **9 November 2025.** MOFCOM suspended its ban on exports of gallium, germanium, antimony and superhard materials to the United States until 27 November 2026[41]. It acted after Donald Trump and Xi Jinping met at the Asia-Pacific summit in October 2025[42].

China repealed none of these controls, and the April licensing rules still apply.

Chinese toolmakers held 10 to 15 percent of their home market before 2022, 25 percent in 2024 and 35 percent in 2025[2]. They hold about 40 percent of the Chinese market for etch and deposition tools[2]. But in September 2025 the Center for Strategic and International Studies (CSIS) judged that making its own extreme-ultraviolet machines remained firmly out of China's reach despite massive state investment[43].

Chinese chipmakers remain behind at the leading edge. SMIC's most advanced process, N+3, reaches the transistor density of TSMC's 6 nm-class process without extreme-ultraviolet scanners[44]. It prints its tightest layers in several passes on deep-ultraviolet tools[44].

In September 2025 the research firm SemiAnalysis conservatively put SMIC's capacity at 7 nm and below at 45,000 wafers a month by the end of 2025[45]. It expected 60,000 in 2026 and 80,000 in 2027[45]. Those figures count wafers, and how many working chips they give depends on yield, the share of chips on a wafer that work.

SemiAnalysis also expected Huawei to ship about 805,000 of its Ascend AI chips in 2025, with dies made by both TSMC and SMIC[45]. It judged that memory would limit China's output of AI chips before logic wafers did[45]. By its count, a stockpile of roughly 13 million memory stacks was enough for about 1.6 million Ascend 910C packages[45]. It expected CXMT, China's own memory

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

37 Ministry of Commerce Notice 2024 No. 46: Notice Concerning Strengthening... Center for Security and Emerging Technology, 3 December 2024. cset.georgetown.edu

38 Announcement No.18 of 2025 of The Ministry of Commerce and The General... Ministry of Commerce of the People's Republic of China, 4 April 2025. english.mofcom.gov.cn

39 Ministry of Commerce Notice 2025 No. 61: Announcement of the Decision to... Center for Security and Emerging Technology, 9 October 2025. cset.georgetown.edu

40 MOFCOM and GAC Announcement 2025 No. 70, suspending the October 2025... Ministry of Commerce of the People's Republic of China, 7 November 2025. mofcom.gov.cn

41 商务部公告2025年第72号 公布调整实施商务部公告2024年第46号的决定. Ministry of Commerce of the People's Republic of China, 9 November 2025. mofcom.gov.cn

42 The U.S.-China Trade Truce Has Not Solved the Gallium Problem. Center for Strategic and International Studies, 11 May 2026. csis.org

43 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

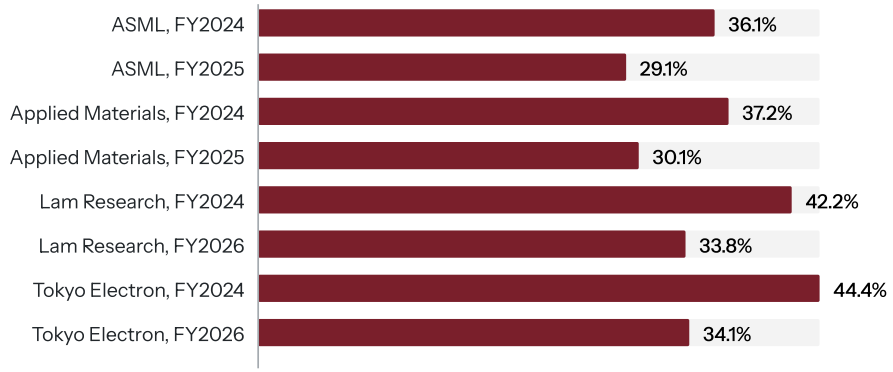
44 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

45 Huawei Ascend Production Ramp: Die Banks, TSMC Continued Production, HBM is The... SemiAnalysis, 8 September 2025. newsletter.semianalysis.com

maker, to make only about 2 million stacks in 2026, enough for 250,000 to 300,000 more[45].

Huawei said in September 2025 that its Ascend 950 chips, due in 2026, would carry high-bandwidth memory of its own design[46]. It gave no figure for how much of that memory it would make[46].

China's share of revenue at the four largest chip-tool makers, peak fiscal year and latest %



Source: ASML 2025 Form 20-F (total net sales), Applied Materials and Lam Research 10-K filings and Tokyo Electron FY2024 and FY2026 consolidated financial statements, each China share computed from the filing's sales by region

Industrial policy: CHIPS, Rapidus, Big Fund and the rest

The CHIPS and Science Act authorized about $52.7 billion[2]. By July 2025 Commerce had awarded $30.9 billion of direct funding to 19 companies for 40 projects, covering a median 14.2 percent of each project's capital cost[48]. Commerce counted a federal tax credit, the Advanced Manufacturing Investment Credit, when it sized those grants[48]. Since the One Big Beautiful Bill Act of 2025, the credit covers 35 percent of qualified investment in an advanced chip plant placed in service after 2025, up from 25 percent[48].

On 22 August 2025 the government agreed to convert $5.7 billion of unpaid Intel grants and $3.2 billion of Secure Enclave defense money into $8.9 billion of Intel stock, a 9.9 percent stake[49].

The United States has also used tariffs to draw chip investment. Under a memorandum signed in January 2026, Taiwanese chip and technology firms are to invest $250 billion in US semiconductor, energy and AI production[50]. In the same memorandum the United States agreed to cap some of its national-security tariffs on Taiwanese goods at 15 percent[50]. On 16 July 2026 TSMC announced another $100 billion for four more Arizona fabs, bringing its total Arizona investment to $265 billion[51].

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

45 Huawei Ascend Production Ramp: Die Banks, TSMC Continued Production, HBM is The... SemiAnalysis, 8 September 2025. newsletter.semianalysis.com

46 Groundbreaking SuperPoD Interconnect: Leading a New Paradigm for AI... Huawei, 18 September 2025. huawei.com

48 GAO-26-107882, SEMICONDUCTORS: Information on Projects Funded to Strengthen... U.S. Government Accountability Office, 11 December 2025. files.gao.gov

49 Intel and Trump Administration Reach Historic Agreement to Accelerate American... Intel, 22 August 2025. intc.com

50 Implementing Certain Tariff-Related Elements of a Trade and Security Agreement... Federal Register, 28 May 2026. federalregister.gov

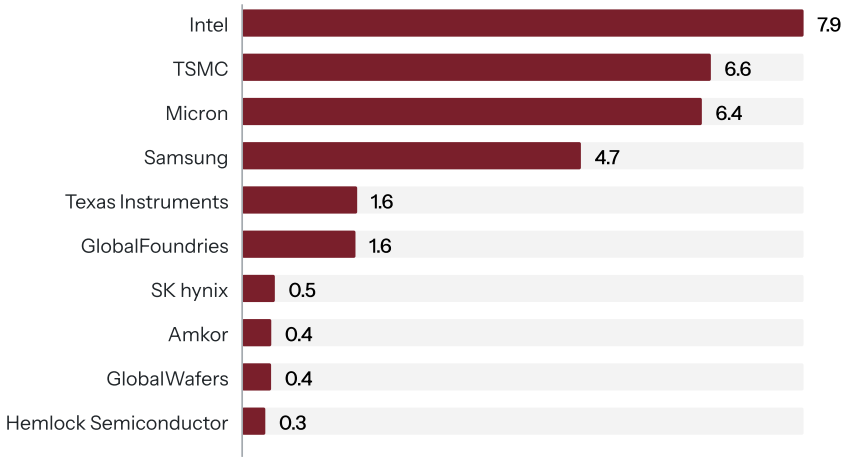
51 TSMC Announces Additional $100 Billion Investment In Arizona. City of Phoenix, 16 July 2026. phoenix.gov

The European Commission says the EU still depends on other countries for advanced chip manufacturing[52]. On 3 June 2026 it proposed a Chips Act 2.0, which would add measures to support advanced chip production in the EU[52].

Japan instead backs a homegrown chipmaker, Rapidus. It began prototyping 2 nm gate-all-around transistors at its IIM-1 plant in July 2025[53]. In February 2026 it raised another 267.6 billion yen, about $1.7 billion, from a government agency and 32 companies, aiming for 2 nm mass production by 2027[54]. By June 2026 it had raised 424.95 billion yen in equity, the latest 150 billion yen from a state agency[55].

China spends more than the United States. The latest round of its National Integrated Circuit Industry Investment Fund, the Big Fund, reached $47 billion[56]. CSIS puts cumulative state funding since 2014 at an estimated $150 billion and calls the CHIPS Act substantially smaller in scale[2].

CHIPS Act direct funding awarded, by firm, as of July 2025 $B



Source: GAO-26-107882, Semiconductors: Information on Projects Funded to Strengthen U.S. Supply Chain

Taiwan

No policy has reduced the world's reliance on Taiwan. Taiwan's foundries held 81.9 percent of the world's contract chipmaking market in 2025, according to the Taiwan Semiconductor Industry Association's August 2026 report[57]. TSMC makes its 2 nm chips only in Taiwan[58]. Taiwan held 69 percent of the world's logic capacity below 10 nm in 2022[4].

Advanced packaging is in Taiwan too. Asked in October 2025 how much CoWoS packaging capacity he would add, TSMC's chairman said only that the

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

4 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

52 Proposal for the Chips Act 2.0. European Commission, 3 June 2026. digital-strategy.ec.europa.eu

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58 Why Taiwan Fears 'America First' Risks Eroding Its 'Silicon Shield'. Stimson Center, 10 October 2025. stimson.org

company was "working very hard to narrow the gap between the demand and supply"[59]. He called wafer and packaging capacity both "very tight"[59].

The silicon shield argument holds that this concentration deters an attack on Taiwan, because so many countries rely on its chips[58]. The Stimson Center calls that open to debate, though many Taiwanese believe it[58].

With an eye to that shield, Taiwan's rules keep TSMC's newest process at home first, so TSMC's Arizona fabs have run a generation behind its fabs in Taiwan[58]. The Stimson Center expected 2 nm production in Arizona in 2028 at the earliest[58]. TSMC said in October 2025 that it was "preparing to upgrade our technologies faster to N2 and more advanced process technologies in Arizona"[60]. The four Arizona fabs it announced in July 2026 are to run at 2 nm or more advanced[51].

The Arizona fabs narrow the gap in wafer production, and TSMC has two advanced packaging fabs planned there[61]. But TSMC's work in Taiwan rests on a local supply chain of more than 300 firms, which the Stimson Center judged extremely difficult to copy in the United States[58]. The US push has raised fears in Taiwan that the shield could weaken[58].

Who gains and who pays

- **Taiwan and Korea.** None of these policies will move leading-edge logic, memory stacking or advanced packaging out of Taiwan and Korea before 2030. Reshoring adds capacity elsewhere without taking any away from them.
- **The Netherlands and Japan.** Their toolmakers build the controlled machines, so the two governments gain a say in who gets them. China accounts for a falling share of the firms' sales. At ASML, China's share of net system sales, which leave out service and field options, fell from 41 percent in 2024 to 33 percent in 2025[62]. Against total net sales, the measure in the chart, it fell from 36.1 percent to 29.1 percent[47]. At Tokyo Electron, China's share of sales fell from 44.4 percent in the year to March 2024 to 34.1 percent in the year to March 2026[63]. The company said China's share of its sales dropped to 26.8 percent in the quarter to March 2026 because spending on leading-edge chips grew faster than spending on older ones[64].
- **The United States.** Its firms took 65 percent of the world's logic chip sales in 2022[65]. The Semiconductor Industry Association, using data from World Semiconductor Trade Statistics, Gartner and Omdia, put US firms' share of

47 ASML HOLDING NV, Form 20-F annual report for the period ended 2025-12-31... U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

51 TSMC Announces Additional $100 Billion Investment In Arizona. City of Phoenix, 16 July 2026. phoenix.gov

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65 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

all chip sales in 2025 at 53.4 percent, the highest since 1984[66]. Commerce expects the CHIPS-funded projects to lift the US share of global leading-edge logic chip manufacturing from 0 percent in 2022 to 20 percent by 2030[48].

- **China.** Mature-node chips, tools below the leading edge, DRAM and NAND are moving to Chinese suppliers. In September 2025 SemiAnalysis expected China, without smuggled memory, to make fewer Ascend chips in 2026 than in 2025[45].

FIVE THINGS TO REMEMBER

- An export control works only when a country's firms sell something the buyer cannot get elsewhere.
- US controls also cover some foreign goods made with American technology.
- The controls have kept China below the leading edge. But Chinese toolmakers' share of China's own chipmaking equipment market rose from 10 to 15 percent before 2022 to 35 percent in 2025.
- China's share of ASML's and Tokyo Electron's sales has fallen since 2024.
- Taiwan held 69 percent of the world's logic capacity below 10 nm in 2022, and TSMC makes its 2 nm chips only there.

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A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

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